



**GREEN
CLIMATE
FUND**

Meeting of the Board
16– 19 May 2022
Antigua and Barbuda
Provisional Agenda Item 05

GCF/B.32/02/Add.03

25 April 2022

Consideration of funding proposals – Addendum III

Funding proposal package for FP186

Summary

This addendum contains the following six parts:

- a) A funding proposal summary titled “India E-Mobility Financing Program” submitted by Macquarie Alternative Assets Management Limited (MAAML);
- b) No-objection letters issued by the national designated authorities or focal point(s);
- c) Environmental and Social report(s) Disclosure;
- d) Independent Technical Advisory Panel’s assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel’s assessment; and
- f) Gender documentation of the funding proposal.

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled “India E-Mobility Financing Program” submitted by Macquarie Alternative Assets Management Limited (MAAML) is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

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Funding Proposal

Project/Programme title: India E-mobility Financing Program

Country(ies): India

Accredited Entity: Macquarie Alternative Assets Management Limited (MAAML)

Date of first submission: [2021/09/13]

Date of current submission: [2022/04/25]

Version number: [V.4]



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the **GCF Information Disclosure Policy**, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in *section G.4*.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-MAAML-India-[2022/04/25]"

A. PROJECT/PROGRAMME SUMMARY							
A.1. Project or programme	Programme	A.2. Public or private sector	Private				
A.3. Request for Proposals (RFP)	If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+. Not applicable						
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.						
		GCF contribution	Co-financers' contribution ¹				
	Mitigation total	Enter number %	Enter number %				
	☐ Energy generation and access	Enter number %	Enter number %				
	☒ Low-emission transport	100.0 %	100.0 %				
	☐ Buildings, cities, industries and appliances	Enter number %	Enter number %				
	☐ Forestry and land use	Enter number %	Enter number %				
	Adaptation total	Enter number %	Enter number %				
	☐ Most vulnerable people and communities	Enter number %	Enter number %				
	☐ Health and well-being, and food and water security	Enter number %	Enter number %				
	☐ Infrastructure and built environment	Enter number %	Enter number %				
	☐ Ecosystems and ecosystem services	Enter number %	Enter number %				
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	9.5 MtCO ₂ ²	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	<table border="1"> <tr> <td>NA</td> <td>NA</td> </tr> <tr> <td>NA</td> <td>NA</td> </tr> </table>	NA	NA	NA	NA
NA	NA						
NA	NA						
A.7. Total financing (GCF + co-finance ³)	1,497 million USD	A.9. Project size	Large (Over USD 250 million)				
A.8. Total GCF funding requested	200 million USD						

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project / programme implementation period.

³ Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8. ☐ Grant Enter number ☒ Equity USD 200 million ☐ Loan Enter number ☐ Results-based payment Enter number ☐ Guarantee Enter number		
A.11. Implementation period	10 years from Funded Activity Agreement (FAA) effective date. This may be extended for a further two consecutive two-year periods following appropriate approvals which will be elaborated in the FAA.	A.12. Total lifespan	Electric buses and four wheelers have a lifespan of ~15 years. Electric Vehicle (EV) charging stations have a lifespan of ~25 years. Given the nascency of the technology, there is wide ranging views on these estimates.
A.13. Expected date of AE internal approval	120 days from GCF Board Approval.	A.14. ESS category	Refer to the Accredited Entity's (AE's) safeguard policy and GCF ESS Standards to assess your Funding Proposal (FP) category. I-2
A.15. Has this FP been submitted as a CN before?	Yes ☐ No ☒	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☐ No ☒
A.19. Complementarity and coherence	Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1. Yes ☐ No ☒		
A.20. Executing Entity information	As per the envisaged transaction structure, there will be multiple Executing Entities across Singapore and India (refer to section B.4 – Implementation arrangements for further details).		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			
Climate change is one of the biggest challenges being faced by India. Between 1901-2018, India's average temperature rose by 0.7°C⁴ primarily led by an increase in Greenhouse Gases (GHG) emissions, which have more than doubled from 1.2 GtCO₂ in 2005 to 2.6 GtCO₂ in 2018.⁵ This rapid growth in GHG emissions has been driven by multiple factors, such as high population growth rate, increasing levels of industrialisation and urbanisation. This increase in average temperature has already affected the livelihood of millions of people, with more frequent sporadic climatic events such as cloud bursts, flash floods, prolonged droughts, etc. becoming a norm. By 2100, it is projected that temperatures will rise by approximately 4.7°C to 5.5°C⁶ resulting in an exponential increase in extreme climatic events, vector borne diseases, food shortage, migration, and loss of livelihoods, all of which will have adverse impact on India's economic development. According to Oxford Economics, India could lose 90% of its Gross Domestic Product (GDP) by 2100 if active steps are not taken to mitigate the GHG emissions⁷ (see mitigation profile of India under section B.1, as well as discussions under section B.2 and D.4).			

⁴ <https://reliefweb.int/report/india/assessment-climate-change-over-indian-region-report-ministry-earth-sciences-moes>

⁵ <https://www.eurekalert.org/news-releases/914976>

⁶ <https://reliefweb.int/report/india/assessment-climate-change-over-indian-region-report-ministry-earth-sciences-moes>

⁷ <https://www.livemint.com/news/india/india-s-gdp-could-fall-90-by-2100-due-to-climate-change-analysis-11594187614729.html>

Whilst the Indian government has been taking proactive steps pursuing various mitigation and adaptation activities in the form of strong policy support for renewable energy and electric mobility sectors to reduce GHG emissions, further support is required to help India achieve its Nationally Determined Contributions (**NDCs**) agreed under the Paris Agreement and COP26 pledges. According to a recent report by Lifegate,⁸ India needs investment of approximately USD 2.5 trillion between 2015 and 2030, or USD 170 billion / year, to meet its NDCs by 2030. Further, a recent study by Climate Policy Initiative (**CPI**) reports that India is currently mobilising less than 25% of the investment needed to reach this target⁹.

Additionally, since early 2020, the onset of the COVID-19 pandemic has pushed the government to change its priorities, shifting substantial focus to creating an effective healthcare ecosystem to deliver quality primary care to citizens. Due to the recessionary impact of the COVID-19 pandemic, along with other socio-economic issues, government finances remain constrained. In FY21, India's fiscal deficit was 9.3%.¹⁰ This is projected to be 6.8% in FY22,¹¹ significantly above its target of 3.0% set under the Fiscal Responsibility and Budget Management (**FRBM**) Act 2003 (a law enacted to institutionalise financial discipline, strengthen fiscal prudence and improve macroeconomic management).¹² This may cause a substantial budgetary shortfall, limiting the government's ability to accelerate and deliver GHG emission reducing projects going forwards.

In the above context, the proposed Programme (the **Programme**) aims to leverage GCF and commercial investor capital to accelerate the decarbonisation of the Indian economy by focusing on **electrification of India's transport sector**. This will be enabled by supporting faster adoption of electric vehicles and growth of associated infrastructure, replacing, and limiting the new purchase of, traditional Internal Combustion Engine (**ICE**) vehicles.

The main objectives of the Programme are as follows:

1. to provide suitable financial solutions (for example, operationally integrated, performance guaranteed leases or structured credit instruments) to electric vehicle owners and operators;
2. to provide tailored financial solutions to ancillary areas to further develop the electric vehicle ecosystem, such as charging infrastructure and batteries;
3. using financial solutions to mitigate current user concerns, including high upfront costs, residual value risk and reliability;
4. to facilitate development of EV ecosystem through investments, partnerships, and capability development of market participants;
5. to set a high benchmark and continually improve the efficiency of services in the Indian EV domain via improvement of systems, processes, and resources;
6. to provide a competitively priced solutions that enhance the long-term Total Cost of Ownership (**TCO**) competitiveness of EVs vis-à-vis ICE vehicles, in alignment with the Indian government's policy; and
7. to mobilise significant amount of nascent institutional capital into the electrification of India's road transport sector.

The Programme is seeking USD 200 million from GCF, and an additional USD 205 million from commercial investors, to set up an India focused E-mobility Financing Platform (the **Platform**).¹³ It will be managed by Macquarie Asset Management (**MAM**), the world's largest infrastructure asset manager,¹⁴ with an established presence in India. If successful, it is anticipated that the Platform will access local and international debt markets for ~USD 1.1 billion to further grow its operations, and may draw in additional equity capital from commercial investors as EV-ICE cost parity approaches transitioning to an unsubsidised enduring commercial enterprise.

The Programme is innovatively structured to a) address some of the key barriers limiting the accelerated growth of the EV sector, b) complement existing initiatives of the government and other private players, and c) align with India's NDCs, National Climate Action Plan (**NCAP**), COP26 pledges, as well various other policy and regulatory initiatives.

⁸ <https://www.lifegate.com/india-climate-finance-sdgs>

⁹ CPI, 2020

¹⁰ <https://www.livemint.com/economy/indias-fiscal-deficit-at-9-3-of-gdp-in-fy21-down-from-revised-estimate-of-95-11622459720813.html>

¹¹ <https://economictimes.indiatimes.com/news/economy/indicators/india-to-breach-fiscal-deficit-target-in-fy22-fitch-solutions/articleshow/82451801.cms?from=mdr>

¹² <https://timesofindia.indiatimes.com/business/faqs/budget-faqs/budget-what-is-frbm-act-why-is-it-in-focus-this-year/articleshow/80454944.cms>

¹³ Establishment of the Platform is subject to Macquarie Group approvals.

¹⁴ IPE Real Assets 2021 – Top 100 Infrastructure Investment Managers.

The Programme aims to introduce **first of its kind financing solutions** (operating and finance leases and other structured credit solutions) into, potentially, one of the largest EV markets in the world, to reduce high upfront capex associated with EVs, impediments around EV charging infrastructure and uncertainty around commercial EV performance. Presently, these are not offered at scale by existing market participants. The constrained availability of suitable financial products is primarily due to a limited understanding of the EV ecosystem, the relatively small current market size and residual value concerns. The proposed Programme is poised to change that. By enhancing the availability of credit and introducing new financial solutions in the EV market that accept operational performance and other risks, the Programme seeks to increase the attractiveness of EV ownership, help Original Equipment Manufacturers (**OEMs**) / sponsors recycle capital faster and cause EVs to become the preferred mode of transport among fleet operators.

Initially the Programme will focus on select segments of the EV ecosystem, such as e-buses, shared fleets and electric charging infrastructure. It will then slowly diversify its portfolio and potentially expand into retail segment as more sub-sectors within the EV value chain gain scale and prominence. Over the implementation period, the Programme is expected to mobilise up to **USD 2.1 billion**¹⁵ of capital into the sector, and targets to reduce **~9.5 MtCO₂e** over its lifetime.

Apart from the above, the Programme will assist India to achieve energy security by reducing its dependence on oil imports, which forms ~80% of the total import bill of the country.¹⁶ It will also help in the creation of new jobs and stimulate economic activity, with a corollary of empowering women and reducing gender bias. Any capital investment and support for the local economy will be extremely critical during the times of pandemic, where millions of people have lost jobs and are pushed to poverty each day. Lastly, it will promote environmental sustainability and increase health benefits through the reduction of pollution. It is also noteworthy that the Executing Entities, together with the Accredited Entity, will promote strong corporate governance standards and manage fiduciary risks in line with GCF policies and requirements (as agreed in GCF's Accreditation Master Agreement (**AMA**) and Funded Activity Agreement (**FAA**)) as well as international best practices for climate impact planning and monitoring, environmental and social management, and gender mainstreamed approach.

¹⁵ Includes redeployment of internal accruals

¹⁶ Economic Times Publication on "India's crude oil import – April 2020".

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Overall climate rationale

The aim of this Programme is to facilitate the reduction of greenhouse gases and other emissions from the road transport sector in India through the accelerated adoption of electric vehicles. This can be enabled through targeted investments across the e-mobility value chain, which can help address some of the key issues impeding the sector, such as high upfront capex, technology / obsolescence risk, range and operational performance anxiety, inadequate availability of charging infrastructure as well as financing solutions through the existing channels. The envisaged Programme can help create an enabling environment for local investment and ultimately catalyse the scaling up of the nascent market.

Climate context

How has India's climate already changed?

India has varied weather and climatic patterns across the country, with an average temperature of 2°C¹⁷ in Jammu and Kashmir (North India) vis-à-vis 29°C¹⁸ in southern India. The amount and frequency of rainfall is equally diverse, with parts of Meghalaya receiving more than 4,000 mm¹⁹ of precipitation annually, compared to parts of Thar Desert in north-western India which receive less than 100 mm²⁰ of rainfall every year. The rainfall in each region is primarily affected by two weather systems, South-West Monsoon (June to September) which brings the majority of the rainfall in the country, and North-East Monsoon (October to December) which plays an important role in southern India.

Given the size, topography, and climatic diversity across India, the impact of climate change on each region is varied. According to a recent Indian government report, the average temperature across the country increased 0.62°C over the last 100 years.²¹ Even though the temperature increase is lower than the global average, severe impacts are already being felt.

Between 1985 and 2009, western and southern India experienced 50% more heatwaves than in the previous 25 years.²² Between 1950 and 2015, central India experienced a threefold increase in extreme rainfall and flood events, leading to death of thousands and displacement of millions.²³ This is in contrast to 6% decline in average precipitation during monsoon events between 1951 and 2015, which has been primarily been offset by melting glaciers of the Hindu Kush region, which have retreated at an average 18 metres per year over the last four decades.²⁴ Further, with rising levels of urbanisation, economic activity, and variations in weather patterns across the country, water is becoming more scarce with each passing day. Today, one billion people in India experience severe water scarcity for at least one month each year, with 180 million facing water scarcity all year round.²⁵

How will India's climate change going forward?

As climate change continues, its impact will be more profound. For India, the intensity of the impact of climate change will increase due to rising levels of greenhouse gases, led by rapid population growth, urbanisation, strong industrial growth, and a significant increase in average household income.

According to a recent report, the level of precipitation, especially during monsoons, and glaciers in the Hindu Kush region will significantly reduce over the next few decades. Consequently, the flow of water in some of the main river systems, such as Indus, Ganges and Brahmaputra, will fall by 8.4%, 17.6% and 19.6% respectively by mid-century compared to the 2000s. This is expected to result in severe water scarcity and displacement of more than 270 million people in the Indus and Brahmaputra basins.²⁶ According to the WWF's Water Risk Filter Tool,²⁷ by 2050, 30 cities in

¹⁷ <https://cdn.odi.org/media/documents/ODI-JR-CostClimateChangeIndia-final.pdf>

¹⁸ As above.

¹⁹ As above.

²⁰ As above.

²¹ As above.

²² As above.

²³ As above.

²⁴ As above.

²⁵ As above.

²⁶ As above.

²⁷ Water Risk Filter: Launched in 2012, the Water Risk Filter is a practical online tool that helps companies and investors assess and respond to water-related risks facing their operations and investments across the globe (<https://waterriskfilter.panda.org/>).

India are expected to suffer the largest rise in water risk,²⁸ and nearly 36 million people in India will be threatened by flooding annually if greenhouse gas emissions are not reduced.²⁹ Additionally, the level of cyclonic storms in the Bay of Bengal region is expected to double by 2070-2100, compared to the baseline period of 1961-1990, leaving millions of low-income households susceptible to food shortage, loss of livelihoods and vector borne diseases.

The economic impact of the above will be significant. Climate change has already reduced the pace of poverty reduction and increased the level of inequality in the country. According to Burke and Tanutama (2019), the districts that have warmed the fastest have seen their GDPs grow 56% less on average than those that have warmed the slowest. On an overall basis, climate change could result in India's GDP falling 2.6% by 2100³⁰, even if the global temperature increase is below 2°C,³¹ and 13.4% if the temperature increase is 4°C.³² Further, without the current costs of global warming, India's GDP would be 25% higher than its current levels³³. Additionally, Nixon predicts that, if temperatures increase by 3°C,³⁴ India's GDP will be 90% lower in 2100 than it would have been without climate change.

These factors put India at a critical inflection point wherein there is a need to balance economic growth, population health, recovery from the pandemic and GHG emissions to limit climate change impact. India's emissions reduction pathway is paramount to the fight against climate change both locally and globally.

India's fight against climate change

As a result of increasing GHG emissions and exposure to the effects of climate change, the Indian government has put several policies and targets in place to support its contribution to reducing global emissions. This is outlined in India's NDCs and COP26 pledges. Furthermore, the government is willing to take on a more ambitious target based on investment and support from both public and private sectors.³⁵

Nationally Determined Contributions (NDCs)

India's Nationally Determined Contributions (Paris Agreement)

Pledge/ Target	Description
2030 unconditional target	33%-35% below 2005 emissions intensity of GDP by 2030 [479-495% above 1990 by 2030 excl. LULUCF] [177-185% above 2010 by 2030 excl. LULUCF]
2030 conditional target	- Non-fossil share of cumulative power generation capacity by 40% by 2030 [371-373% above 1990 by 2030 excl. LULUCF] [126-127% above 2010 by 2030 excl. LULUCF]
Conditions	Transfer of technology and low cost international finance
LULUCF	Additional cumulative carbon sink of 2.5-3.0 GtCO₂ by 2030 through additional forest & tree cover

COP26 pledges

At the COP26 meeting in Glasgow in November 2021, India pledged itself to five unconditional targets in order to improve its commitments to climate change and lead the world's decarbonisation drive. These include:

1. increase non-fossil energy capacity to 500 GW (gigawatts) by 2030;
2. meet 50 percent of energy requirements from renewable energy by 2030;

²⁸ 30 Indian cities likely to face acute water risks by 2050: WWF (livemint.com).

²⁹ <https://www.deccanherald.com/science-and-environment/36mn-in-india-to-face-threat-of-annual-floods-by-2050-771934.html>

³⁰ Kahn et al, 2019

³¹ <https://cdn.odi.org/media/documents/ODI-JR-CostClimateChangeIndia-final.pdf>

³² Ibid.

³³ Nixon (2020)

³⁴ Ibid.

³⁵ "Reducing Carbon Emissions from Transport Projects" Evaluation Knowledge Brief, ADB, July 2010.

3. reduce the total projected carbon emissions by 1 billion tonnes by 2030;
4. reduce the carbon intensity of the economy by less than 45 percent; and
5. achieve net zero carbon by 2070.

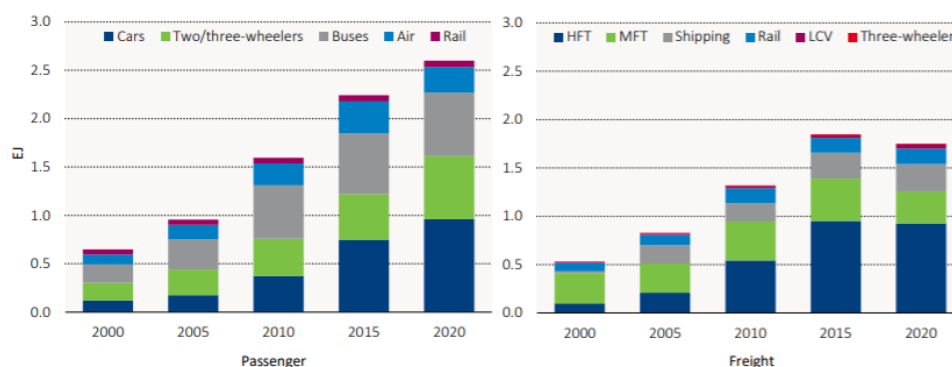
The role of transport sector in emission generation

Present status

According to an Asian Development Bank (ADB) report, the transport sector accounts for 23% of all energy-related CO₂ emissions globally.³⁶ However, in India, it contributes only 13% of total carbon dioxide emissions despite being the third largest GHG emitting sector in the country. Further, as per IEA, carbon-dioxide emissions from India's transport sector have more than tripled since 1990,³⁷ of which road transport contributes ~90%. This is driven by:

- *a significant increase in passenger and freight transport activity* – Passenger and freight traffic by inland mode have both tripled between 2000 and 2020 to 6 trillion passenger kms and ~1.8 trillion tonne kms respectively, which has been predominately driven by the road sector,³⁸ and;
- *the sector's high reliance on fossil fuels* – The Indian transport sector has grown 50% since 2010, and now accounts for 20% of the country's final energy use (in terms of exajoules). Oil products contribute 95% of final energy used in the transport sector, with natural gas, biofuels, etc. comprising of the remaining portion. Further, as the energy use has increased over the last two decades, GHG emissions too have increased, implying a strong causal relationship between the two (refer to the below charts for more details).

Figure: Energy use by mode in passenger and freight transport in India (exajoules), 2000-2020³⁹



Note: LCV= light commercial vehicle; MFT= medium freight truck; HFT= heavy freight truck.

Figure: CO₂ emissions by passenger and freight traffic in India (MtCO₂e), 2000-2020⁴⁰

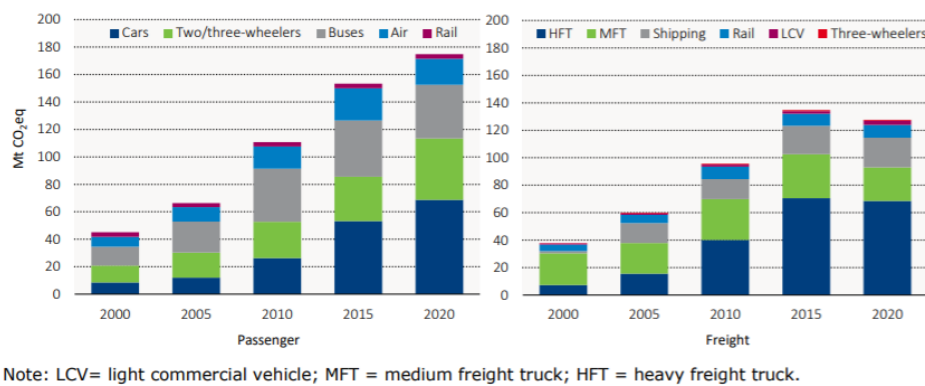
³⁶ Reducing Carbon Emissions from Transport Projects" Evaluation Knowledge Brief, ADB, July 2010 pp 01.

³⁷ <https://www.iea.org/countries/india#overview>

³⁸ International Transport Forum (ITF), "Decarbonising India's Transport System", 2021. ITF analysis prepared with the support of the IEA Mobility model.

³⁹ Sources: ITF analysis prepared with the support of the IEA Mobility Model. Results for freight transport exclude domestic aviation. ITF analysis based on activity data outlined in earlier figures and energy use (in 2018) of 2 litres of gasoline equivalent (Lge) per 100 km for two-wheelers, 3 Lge per 100 km for three-wheelers, 6.4 Lge per 100 km for cars, 36 Lge per 100 km for buses, 7 Lge per 100 km for LCVs, 26 Lge per 100 km for MFTs and 46 Lge per 100 km for HFTs. Energy intensities are 0.04 MJ/pkm for passenger rail services, 0.2 MJ/tkm for freight rail and 1.35 MJ/pkm for passenger aviation. Energy intensities for cars in earlier years reflect variations in energy intensity. Data for 2020 rely on estimations that have a higher degree of uncertainty than earlier years. See also the note to Figures 3 and 4 regarding other data uncertainties. Data for rail are informed by IEA (2019a); for cars, by IEA (2019b); for medium and heavy road vehicles, by IEA (2017) and Delgado et al. (2016); for LCVs, estimates are consistent with data available from Deo (2021); for aviation, data are informed by IEA (2020c) and ICAO (2020).

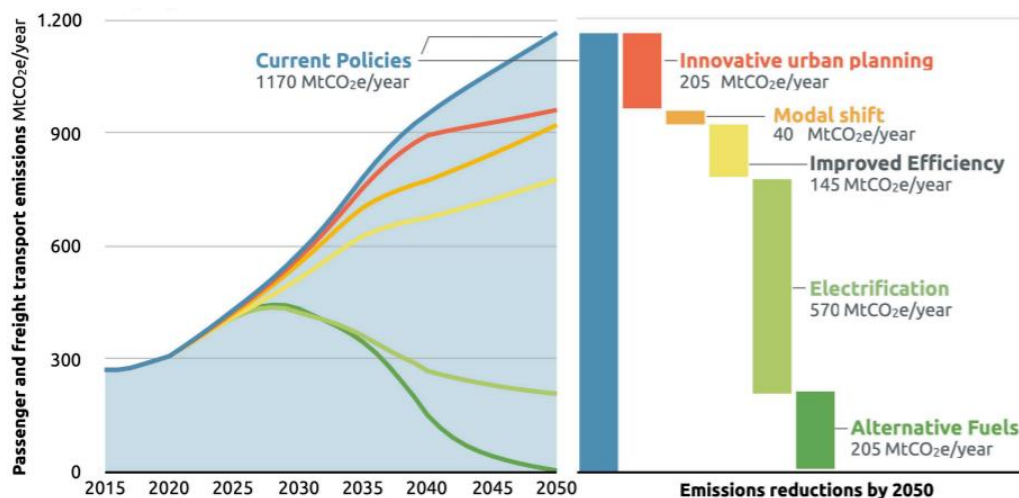
⁴⁰ Sources: ITF analysis prepared with the support of the IEA Mobility Model. Results are based on energy use data outlined in earlier figures. Emissions related with biofuels are informed by the analysis developed in Prussi et al. (2020) with simplified assumptions about attribution to different production pathways. Data for 2020 rely on estimations that have a higher degree of uncertainty than earlier years. See also the note to Figure 1 regarding other data uncertainties. CO₂ intensities for fossil fuels are from IPCC (2019); carbon intensities of electricity generation are from IEA (2020c).



Going forward

To limit the impact of growing GHG emissions from India's road sector, a multi pronged approach is required. According to a report by Climate Action Tracker, the electrification of vehicles is "absolutely essential" if India is to achieve net zero emissions from its road sector by 2050. The electrification of vehicles has the potential to reduce emissions by 570 MtCO₂e/year, which is 2.8 times more of a reduction than can be achieved through innovative urban planning and the use of alternate fuels.⁴¹ However, to achieve this ambitious target, all new vehicles sales will need to be electric by 2035, a mammoth task given the electric vehicle penetration was just 0.7% of total vehicle sales in FY20.

Figure: Annual emissions from the transport sector until 2050 (MtCO₂e)



Whilst both central and state governments have introduced policies and initiatives to promote electric vehicles over the last few years, such as the Faster Adoption and Manufacturing of Hybrid and EV (**FAME**) subsidy scheme I and II, the National Electric Mobility Mission Plan (**NEMMP**), and the vehicle scrappage policy, a lot more needs to be done to achieve the emission reduction targets mentioned above. According to a report by Niti Aayog, Govt. of India's think tank, an estimated USD 266 billion of capital is required to be invested in the EV industry over next 10 years.⁴² To date, only USD 5-6 billion has been invested, the majority of which has been over the last 12 months. There are several key barriers that need to be addressed before significant private capital starts to invest. These include the high upfront cost of EVs, unproven operational performance, limited availability of financing, low product offerings from OEMs and inadequate infrastructure buildout, primarily charging stations (*refer to section B.2 – Theory of Change for more details*), all in the context of a very low operating margin business model. Robust solutions are required to address these issues before the EV industry can fulfil its role as a significant contributor to India's climate change mitigation story.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

⁴¹ Decarbonising the Indian transport sector pathways and policies, Climate Action Tracker, December 2020.

⁴² <https://rmi.org/insight/mobilizing-finance-for-evs-in-india/>

The Programme intends to support the reduction of greenhouse gas emissions in the transport sector in India, aligned to GCF's results management framework. This is expected to be achieved by providing enhanced and innovative financing (leasing and other financing products) across the e-mobility value chain, which will directly or indirectly overcome the current barriers to increased electrification of vehicles in India.

Barriers

There is a need to address the various challenges that have limited the e-mobility sector's growth potential and prevented it from becoming a mainstream sector. Some of the key factors affecting the EV adoption are:

1. *Availability of financing:*⁴³ Improved access to attractive financing products will be a key growth driver for EV demand. High financing costs and uncertainty around long-term economics, including residual value, remain both real and perceived issues for financial investments. Risks associated with the nascency of the electric mobility ecosystem have given rise to investment barriers, such as high interest and insurance rates, short tenure, low loan-to-value ratios, and requirements for high collateral value or corporate guarantees, all limiting retail / corporate customer interest. These issues are most pronounced for electric buses and four wheelers, and charging infrastructure, all of which have longer gestation periods and high per unit capex involvement. The envisaged Programme will seek to provide bespoke financing products in the target segments across the value chain to overcome these barriers.

The investments made by the Platform (activity II below) will provide new financing options, improving the availability of, and risks assumed by, financing.

2. *Upfront cost:* The high upfront cost of EVs is slowing down its adoption rate, despite lower running and maintenance costs associated with EVs vis-à-vis ICE vehicles. Demand for more affordable EV products is also expected, as consumers and businesses reduce spending in the short term due to COVID-19. This potential shift in consumer preferences may affect manufacturers' investment and production decisions,⁴⁴ which may slow the entire EV value chain evolution in India. Providing attractive financing solutions for better adoption by lowering the upfront capital outlay is required to accelerate growth of the sector, which is in its nascency in India. Furthermore, increased investment in the sector is expected to support a greater flow of capital into research and development, which should lower the cost of new EVs over time.

The financing provided by the Platform (activity II below) is intended to target a substantial reduction in the upfront cost paid by the customers of EVs. The investments in charging infrastructure section (activity II) will increase India's capacity to support more growth in EVs. Similarly, the increased EVs financed by the Platform will support the development of charging infrastructure, by the Platform or otherwise.

3. *Infrastructure buildout:* Charging infrastructure is the backbone of electric mobility, but is also one of the key perceived barriers to EV adoption in India given its limited availability and long charging times. High operating costs, uncertain utilisation rates and unproven business models are some of the key challenges holding back existing and new EV charging stations operators from expanding their current presence. However, as technology develops, EV adoption continues, and financing becomes available, electricity distribution companies (**DISCOMs**), charging service providers, and other stakeholders can focus on building robust charging infrastructure networks. The Platform will seek to work with these stakeholders to finance charging infrastructure projects to propel overall EV adoption.
4. *Manufacturing and supply:* EV battery manufacturing in India is still largely dependent on imports due to a lack of domestic supply of lithium, nickel and cobalt, the key raw materials and major cost drivers of EV batteries. Domestic manufacturing of advanced batteries and cells, electric motors, and other EV components are required to reduce the high cost of importing these critical components and equipment from international markets. OEM capital has recently focused on the migration of ICE vehicles to Bharat Stage VI (**BSVI**) standards,⁴⁵ whilst the industry is experiencing lower sales due to COVID-19. This is hampering supply-side investment in EVs. The Platform's investment in the EV space will provide OEMs / operators the necessary liquidity to recycle their capital and invest in enhancing their manufacturing and research and development (**R&D**) capabilities. The Platform will encourage this to be done in accordance with latest Environmental, Social and Governance (**ESG**) standards.

⁴³ Mobilising Finance for EVs in India: A Toolkit of Solutions to Mitigate Risks and Address Market Barriers, Niti Aayog and Rocky Mountain Institute, January 2021 PP 11.

⁴⁴ Mobilising Finance for EVs in India: A Toolkit of Solutions to Mitigate Risks and Address Market Barriers, Niti Aayog and Rocky Mountain Institute, January 2021 PP 1.

⁴⁵ On 19 Feb 2016, the Ministry of Road Transport and Highway issued a draft notification of Bharat Stage (BS) VI emission standards, equivalent to Euro VI. The standards, as proposed, will take effect throughout the country for all light-duty and heavy-duty vehicles as well as two and three-wheelers manufactured on or after 1 Apr 2020. The BS VI standard specifies 10 ppm sulphur for gasoline and diesel fuels.

Establishment of the Platform and the financing provided by it, is expected to increase the penetration of EVs, support local manufacturing of vehicles in an inclusive and sustainable manner, as well as reduce upfront capital commitment from OEMs to invest further into R&D in the e-mobility space.

5. *Investor risk appetite and availability of investment opportunities:* Increased investment across the e-mobility value chain is expected to support market development, grow existing business models, and encourage additional research and development. Without this investment, the barriers outlined above are expected to persist. Investment into the e-mobility sector in India has not yet achieved scale due to high perceived and real investment risks, and a limited number of investment opportunities that meet the risk appetite of large investors. Despite the increase in the quantum of investment over last few months, the aggregate investment amount is minimal compared to the potential size of the Indian EV market. Risk-taking capital is required to support commercial investors to invest and develop the sector to the point where there is increased commercial and maintenance risks appetite for the sector.

Establishment of the Platform (activity I) provides an investment opportunity for commercial investors to invest in the e-mobility sector in India. Further investment and development of the sector through other activities will also help to reduce investment risks (real and perceived) associated with the sector.

Activities and Outputs

The activities of the Platform, as shown in the diagram below, will immediately and permanently increase the flow of private sector investment into the EV sector, assisting to overcome the barriers outlined above and enable delivery of the intended programme results and desired outcomes.

1. *Platform Establishment and Fundraising:* The E-mobility Financing Programme will facilitate investment from GCF to the end beneficiaries whilst also crowding-in investment from other investors. MAM will use its vast and experienced capital raising capabilities to attract funds from commercial investors alongside GCF, which will be used to capitalise a “first of its kind” e-mobility focused financing platform in India. This platform will adopt best in class risk management, ESG, Diversity and Inclusion and IT and HR policies, along with other industry leading asset management frameworks. GCF’s capital will act as a risk-mitigating buffer providing downside protection to commercial investors. This is expected to help meet the risk-return expectations of commercial investors, consequently resulting in increased interest and higher availability of capital for the e-mobility sector.

Relevant Output	Description
I. Commercial capital crowded-in	By establishing the Platform, investors will have an avenue to invest in the e-mobility sector in India which meets their risk / return requirements.

2. *Platform Investments, Monitoring and Exit:* The Platform will provide operationally integrated financing products to overcome the barriers outlined above, either through scaling-up access to existing products, such as vehicle and project financing, or developing bespoke structures, such as operating and finance leases in the EV domain, to meet the needs of market participants. Financing will be provided across the e-mobility sector which will support an increased number of EVs, charging stations and other associated infrastructure across the e-mobility value chain. With the Platform being a MAM-managed vehicle, it will adopt best in class investment monitoring, workplace health and safety, environmental and social governance and risk management practices, hire an experienced and professional management team and leverage MAM’s industry leading standards for portfolio companies to deliver the desired results for investors. Once proof of concept is established using equity from GCF and commercial sources, it is anticipated that further capital will be mobilised from local and international debt markets, with the potential to raise further equity capital to enable continued growth.

Relevant Output	Description
I. Commercial capital crowded-in	The Platform will support investment made by OEMs / operators / sponsors, enabling capital to be crowded-in.
II. Electric vehicles financed by the Platform	The Platform will finance a number of electric vehicles based on transaction opportunities and market demand across different vehicle segments.

III. Charging stations built or financed by the Platform	The Platform will finance several charging stations based on transaction opportunities and market demand.⁴⁶
III. Investments made in other e-mobility sectors	The Platform may make investments in other e-mobility sectors and EV technologies based on transaction opportunities and market demand.

3. *Wind up of corporate structure:* From an exit point of view, as outlined in section B.6, any partial or full exit from the Platform will be possible only when the EV financing market in India has or is becoming mature and offers high growth potential. At the time of exit, depending upon the market conditions, past performance and future growth estimates, the Platform is expected to attract significant interest from strategic and financial investors. Once either the SG Master Co. sells its stake in the SG Co, or the SG Co sells its stake in the Platform, within the exit period to a new investor, the exit proceeds received will be distributed to the investors, depending upon the cashflow distribution waterfall as provided in the Term Sheet. However, the winding up of the structure may take time, depending upon the nature of indemnities and warranties that may be offered to the buyer at the time of sale.

Relevant Output	Description
I. Commercial capital crowded-in	The sale of the Platform is likely to attract new or existing market players who are looking to expand their investment in the e-mobility space.

Outcomes

This Programme is focused on crowding-in private sector capital to support the scaling-up of investment in e-mobility sector across India. There are two programme level outcomes:

- Enabling environment for EV market established:* Through the demonstration of innovative financing products, increased EVs on the road, and more charging infrastructure, it is expected that there will be an enabling environment for EVs to accelerate further, including an increased number of participants, new financial products, greater infrastructure, and developed domestic manufacturing and supply capabilities.
- Increased penetration of electric vehicles in India:* This is the fundamental outcome sought as a result of all the activities. This will be achieved both directly through investments made by the Platform (increasing number of e-vehicles and charging stations) but also indirectly through crowding-in commercial capital that is expected to expand operations and make independent investments (not linked to the Platform) which in turns will also increase the penetration of e-vehicles.

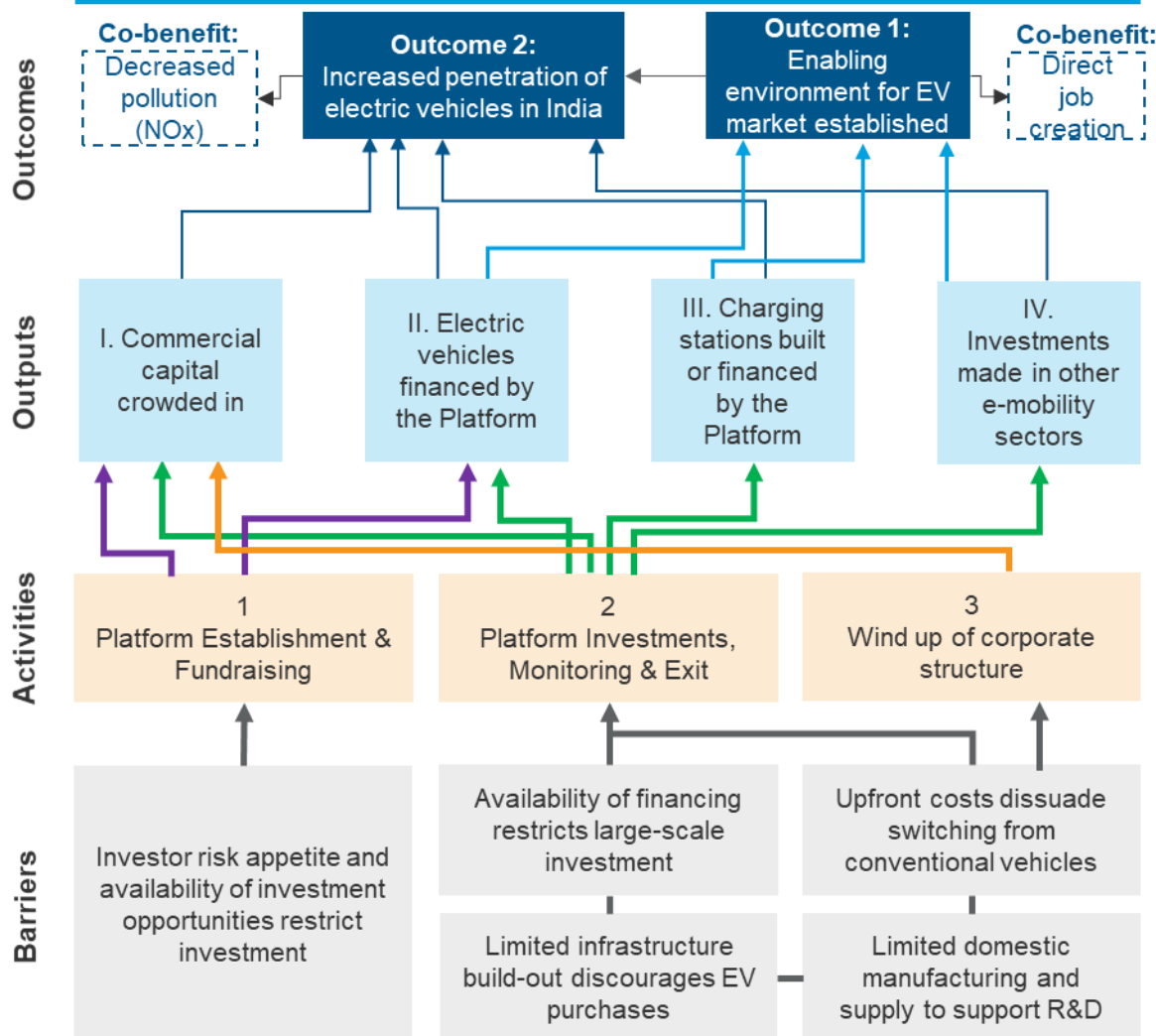
Co-benefits

- Decreased pollution:* As outlined in section D.3 (co-benefits), greater penetration of EVs is expected to have a significant impact on air quality and related health benefits due to lower pollution in Indian cities.
- Direct job creation:* There is expected to be significant growth in the EV sector and a corresponding increase in financing solutions for such EVs. The Platform will create new jobs as part of this broader sectoral growth.

Ultimately these outcomes contribute to achieving the GCF Results Area of reducing emissions through increased access to low-emission transport.

⁴⁶ Please note that the Platform will encourage the potential counterparties to use renewable energy as the primary source of energy to power the EV charging stations. To incentivise the use of renewable energy, Platform may provide certain incentives to potential counterparties if renewable energy use related targets are met. Irrespective of this, India's electricity grid is getting increasingly decarbonised. Over the last 5 years, the contribution of renewable energy in the overall energy mix has increased from 50 GW in January 2017 (15.9% of total capacity installed) [Source: CEA Monthly Installation Report, January 2017] to 105 GW in December 2021 (26.6% of total capacity installed) [Source: CEA Monthly Installation Report, December 2021], an average increase of ~10.8 GW per annum. Further, the central government has set a target to achieve 500 GW of renewable energy by 2030 (50% of all its energy requirements through renewable energy sources), as it looks to achieve its commitments under Nationally Determined Contributions (NDCs) as per the Paris Agreement and meet its COP26 pledges. This robust government focus on decarbonisation of India's power sector will help to reduce the overall level of GHG emissions by the country and assist the Program in achieving part of its additionality targets.

IF the project platform is set up and provides financing across the e-mobility value chain in India, **THEN** the e-mobility sector in India will be scaled up sustainably and contribute to the national target of low GHG emission **BECAUSE** there will be greater penetration of EVs and an enabling environment for further growth in e-mobility.



- Assumptions**
- Continued strong government support for electrification of transport, including maintaining and expanding the enabling environment for investment in e-mobility through regulations, policies, financial support, or other interventions where necessary.
 - Regular government tenders for EV buses, charging infrastructure and other e-mobility sub-sectors.
 - Decreased investment in ICE vehicles over time and the TCO of an EV will be less than that of ICE vehicles over time.
 - Capital raised by OEMs will be invested into research & development in order to reduce the high upfront EV costs over time and increase domestic capabilities.
 - GCF risk-taking capital provides sufficient buffer to meet risk / return metrics of investors.
 - Enabling macroeconomic environment which supports domestic and international investment.
 - National and international trends in electrification of transport increase and support the development of domestic manufacturing, supply of materials, business models and innovation, as well as overall competition.
 - Increased penetration of renewables on the electricity grid, and/or availability of charging sites using only renewable energy.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Enabling environment for EVs established	☐	☒	☐	☐	☐	☐	☐	☐
Increased penetration of EVs in India	☐	☒	☐	☐	☐	☐	☐	☐

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Decreased pollution	☒	☒	☐	☐	☐	☐
Direct job creation	☐	☒	☒	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

The Indian E-Mobility Financing Programme's aim is to accelerate decarbonisation of the Indian economy by focusing on **sustainable and electrified road transport**, enabled through faster adoption of electric vehicles and growth of associated infrastructure in the country.

The main objectives of the Programme are:

1. to provide suitable financial solutions (for example operationally integrated, performance guaranteed leases or structured credit instruments) to electric vehicle owners and operators;
2. to provide tailored financial solutions to ancillary areas to further develop the electric vehicle ecosystem, such as charging infrastructure and batteries;
3. using financial solutions to mitigate current user concerns, including high upfront costs, residual value risk and reliability;
4. to facilitate development of EV ecosystem through investments, partnerships, and capability development of market participants;
5. to set a high benchmark and continually improve the efficiency of services in the Indian electric vehicle domain via improvement of systems, processes, and resources;
6. to provide competitively priced solutions that enhances competitiveness of EVs in terms of long-term cost of ownership vis-à-vis ICE vehicles, aligned with the Indian government's policy; and
7. to mobilise significant amount of nascent institutional capital into electrification of India's road transport sector.

The Programme is uniquely designed to offer ‘first of its kind’ financing solutions in the Indian e-mobility sector. The Programme is expected to create a wide impact in this sector by addressing the key gap hindering the uptake of electric vehicles viz. access to scalable and competitive financing solutions. The Programme will provide financing products and solutions at competitive rates to multiple domestic businesses and entrepreneurs, facilitating healthy competition in the space to further drive down costs and enabling “crowding-in” of commercial capital.

The Programme is designed with a blueprint for success followed in other nascent technologies at early stages in their evolution. For example, in the renewable energy space, formation of Indian Renewable Energy Development Agency (IREDA) as the pioneer for financing and promoting self-sustaining renewable energy has vastly contributed to the high growth and mature renewable energy market that operates in India today.

The Programme is also in line with India’s NDCs as well as policy initiatives as outlined in *section B.1*.

Part A – Programme description

Legal Form	Under the programme, a dedicated E-Mobility Financing Platform will be setup in India, that will be managed by Macquarie Asset Management (MAM) (more details provided below). This Platform (<i>refer to section B.4 – Implementation Arrangements for more details on the proposed corporate structure</i>) will adhere to all local licensing and regulatory requirements, including being registered as a Non-Banking Financial Company (NBFC) with India’s central bank, the Reserve Bank of India (RBI). A NBFC license will be essential for the Platform to provide leasing and other credit solutions in the Indian market to both business and retail customers. ⁴⁷
Currency	While the investments from GCF and commercial investors will be in USD at the Limited Partner (LP) level (<i>refer to section B.4 – Implementation Arrangements</i>), financing provided by the Platform to counterparties in India will be in Indian Rupees (INR).
Target Initial Equity Capital Commitment (Proof of Concept stage)	USD 405 million, including USD 200 million of ‘first loss’ capital from GCF and USD 205 million from commercial investors. The first loss coverage of GCF will be up to its share of capital commitment for all commercial investors coming with preferential shares across the proposed structure (<i>refer to section B.4 – Implementation Arrangement for more details</i>). Further, any capital commitment by commercial investors across the chain will be counted as part of their target commitment of USD 205 million. Given the ‘market making’ nature of the program, GCF will invest a minimum of USD 50 million on the FAA effective date (First Close). These funds will be used to capitalise the above legal entity and enable its initial formation and growth as a proof of concept. If this is done successfully, it is envisaged that the next phase of growth will be funded by capital raises of ~USD 1.1 billion from local and international debt markets. The current business plan does not incorporate additional equity injections from new investors during the implementation period. However, if the Platform successfully achieves its base case business plan, it is anticipated that additional equity will be secured from commercial investors to enable continued corporate growth. Determination in this regard will be taken later depending upon market conditions and future growth prospects.
Investment Period	5 years from the FAA effective date. However, note that the 5-year period for GCF to make disbursements is an approximation and, in reality, it will be driven by actual traction on the ground (the actual timelines may be

⁴⁷ Securing NBFC license from RBI is an iterative and collaborative process, which needs engagement with RBI. MAM will work with its advisors to prepare the right information package, which would include details about the shareholder, capitalization information and background on the management team, among other information, which puts it in the best position to secure this license. The process to secure NBFC license from the central bank will be initiated only after signing of the FAA. As per advice provided to MAM, the timelines to get this approval varies between 6 to 18 months subject to RBI requirements and internal processes

	faster or slower). GCF will need to set aside the entire committed amount from the beginning and disburse the same as per the disbursement mechanism that will be detailed in the FAA.
Implementation Period	Implementation period will be 10 years from FAA effective date. It can be further extended by two consecutive two-year periods based on customary approvals which will be elaborated in the FAA.
Platform's guiding principles	The Platform will be driven by a few guiding principles as below: 1. Build a business that will provide “customer-centric” financing solutions to e-mobility companies in India. The Platform will build unique capabilities and partnerships that will underwrite the technology risks associated with electric vehicles and develop the ability to own and redeploy assets in the market if required. 2. Focus primarily on the Business to Business (B2B) space. This will allow the Platform to address the key problem in the sector, which is access to sustainable and scalable capital for growing businesses. However, the Platform may strategically expand its coverage to the retail customer segment if there is a strategic requirement and requisite market conditions. 3. In the B2B space, the Programme's focus will be on the entire electric mobility value chain. However, given the nascency of the industry, few key sectors within the overall value chain, such as electric buses, shared 4W fleet and electric charging infrastructure, are expected to act as the anchor for the Programme. Other segments may include light and medium commercial vehicles (LCVs and MCVs), commercial 2W and 3W fleet, old vehicle scrappage, battery recycling, or others. Additionally, the Platform will have flexibility to re-orient its financing focus depending on the emerging requirements and its ability to drive high impact. 4. The Platform will seek to develop long-term partnerships with a few large OEMs / sponsors / operators who are actively present in the e-mobility space. It will work with them to promote strong Work, Health, Safety and Environment (WHSE) practices and focus the portfolio deployment in partnership with these large organisations, whilst seeking to create alignment around vehicle reliability and residual value.
Target Sub-Sectors	As mentioned above, while the mandate of the Platform will be the entire e-mobility value chain in India, there are a few key sub-sectors that will be of higher focus initially. These include electric buses, shared 4W fleet and electric charging infrastructure. These sectors all have a substantial growth potential with an estimated USD 50 billion of financing required by 2030.⁴⁸ A. Electric Buses Electric buses are capital intensive, with a concentrated OEM base and a fragmented operator base. The Platform will focus on this space and partner with a few key large OEMs actively seeking to migrate to electric buses. The Platform will provide innovative financing solutions to the key players targeting the intra-city, government led auctions as well as to key players operating in the private, inter-city bus space. The financing solutions will include products such as leasing, project financing, receivables financing, and other innovative solutions that best help addresses the market need. The Platform will target all segments of bus deployment including intra-city public transport, inter-city transportation, school busses, office and other commercial applications. B. Shared / Corporate 4W Fleet Migration to electric vehicles is a key theme for companies that provide mobility as a service to private corporates. However, this space faces key challenges of limited financing availability from banks due to a lack of suitable business models and / or the inability to achieve the utilisation thresholds required to run a sustainable business. This sector offers the investment themes preferred by the Platform of a key financing and viability gap, as well as of working with a few large players and concentrating the portfolio to a limited number of key relationships. The Platform will work towards enabling migration of

⁴⁸ https://niti.gov.in/sites/default/files/2021-01/RMI-EVreport-VF_28_1_21.pdf

	shared fleet service providers to electric vehicles across buses and other 4W light commercial vehicles. C. Electric Charging Infrastructure To achieve the desired electrification targets for vehicles, charging infrastructure (public and private) needs to be rapidly scaled up from current levels. This space is currently plagued by supply chain issues, policy uncertainty and limited financing, coupled with high upfront costs. The Platform will seek to alleviate a major bottleneck in growth of e-mobility space in India by making financing solutions available for aggressive and pro-active deployment of charging infrastructure as a shared service for multiple owners / operators of electric vehicles. Further, to enhance the additionality aspect of the program, the Platform will both encourage and incentivise electric charging station operators through appropriate financial incentives (to be decided by the Platform management) to directly source renewable energy to meet the power requirement of EV charging stations. The balance of the portfolio will comprise other segments such as OEM or vendor supply chains, vehicle scrappage, battery swapping, commercial 2W and 3Ws, or others. Over time, the Platform may also expand into the B2C / retail segment. Note that funding of hybrid vehicles will be out of Programme's scope of target sub-sectors. Moreover, the Platform will continuously look at realigning its business focus as the e-mobility sector evolves and new business opportunities emerge in this space.
Leasing and Financing Solutions	The leasing and financing solutions to be provided by the Platform will be based on the requirements of its customers as no one solution can cater to, and address all the needs of, the counterparties in the sector. By offering leasing and / or credit solutions (noting that no equity investments will be made by the Platform), the Platform will be able to support a larger number of counterparties within the same sector without giving rise to any conflict-of-interest issues. Some of the main financing products to be provided by the Platform: 1. <i>Leasing (Operating and Finance leases):</i> The Platform's strong preference will be to provide leasing solutions to its customers. Most of the financial institutions in India have focused on providing credit offerings rather than leasing solutions which require developing strong underlying product and technology capabilities. The lack of depth of leasing offerings in India force the operators / vehicle owners to invest substantial upfront capital as part of overall capex. The problem is even more acute in case of electric vehicles, where the upfront capex requirement can be much higher than the ICE vehicles. This leasing offering will be most pertinent for customer segments like e-busses (private counterparties), 2/3/4W fleets, LCVs and MCVs. We expect that, as the Indian e-mobility market matures, the Platform will emerge as one of the leading leasing companies in India with end-to-end capabilities on asset sourcing, asset deployment / financing, asset maintenance and redeployment at the end of the leasing tenure, which also addresses a key barrier of residual asset value. 2. <i>Project and vehicle financing (and other relevant credit solutions including Revenue Discounting):</i> Project and vehicle financing solutions generally have shorter tenures and have higher risk premium as the asset ownership continues to remain with the borrower. However, there are additional accounting benefits (such as depreciation and interest) that can be valued by the borrowers. The project and vehicle financing markets for traditional vehicles are highly competitive due to the presence of large number of banks / NBFCs and other financial institutions. However, Indian financing markets do not currently have any scalable offering for electric vehicles and so, depending upon customer's requirement, the Platform will look to utilise its strong sectoral understanding and experience to offer a tailored solution for EVs that is beneficial

	for the counterparty and seek to establish a long-term relationship and commitment. Irrespective of the type of financing solution, the key element for any financial product offered by the Platform is GCF's first loss capital, which will allow the Platform to offer the necessary product for the higher upfront cost at a lower financing rate and assume right residual value risk. This will allow the provision of electric vehicles at competitive cost to ICE alternatives in order to scale up the sector. On the back of this structure, the Platform can attract other commercial capital that would otherwise be hesitant to invest in this nascent sector. By raising a large amount of capital, the Platform can deploy funds across financing solutions at scale, thereby driving down costs of electric vehicles in India and building a vibrant ecosystem across the value chain.
Select Financing Criteria	Financing solutions provided by the Platform will be subject to various financing criteria, such as funding size per counterparty account, single counterparty limit, credit quality, coupon type, repayment terms, etc.
Macquarie Asset Management (MAM) Overview	The Platform as indicated in the structure in <i>section B.4</i> will be managed by offshore and onshore management companies that are owned and controlled by MAM, the asset management division of Macquarie Group. MAM has a significant track record in managing investments, with >140 portfolio companies across its managed funds globally.⁴⁹ <u>MAM overview</u> 1. <i>MAM's expertise in asset management:</i> MAM is a global specialist asset manager with AUD 735.5 billion of assets under management (AUM),⁵⁰ aiming to deliver positive outcomes to everyone. MAM provides investment solutions to clients across a range of capabilities, including infrastructure and renewables, real estate, agriculture and natural assets, assets finance, private credit, fixed income, and multi-asset solutions. Macquarie Group is a diversified financial services company headquartered in Sydney and listed on the Australian Securities Exchange (ASX), with a market capitalisation of AUD 74.4 billion.⁵¹ Macquarie Group provides asset management and finance, banking, advisory and risk, and capital solutions across debt, equity, and commodities. Through its infrastructure focused private equity arm within the Private Markets division, MAM is the world's largest infrastructure asset manager⁵² with equity under management (EUM) of AUD 160.1 billion,⁵³ receiving commitments from pension funds, insurance companies, and sovereign wealth funds for its products and broader platform. 2. <i>MAM's expertise in India:</i> MAM has been present in India since 2009 when it established a ~USD 1 billion infrastructure focused fund in partnership with State Bank of India (SBI), India's largest bank.⁵⁴ Today, MAM has invested across multiple assets in roads, power, telecom towers and airports sectors in India. It is led by a team of highly experienced professionals who have robust record of establishing new businesses in the country. For example, in March 2018, MAM won the first auction of toll roads under the Toll Operate Transfer (TOT) model from National Highway Authority of India (NHAI) wherein it was awarded the concession to operate 687 kms of toll roads over a period of 30 years at a cost of ~USD 1.5 billion.⁵⁵ <u>Macquarie Group's track record in e-mobility sector</u> 3. <i>Expertise in e-mobility sector:</i> Since 2018, Macquarie Group has been active in the e-mobility sector undertaking both debt and equity positions in some of the leading e-mobility companies across the world.

⁴⁹ As at 31 December 2020.

⁵⁰ As at 30 September 2021.

⁵¹ As at 15 February 2022.

⁵² IPE Real Assets 2021 – Top 100 Infrastructure Investment Managers.

⁵³ As on 31 December 2021.

⁵⁴ <https://www.macquarie.com/au/en/about/news/2009/sbi-and-macquarie-launch-indian-infrastructure-fund-usd1billion-raised.html>

⁵⁵ <https://www.infrastructureinvestor.com/miras-second-asia-fund-wins-indias-1-5bn-highway-privatisation/>

	MAM plans to leverage the above expertise and experience across its various business divisions to establish a high quality e-mobility financing platform. <u>Macquarie Group's track record in leasing platforms</u> 4. <i>Expertise in building and exiting leasing platforms:</i> Since 2006, Macquarie Group has successfully built multiple leasing platforms across sectors such as aircraft, helicopters, engines, rolling stock, passenger vehicles, computer equipment and utility meters.
Platform Management	MAM will hire an experienced management team to manage the Platform once it has been established post GCF's Board approvals. Being a MAM managed vehicle, the Platform will also be able to leverage Macquarie's global network, expertise, and brand value across the world, thereby increasing its visibility and ability to generate inbound approaches from market participants.
Governance	MAM's main responsibility will be to approve new financing opportunities, have robust oversight over the Platform's performance and drive innovative, leading capital sourcing and management solutions, including exit at an appropriate time. Through the Accredited Entity (AE), it will apply its fiduciary duties as per GCF's AMA and the best practice standards it applies across its global operations. It will report on the Platform's performance on a quarterly and annual basis. The audit of annual financial statements will be performed by an independent auditor, to ensure transparency and control of flow of funds. Impact reporting will be performed annually, either in-house or by a third party consultant, with the two GCF specific programme independent evaluations (mid-term and final) to be undertaken by the AE and shared with GCF. ESG will be carried out as per the Environmental and Social (E&S) Document (<i>refer to Annex 6 – Environment and Social Management System</i>) and IFC Performance Standards. Further, the Platform will be subject to MAM's best in class asset management practices, which provide guidance on Governance, Money Laundering, Bribery and Corruption, Diversity and Inclusion, Cyber-security, etc. related aspects.
Economics	The Platform will target investment in opportunities that balance the long-term risk adjusted returns for investors with market development and innovative financial solutions for counterparties, aiding the accelerated growth of the Indian e-mobility sector. Distribution of proceeds to investors will be in accordance with the distribution waterfall as detailed in the term sheet provided to GCF.

Part B – Key activities of the Program

As mentioned in *section E.6*, the Programme will have 3 key phases:

1. Activity Set I – Platform Establishment and Fundraising
2. Activity Set II – Platform Investments, Monitoring and Exit
3. Activity Set III – Wind up of corporate structure

Activity Set I – Platform Establishment and Fundraising

This primarily includes the establishment of entities / structure in Singapore and India for efficient inflow / outflow of capital to establish India's 'first of its kind' 'E-mobility Financing Platform'. Some of the key activities include:

1. setup of LP – GP structure (*refer to section B.4 – Implementation Arrangements for more details*) for the transfer of GCF capital into the e-mobility programme through the Accredited Entity;
2. establishment of structure and entities in India, with necessary regulatory and legal approvals, to start providing financing solutions in the e-mobility space in India;

3. crowding-in of commercial investors, including fundraising efforts and legal execution of their investments, either into LP or SG Master Co. or SG Co. or the Platform;
4. ensuring compliance of the entire structure with all applicable local rules and regulations;
5. setup of systems and processes across the structure, which includes adoption of risk management, HR, cybersecurity, third party vendor management, E&S and Diversity and Inclusion (**D&I**) frameworks and policies; and
6. hiring and appointing the management team and the Board of Directors (the **Board**) at various levels, to ensure proper functioning of each entity.

Activity Set II – Platform Investments, Monitoring and Exit

This primarily includes screening and evaluating transactions and counterparties, application of best business practices, WHS, gender and environmental standards, diligent monitoring, establishing innovative financial and capital management solutions and, ultimately, exit on behalf of investors from the Platform.

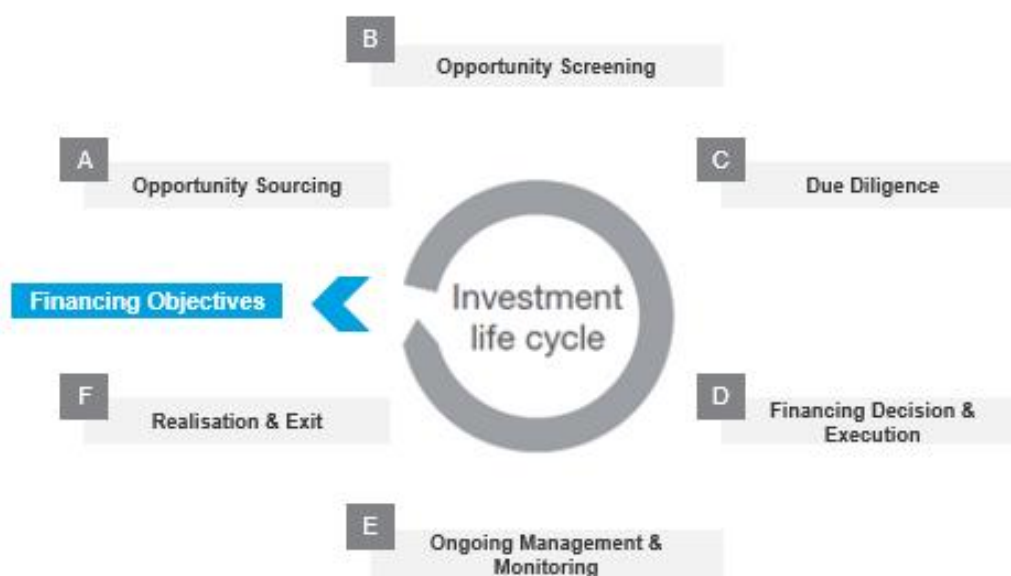
Financing Process

The Platform will implement its strategy through a well-defined policy of identifying, evaluating, investing, managing, and realising the financing opportunities (where applicable).

Key investment and financing decisions of the Platform will be approved by SG Co. (as shareholder) with guidance from the Singapore Manager. This will be supported by:

1. day-to-day management of the Platform by the experienced professionals employed by the Platform; and
2. strategic support and oversight to the Platform by the Singapore Manager in consultation with Macquarie India Advisory Company (**IAC**).

Throughout each stage of the financing process, risk assessment will be a key focus for every financing solution provided by the Platform. This risk focused approach will begin at the earliest possible stage, even before transaction screening and will continue throughout the life of each financial solution.



A. Opportunity Sourcing

Access to deal flow will be one of the key factors in the success of the program. Access to a wide range of potential opportunities will enable the Platform to assemble an appropriately diversified portfolio. The Platform expects opportunities to arise from several sources, including:

<i>Management team</i>	Management team members will have many years of work experience in infrastructure, vehicle financing and leasing / credit sectors across the world. They will therefore have long-standing relationships with sponsors, operators, banks, and intermediaries who will provide a regular flow of opportunities.
<i>Macquarie relationships</i>	Macquarie is a leading infrastructure advisor and investor with global coverage. It holds significant roles on a number of globally relevant forums, such as the Climate Finance Leadership Initiative and the Global Committee for Adaptation. This will bring opportunities and provide personal introductions for the Platform to prospective financing counterparties.
<i>Strategic partnerships</i>	Macquarie has unparalleled group-level relationships with a range of infrastructure sponsors, operators and banks, all of which will enhance Platform's access to deal flow. MAM has developed strategic partnerships with several domestic and international financial institutions (banks and NBFCs) through its existing portfolio of infrastructure assets in India. It also has close relationships with key individuals within the e-mobility, automotive and energy sectors, which will enable detailed discussions that are important for originating and refinancing transactions.

Further, these opportunities are expected to arise through circumstances which may include companies:

- bidding for new central / state government tenders;
- partnering due to the large amounts of capital required to construct or manufacture the assets;
- looking to recycle capital to expand the existing business in the sector; or
- restructuring or repositioning their business activities.

In certain cases, due to MAM's position and relationships in the market, the Platform may have the ability to finance assets / counterparties on an exclusive basis.

MAM executives and the Platform management team will develop / have strong relationships with many of the key stakeholders in the sector, including corporate sponsors (such as utilities and suppliers), professional advisors and lenders. These relationships are important, not only for deal origination, but also execution, on-going management, and exit.

B. Opportunity Screening

After becoming aware of a financing opportunity, the Platform will apply its Opportunity Screening process to test the characteristics of the opportunity against the Financing Selection Criteria of the Platform.

The Platform's investment screening process will typically include assessment of:

- the financing thesis and alignment to Financing Selection Criteria;
- the creditworthiness, operational capability and ESG performance of the counterparty (lessee / borrower), with particular regard to their financial position, expected outlook, and sensitivity to external factors;
- the capability of existing management of the lessee / borrower, including recent performance, expertise, experience, and incentives to perform;
- the revenue model of the business (including whether underpinned by contracts or regulation), the stability of the cashflows of the business and possible factors influencing future revenue and cashflows;
- an initial assessment of the environmental and social categorisation of transactions (*refer to section G.1 – Environment and social risk assessment for more details*);
- a high level overview of expected terms to be provided under the financing; and
- an initial assessment of GHG emissions reduction, ESG risks and other investment risks.

Based on the Financing Selection Criteria and in accordance with delegation of authority within the Platform (to be established once the Platform is operational), the Platform management team will negotiate the term sheet with the counterparty. Once there is broad alignment on the commercials and additional analysis has been undertaken by the

Platform management team, a detailed discussion paper will be presented to the Board of the Platform (or an empowered team as mandated by the Board) for approval and advice. The Platform management will also regularly seek advice from Macquarie's IAC and necessary approvals from the Singapore Manager as required.

C. Due Diligence (DD)

The management team will engage with the counterparty and undertake, in conjunction with specialist experts and advisers, due diligence as required to ensure all aspects of the transaction are sufficiently reviewed. The management team will also take an active role in the negotiation and setting of lease or loan terms and conditions as reflected in the documentation.

Key risk considerations will typically include:

Due Diligence (DD) Items	Description
Enhanced Counterparty / Promoter DD	Assess Counterparty / Promoter's financial strength, reputation, ESG performance and track record of honouring contractual commitments.
Financial Modelling	Counterparties typically provide a financial model which can be used for future cash flow and financial statement analysis. The financial model will be stressed for several different scenarios based on the management team's view of key drivers, with input from Platform's professional advisors if required.
Financial DD	Tax, accounting, and financial statement diligence as required to ensure financial robustness.
Technical DD	Third party consultants and other expert reports (either undertaken in-house or provided by the counterparty) providing details around technical aspects of the asset, including whether technology is commercially proven, environmental exposures, asset condition, construction risk (including network expansion), operations and maintenance complexity and experience, contracts, and performance damages and ESG.
Legal DD	Dependent on the nature of a transaction and contract. Typically, would include diligence on key entities, contracts, regulatory regime, security, and enforcement analysis. Additionally, legal advisers / in-house experts will draft robust leasing / loan documents that ensure strong enforceability of contracts, preserve the rights of the lessor / creditor in the event of asset deterioration and provide an appropriate redressal mechanism. It is possible that these markets may form the basis of market standards in the future.
Insurance DD	Summary of insurances in place and adequacy of the insurances (construction risk, business interruption, operational risk, third party liability insurances). Often the lease / credit agreement will specify a required level of cover.
Commercial or Market DD	Economic and market reports (comprising of industry structure, barriers to entry, product substitution and certainty of supply). Only applicable for first time financing in a segment where there is an underlying demand risk, tariff risk or where there is a significant cost component that may be subject to market dynamics.

Note that the above list of due diligence items may not be applicable for all transactions. Any decision in relation to the due diligence process (and appointment of relevant third party advisers) will be undertaken by the management team in consultation with Macquarie India Advisory Company and the Singapore Manager.

D. Financing Decision and Execution

The results of the due diligence process will be set out in a financing paper and presented to an empowered team within the Platform (as per the delegation of authority) and the Singapore Manager (for specific cases based on size of the transaction) for guidance and decision regarding the financing. Once a decision is made and financing is approved, the management team and its legal advisers will work with the counterparty to negotiate and agree the leasing / loan documents. These documents will detail out the various rights of the lessor / lender, lessee / borrower's reporting requirements and clearly layout the redressal mechanism in case of any default or conflict.

E. Ongoing monitoring and management

The Platform will take a proactive approach to asset management by following a structured framework as approved by the Platform Board, with guidance from the Singapore Manager and Macquarie India Advisory Company, which protects the best interests of the investors and ensures that Platform is protected against any risk (including fiduciary or reputational risks). All of Platform's financing solutions will be continually monitored and maintained. The general principles of the finance and impact monitoring process are:

- The Platform's management team will take the primary responsibility for monitoring performance of the financing instrument and the impact created by it. The team will be responsible for tracking both financial and impact performance of the instrument and will update the Board of the Platform and the Singapore Manager on a quarterly basis on lessee / borrower's financial and non-financial covenants.
- Monitoring by the management team will be dynamic (following general relevant market data via information subscription services), and periodic (prompted by ongoing proactive engagement with the lessee or borrower as well as the lessee / borrower's reporting obligations) including internal quarterly monitoring meetings and reporting.
- Material changes to the lessee / borrower's business or credit quality, and the impact created by it, will be discussed internally and escalated to the Singapore Manager where it will be determined whether it is sufficiently material or not.
- Requests for waivers or amendments to facility terms, if any, will be reviewed and approved post detailed discussions with the Singapore Manager as and when required. Further, at the quarterly meeting, the Singapore Manager will be notified regarding all waivers and amendments granted in the previous period.

Further, as part of the monitoring process, the Platform will utilise a heightened review process called Watchlist. If a management team member sees evidence of potential stress, business practice concerns or other factors that could have a material impact on the financing solution provided to the counterparty, then it will decide whether to recommend that lease or loan be placed on Watchlist.

The counterparty will be placed on Watchlist when it is deemed, they are, or are expected to come, under stress. The Watchlist process described above will seek to ensure heightened scrutiny and early action in the case of counterparties that are at high risk of a material change in credit profile or loan / lease restructure. In an event of default, the Platform will take the necessary action, which is likely to result in the best return to the investors. These actions could include managing the financing instrument through the work out process, asset repossession or sale of the instrument.

F. Realisation and exit

During the term of the Platform, the Singapore Manager will consider appropriate strategies to provide liquidity from the Platform which achieves an exit, provides liquid return to the investors and generates capital for re-investment (*refer to section B.6 – Exit and Sustainability for more details*). Within its Investment Period, the Platform will seek to create a sustained, positive impact through its robust deal sourcing, execution and monitoring capabilities, focus on sustainability and good corporate citizenship and high performing product portfolio, all of which is anticipated to be attractive to a future investor. The Platform is expected continue to make a positive impact on the Indian road transport sector, even after the exit of the initial set of investors in case the potential buyer(s) continue the mandate in the future.

Activity Set III – Wind up of corporate structure

Once the General Partner (GP) of the Singapore LP has distributed capital to the investors, it will need to wind up the corporate structure. However, wind up of the corporate entities may take time in case the Singapore Manager offers any indemnities or warranties to the buyers at the time of exit.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

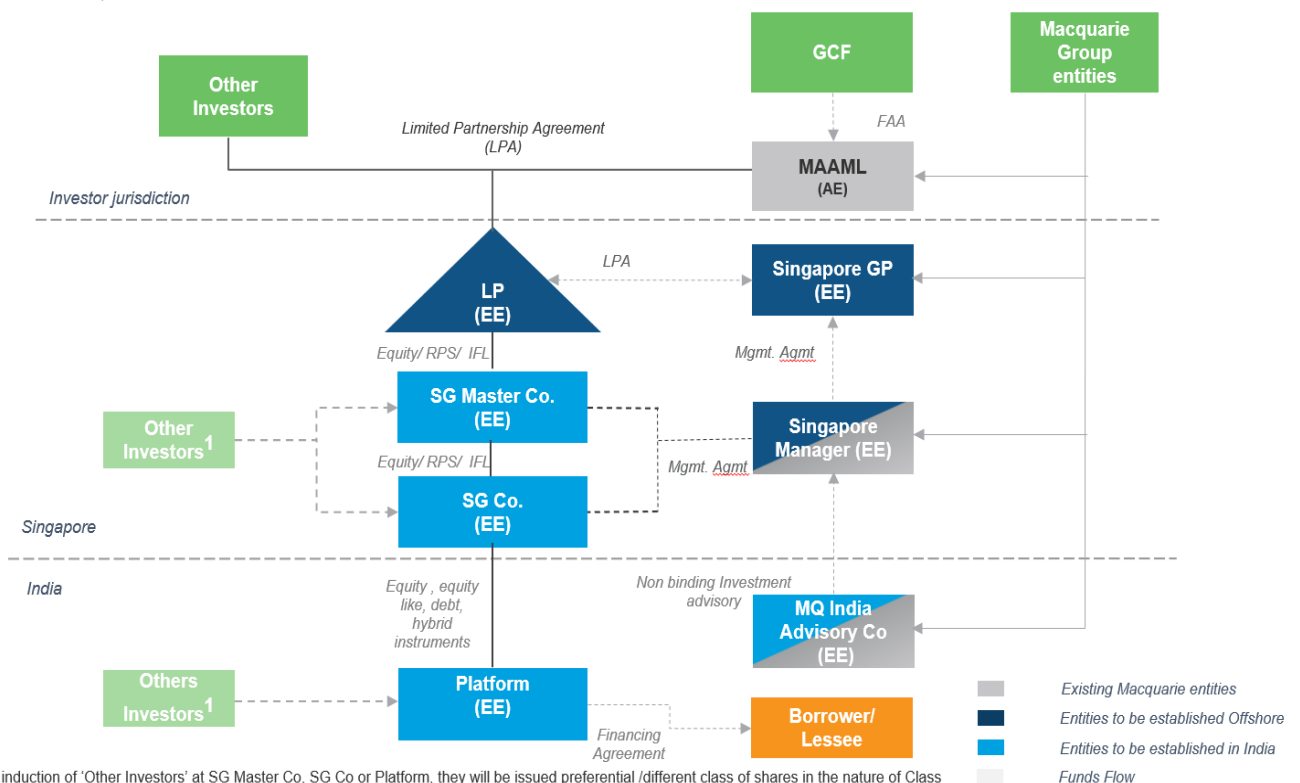
Platform structure, governance, and flow of funds⁵⁶

Proposed Structure and Governance

An offshore pooling vehicle (LP) is proposed to be set up in Singapore to aggregate funds from GCF and commercial investors via a Limited Partnership Agreement (LPA). This Singapore LP will set up a Singapore private limited company (SG Master Co.) which will, in turn, establish another wholly owned Singapore private limited company (SG Co.) which will invest in the Platform in India by way of equity, equity like, hybrid securities, debt instruments and other similar instruments⁵⁷ from shareholders for efficient capital structuring.⁵⁸

Proposed Structure

All 100% ownership unless noted otherwise



Accredited Entity

GCF will enter into a Funded Activity Agreement (FAA) with MAAML, the AE, which will detail the key implementation arrangements, commercial terms, and legal obligations, including mechanics to pass down GCF policies and requirements to the offshore entities in Singapore and onshore entities in India. These key terms are summarised under the Term Sheet. The AE will transfer the funds to the LP on behalf of GCF, as the investor of record.

The AE will oversee the implementation of the Programme on behalf of GCF. It will ensure that:

1. all GCF's policies are implemented at the Platform levels and in underlying financing documents;

⁵⁶ Subject to Legal and Tax Due Diligence which will be completed prior to FAA execution.

⁵⁷ The decision regarding the use of specific instruments for investing in the Platform will depend on the regulatory requirements and prudent capital structuring opportunities. It is noteworthy that any capital repaid from any equity like or debt instrument injected by SG Co into the Platform will either be available for distribution to the investors or re-invested for business expansion subject to the Singapore Manager's discretion..

⁵⁸ The proposed structure is subject to regulatory approvals.

2. the GCF account is managed and maintained in compliance with the requirements under the AMA and FAA; and
3. GCF is provided access to the Platform's performance vis-à-vis the Programme's targeted outcomes and results.

In order to satisfy its obligations under the AMA and the FAA to be entered into with respect to this Programme, MAAML will enter into a deed of undertaking with each Executing Entity).

LP

The offshore pooling vehicle will be domiciled in Singapore as a Limited Partnership (**LP**). The Singapore GP, as general partner, together with GCF and the other Investors, as limited partners, will enter into a partnership agreement. The capital commitment / contribution of each limited partner in the LP will have different class of interests:

1. Class A interests for commercial investors. Class A would benefit from first-loss benefit provided by Class B investor. Class A investors would not have participated in the LP without this first-loss benefit due a perceived unfavorable risk / reward dynamic;
2. Class B interests for GCF, against which GCF would provide first-loss benefit to the Class A investors, making the risk / reward dynamic for Class A investors acceptable; and
3. Class C interests for Singapore GP / Singapore Manager or provision by way of management agreement, for the purposes of receiving carried interest distributions from the LP.

The LPA will detail terms of investment (including the ownership interest of each LP), in line with the FAA signed between the GCF and the AE. Note that the proposed structure will not change as commercial investors become part of the LP, SG Co. or the Platform, on or after First Close. The AE will analyse the LP's compliance with GCF's business model by reviewing several documents provided by other Executing Entities (**EEs**) in the structure.

Singapore GP and Singapore Manager

A 100% Macquarie Group owned, Singapore private limited company will be established to act as the GP to the LP. The GP, under a management agreement, will delegate its investment management responsibilities to the Singapore Manager, which will be a Capital Markets Services licensed entity. The Singapore Manager will have an overall responsibility for the management and operation of the LP under the management agreement. Some of its key responsibilities include:

1. managing LP's investment, including identifying representatives for the Board of the Platform (where appropriate);
2. identifying and assisting with the implementation of strategic enhancements at the Platform level;
3. undertaking other matters involved in the management and administration of the Platform; and
4. determining appropriate risk and liquidity strategies for the LP and the Platform.

To ensure that Singapore Manager's responsibilities are appropriately resourced, the Singapore Manager will enter into a non-binding investment advisory agreement with Macquarie India Advisory Company (**IAC**).⁵⁹

Singapore Master and SG Co.

The LP will establish a Singapore private limited company (**SG Master Co.**) as a wholly owned subsidiary, which will in turn incorporate another Singapore private limited company (**SG Co.**) to make investments via equity, equity-like hybrid securities, debt instruments and other similar instruments from shareholders in the Platform. SG Master Co. is primarily being set up to provide flexibility to investors to exit the investment at Singapore level and avail India – Singapore tax treaty benefits, subject to meeting relevant conditions. SG Master Co. and SG Co. will also be a party to the GP management agreement, or enter into a separate management agreement with the Singapore Manager.

Platform

⁵⁹ Note that there will be no direct legal arrangement between the Platform, Singapore Manager and Macquarie India Advisory Company. SG Co pursuant to shareholder rights in the Platform will review the information received and based on the advice from the Singapore Manager, approve relevant shareholder matters. All recommendations to the SG Co's Board would be reviewed and proposed by the Singapore Manager under the management agreement. Further Singapore Manager will engage Macquarie's India Advisory Company to provide non-binding investment advisory services.

A company incorporated under the Indian Companies Act will be established by SG Co. as the Platform.⁶⁰ The Platform will apply for necessary licenses and approvals in India to undertake a lease and loan business in India. To enter into lease (finance lease) and lending arrangements, the Platform will obtain NBFC license as prescribed by the Reserve Bank of India. A professional management team will be identified and hired to manage the Platform with support / approval from the Singapore Manager. SG Co. will nominate its representative (MAM staff) on the Board of the Platform to participate in the key investment, operational, and governance decisions. The Singapore Manager will also provide strategic oversight to the Platform through SG Co.'s shareholder rights, including supporting the Board and management with access to subject-matter experts and support on operational and governance expertise within Macquarie Group. The Platform's management team will be responsible for day-to-day screening and evaluation of investment opportunities, performance monitoring and compliance. It is this entity that would likely engage with debt markets to source the anticipated ~USD 1.1 billion of debt over the implementation period, if proof of concept is successful.

Macquarie India Advisory Co (IAC)

IAC, a 100% Macquarie Group owned entity incorporated under the Indian Companies Act will provide non-binding investment advisory services to the Singapore Manager. Its role inter-alia includes:

1. analysing the progress of the transactions in Platform, monitoring performance, performing the necessary compliance checks, and providing reports to the Singapore Manager;
2. providing assistance in relation to the valuation of SG Co.'s investment in the Platform; and
3. providing assistance in relation to adequate implementation of risk management systems in order to identify, measure, manage and monitor appropriately all risks relevant to the LP's investment strategy.

Roles and Responsibilities

The roles and responsibilities of various Executing entities in the above structures are laid down below:

Category	Name	Role
Accredited Entity	MAAML	- Execute an FAA with GCF - Act as an investor on record in the LP on behalf of GCF - Enter into the LPA (see below) on behalf of GCF - Execute the FAA that incorporates all agreed GCF requirements set out under the AMA and the term sheet for the programme, and to be implemented down the chain - Enter into a compliance undertaking with each of the Executing Entities - Crowding-in of commercial investors, including fundraising efforts and legal execution of their investments into the LP
Executing Entity	Singapore LP	- Pool GCF capital (via MAAML) and other commercial investor capital - Enter into the LPA with Limited Partners (LPs or Investors) and General Partner (GP) - GCF requirements as set out under the AMA and FAA, to be incorporated in the constitution documents of LP. Enforcement of the policies to be a limited partner's right for MAAML acting as an investor on record for GCF

⁶⁰ At least two shareholders are required for incorporating a private company in India. Apart from SG Co., the second entity will be a nominee shareholder (can either be a person or a company) holding one share on behalf of SG Co. Second nominee will be finalised once the Platform is being incorporated.

		• Ensure compliance with all applicable local rules & regulations
Executing Entity	Macquarie Group entity that will set up GP	• Incorporate Singapore GP to act as a General Partner to the above LPA
Executing Entity	Singapore GP	• Party to the above LPA as a GP • Responsible for managing the LP • Singapore GP to be a Macquarie Group Limited controlled entity • Ensure compliance with all applicable local rules & regulations
Executing Entity	Singapore Manager	• By way of Management agreement, GP to appoint Singapore Manager as an Investment advisor and incorporate GCF requirements in AMA and FAA, in the Management Agreement • Investment Committee (IC) comprising of Macquarie employees and/or industry experts to take investment decisions on behalf of the Singapore LP/ SG Co. • The Manager to provide investment advisory service to the GP, SG Master Co. and SG Co. • Investment advisory service to comprise of advice around investments in Platform, recruitment of key management at the Platform and key business activities of the Platform • Singapore Manager to enter into an agreement with onshore IAC for receiving non-binding advisory services • Provide quarterly investment report to Investors based on information received from IAC • Engage with potential investors in the market with the assistance from IAC to explore and identify exit opportunities. Summary of any relevant exit process to be provided in quarterly investment reports. • Lead all documentation discussions with the assistance of IAC. • Ensure that appropriate sale value is received on time and subsequently distributed back to the limited partners as per the LPA • Distribute capital and returns to investors as per the LPA • Ensure compliance with indemnities & warranties, if any • Wind up of corporate entities (including the Platform) Ensure compliance with all applicable local rules & regulations

Executing Entity	SG Master Co.	• SG Master Co to act as a pass-through entity between the LP and SG Co. • SG Master Co board to comprise of employees of Macquarie and/or industry experts • Constituent documents of SG Master Co, as well as shareholder rights with LP, to include implementation of GCF requirements as set out in AMA and FAA down the chain • Establishment of SG Co. to make investment into the Platform in India • Ensure compliance with all applicable local rules & regulations
Executing Entity	SG Co	• SG Co to act as a pass-through entity between the SG Master Co. and the Platform • SG Co board to comprise of employees of Macquarie and/or industry experts having diverse industry and finance experience • Shareholder rights with SG Master Co to include implementation of GCF requirements set out under the AMA and FAA. • Establishment of the Platform in India • Ensure compliance with all applicable local rules & regulations
Executing Entity	Macquarie India Advisory Company (IAC)	• Macquarie India entity controlled by Macquarie Group Limited • To provide non-binding advisory services to the Singapore Manager • Help in identification of the management team and relevant support staff for the Platform • Reporting on opportunities considered, screened, approved, and executed. • Provide quarterly investment report to Singapore Manager • Engage with potential investors in the market to explore and identify exit opportunities under the guidance of Singapore Manager. Summary of any relevant exit process to be provided in reporting to Singapore Manager. • Lead all documentation discussions with the assistance of the legal adviser under the guidance of Singapore Manager • Ensure compliance with all applicable local rules & regulations

Executing Entity	Platform	• SG Co to establish and control the Platform in India • Platform to comprise of a management team and a board • The board to comprise of Macquarie employees, senior management team members and/or industry experts having diverse industry and finance experience • Platform to procure NBFC license to provide financing and leasing solutions to counterparties. • Participate in workshops and/or stakeholder consultations organized by local, state or national governments, or affiliated think tanks and/or industry bodies to promote the development of EV financing in India • Use learnings, data or market knowledge from the Platform to inform new policies or regulations to promote development of the EV sector • Source transactions & implement the mandate of the platform • Undertake ongoing monitoring of financial solutions provided to counterparties, including compliance with financing covenants, meeting of agreed reporting requirements, and credit assessment of counterparty. • Undertake climate impact assessment of the portfolio on annual basis. • Based on delegation arrangements, Platform to seek approvals from the SG Co. on shareholder matters (and correspondingly up the chain) • GCF requirements as agreed in AMA and FAA to be incorporated in the shareholder rights with the SG Co. • SG Co, pursuant to shareholder rights in Platform, as the case may be, would review the information received and based on advice from the Singapore Manager, approve the relevant shareholder matters • Ensure compliance with all applicable local rules & regulations
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Flow of Funds

The capital invested by LP into SG Co. will be used to create the Platform which will seek to secure the necessary regulatory approvals in India to provide a different kind of financing in the market.

The AE on behalf of GCF will channel funds to the Platform through the LP structure.⁶¹ The GCF funds will be channeled by the AE as follows:

Fund flows for investments

⁶¹ Please note that arrangement of MAAML / AE acting as trustee for GCF's bank account is subject to regulatory advice.

1. GCF and the AE will contractually agree that GCF will commit an amount of funding, as detailed in the Term Sheet, with the single stated purse of the AE utilising it for investment in the Singapore LP, for the purpose defined in this FP.
2. The AE will request GCF to remit funds to its fiduciary GCF account. GCF funds will be kept in this account without being co-mingled with any other sources of capital with AE.
3. Once the Platform is established and additional funds are required, the Platform, subject to approval by the Singapore Manager and Singapore GP, will issue a drawdown notice to the LP and the LP will in turn issue a drawdown notice to the AE (for GCF as prescribed under FAA).
4. Upon the receipt of drawdown notice, the AE will remit the funds to the LP from its fiduciary GCF account, which, in turn, remit the funds to SG Co. through SG Master Co.
5. SG Co. will utilise the funds received to capitalise the Platform.
6. Note that the AE will drawdown the capital as committed by GCF over the Investment Period as defined in the Term Sheet.

Fund flows for distribution

Upon receipt of distributions from the Platform, SG Co. will remit the funds to SG Master Co. which, in turn, will remit the funds to the LP. The LP will then make distributions to fiduciary GCF account as per the distribution waterfall mechanics in the LPA.

Please note that we do not expect any conflict of interest issue to arise between the executing entities and the AE, as the mechanics for payment rights of GCF via the AE as against those of commercial investors will be well defined and documented as part of the FAA and LPA.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Climate vulnerabilities and fiscal situation

According to Climate Action Tracker, India's GHG emissions, excluding Land Use Change and Forestry (LULUCF), are estimated to be 2.8-2.9 GtCO₂ in 2020. They are expected to increase by ~3.5% over next 10 years, reaching 3.8-4.1 GtCO₂ by 2030. This represents a 27-35% increase in emissions from 2010 to 2020 and a 76-87% increase by 2030.⁶² Further, India's average temperature is expected to increase by >4.4°C by 2100 (relative to average temperature between 1975-2006) in the worst-case scenario, according to a report by the Indian government's Ministry of Earth Sciences.⁶³ This leaves India highly vulnerable, prone to extreme climatic events such as flash floods, cyclonic storms, droughts, heatwaves, etc. and more susceptible to crop damage, forced migration, increases in social issues and poor economic growth. Whilst India's NDCs under the Paris Agreement are 2.0°C compatible (indicating that India's climate commitment in 2030 is a significant share of global effort based on its responsibility and capability), a significant amount of capital is required to achieve it. According to the Indian government's preliminary estimates, India requires ~USD 2.5 trillion of capital between 2015 – 2030, which is equivalent to annual requirement of USD 170 billion per year at 2014-15 prices (8% of India's FY15 GDP). Further, as per IFC, total capital required to achieve the climate targets was much higher at USD 3.1 trillion between 2018-2030. Given the size of the required investment, a robust support from domestic and foreign investors is essential.

Additionally, with the onset of COVID-19 pandemic since March 2020, the finances of the Indian government have become further stretched (fiscal deficit was 9.5% of India's GDP in FY21 vs 4.6% in FY20⁶⁴ and is expected to be in 8.3% in FY22 according to Fitch⁶⁵), forcing the Indian government to prioritise investments in healthcare infrastructure and research and development of affordable medicines and vaccines. As a result, the availability of capital from GCF during these testing times for the Indian population and economy will be significantly important as it will 1) assist the country to be on track to its achieve its set targets under the Paris agreement, 2) reduce pressure on the government's limited resources while it focuses on keeping its citizens safe and healthy, and 3) provide the necessary capital to the nascent and cash starved e-mobility sector to drive innovation and achieve scale.

GCF value add

⁶² <https://climateactiontracker.org/countries/india/current-policy-projections/>

⁶³ <https://link.springer.com/content/pdf/10.1007%2F978-981-15-4327-2.pdf>

⁶⁴ <https://www.livemint.com/economy/indias-fiscal-deficit-at-9-3-of-gdp-in-fy21-down-from-revised-estimate-of-95-11622459720813.html>

⁶⁵ <https://economictimes.indiatimes.com/news/economy/indicators/india-to-breach-fiscal-deficit-target-in-fy22-fitch-solutions/articleshow/82451801.cms?from=mdr>

Under the proposed transaction structure, GCF's USD 200 million investment will be used to provide "first loss" capital protection to commercial investors, de-risking them by absorbing higher risk in case of the Platform's underperformance due to the multiple risk factors as highlighted in *section F – Risk Assessment and Management*. GCF's concessional capital is expected to have a multi-fold impact on the entire e-mobility value chain in terms of:

1. *Crowding in capital*: Catalysing investments from various institutional investors who otherwise would have been reluctant to participate in the nascent e-mobility sector in India.
2. *Drive TCO parity*: Help achieve TCO parity for electric vehicles at a faster rate than currently estimated, thereby assisting the government in limiting its financial subsidy support for the sector.
3. *Improve availability of capital*: Assist in easing the availability of finance for electric vehicles and e-mobility focussed companies and / or customers, which is presently one of the biggest barriers impeding the growth of the sector. Successful implementation of this enterprise will likely lead to competitors in time.
4. *Help offer differentiated products*: Enable the Platform to offer bespoke leasing and structured credit solutions at attractive terms such as long tenure, competitive interest rates, high Loan to Value (**LTV**) ratio, reduced requirement of collateral or other forms of security and acceptance of operating and residual value risks, thus addressing several requirements of customers which are unmet by current product offerings in the market.
5. *Assist in capital recycling*: Help e-mobility focussed companies recycle their capital and invest either in technological advancements or expansion of existing manufacturing facilities (*refer to section C.3 – Capacity building and technology development / transfer for more details*).
6. *Drive replication in other geographies*: Demonstrate a business model that could be deployed in other markets to accelerate EV adoption.
7. *Establishment of sustainable, inclusive business practices*: The Platform will encourage counterparties to adhere to GCF standards and policies, on environmental and gender aspects. Increasing sensitisation and proactiveness on such aspects by counterparties is expected to have a profound impact on the local community and environment, and set a good example for other businesses / competitors in the industry. *For more details refer to Annex 06 – Environment Social Management System and Annex 08 – Gender Assessment Report*.

Apart from raising equity capital from institutional investors, the Platform will leverage on its Macquarie Group affiliation and networks to identify and hire an experienced management team and raise large amounts of third party debt capital from domestic (INR denominated) and / or foreign banks / financial institutions (USD denominated) depending upon market conditions, thus compounding the effect of GCF's concessional capital.

The Platform is anticipated to have a positive impact on India's post pandemic economic recovery, carbon emission reduction targets, air quality, health outcomes, new job creation and gender inequality.

B.6. Exit strategy (max. 500 words, approximately 1 page)

Exit strategy

The aim of the Programme is to build a leading E-mobility Financing Platform which can continue as a 'going concern' post the exit of investors. GCF and commercial investors' ('Investors') capital commitment will be for a period of up to 10 years, subject to defined extension options. The form of exit from the investment will be dependent upon the global and India's macro-economic environment, growth prospects, competitive landscape and status of rules and regulations governing the EV sector, future business potential of the Platform, ease of exit and returns offered to the Investors.

Exit can be in the form of:

1. Partial or Full Platform trade sale to:
 - **Strategic investors** such as banks and NBFCs who are looking to expand their exposure to the EV sector and / or are looking to diversify their existing asset portfolio.
 - **Financial investors** such as impact funds, infrastructure or sector agnostic private equity funds, pension and sovereign wealth funds that may be looking to enter the market.
2. Initial Public Offering (**IPO**): The Singapore Manager may also look to provide exit to existing investors through the IPO route which will provide opportunity for retail investors to participate in the company's long term of objective of decarbonising the transportation sector of India.

Sustainability

MAM, with support from GCF and commercial investors, is looking to build 'first of its kind' E-mobility Financing Platform in India, with the aim to create long term value and provide a sustainable and successful pathway for other companies and investors in the EV sector, even after GCF and commercial investors have exited the Platform.

To achieve this, the Programme has identified key areas related to building of 1) a sustainable business by instilling best practices (ESG and financial processes including monitoring, reviews, etc.), and 2) a sustainable EV ecosystem as critical for long term value creation and development of future investments' foundation.

A. Building a sustainable business by instilling best governance practices

MAM is recognised as the largest infrastructure asset manager globally⁶⁶ with an AUM of AUD 735.5 billion as at 30 September 2021.⁶⁷ It employs 1,921⁶⁸ investment and portfolio management professionals across the world who have relevant experience in building successful platforms and unique business models leading to scaled operational platforms.

As an entity managed by MAM, the Platform will leverage on MAM's experience to implement best practices and frameworks developed over several decades to build a sustainable company. This will include establishment of an empowered and effective Board and management team, operational checks and balances, effective systems of internal control and risk management, promoting a 'governance culture' of transparency and accountability, and putting in place remuneration policies that reward achievement of objectives and long-term shareholder value creation.

Further, post initial investment, MAM will be actively engaged with the Platform, driving long term sustainability and value creation. This will happen through active performance oversight, which includes monthly performance monitoring, news tracking, periodic assessment of counterparty's credit quality, active engagement and assistance to Platform management and impact assessment (*refer to section B.3 – Programme Description for more details*).

Additionally, the Platform will also implement a 'Watchlist' framework which will ensure that any stress or other factors that could have a material impact on the counterparty viability, are highlighted in advance so that appropriate remedial measures can be taken where possible.

Establishment of a robust oversight culture will seek to ensure that there is low probability of default and / or fraud among the lessees / borrowers which, in turn, will result in:

1. a strong sense of comfort to new players, investors and financiers looking to enter the market; and
2. 'proof of concept' affirmation for existing portfolio companies' business models.

B. Building a sustainable EV ecosystem

Development of new financial products

The Platform will look to offer bespoke financial products in the EV sector, which will include operating and finance leases. Apart from this, the Platform will consider offering other structured credit / financing instruments depending upon the requirements of the counterparty. As these products are scaled-up, there will be increased capacity from existing financial institutions, such as banks and NBFCs, to understand the key risks in EV financing, assess the performance of the financial products and develop relationships with key market players. All this will support potential replication of the financing products and development of newer products as the market heads towards maturity.

Reduce upfront EV costs through long-term financing at attractive rates

Currently, most EV companies generally find it difficult to raise capital from financial institutions. Even if companies can raise capital, the terms of that capital are usually unattractive and biased towards the financier making it a significant barrier impeding the growth of the sector. This Platform aims to change this paradigm through:

1. *differentiated product offering*: The Platform will look to provide differentiated financing products (operating and financial lease including variations that assume certain reporting and performance risks) at attractive rates that help reduce upfront capex and ongoing costs for OEMs, operators, and sponsors;
2. *focusing on the entire value chain*: Support the roll-out of EVs, charging infrastructure and other e-mobility infrastructure;

⁶⁶ IPE Real Assets 2021 – Top 100 Infrastructure Investment Managers.

⁶⁷ <https://www.macquarie.com/assets/macq/investor/reports/2022/macquarie-group-hy22-interim-report.pdf>

⁶⁸ As at 31 March 2021, <https://www.macquarie.com/assets/macq/investor/results-and-presentations/2021/macquarie-group-fy21-annual-report.pdf>

3. *longer tenure*: Make long tenured financing products a norm rather than an exception;
4. *acceptance of residual value risk*: Accept residual value risk to clearly demonstrate the useful life of EVs (>10 years in most cases) and promote greater market acceptance; and
5. *increasing private sector participation*: Increase participation of OEMs, operators, and sponsors in government tenders (wherein the concession agreements are usually awarded for 8-10 years).

Catalysing further investment through capacity building and replicable models

The risk-sharing models developed with local OEMs and financiers will have compounding effects outside the direct impact from the Platform itself:

- Increased investment in the sector is expected to instil greater confidence in the market, increase competition and drive down upfront and operating costs, thereby encouraging a cycle of further investment and market maturation.
- Increased capacity of local OEMs will support higher investment from the private sector in enhancing the manufacturing capabilities and developing new business models that propel long-term presence in the sector.

Lastly, the Platform will ensure continuous engagement with various stakeholders such as Niti Aayog (nodal agency for electric vehicles in India), Ministry of Environment, Forest, and Climate Change (**MoEFCC**), OEMs, Operators, other financing companies (such as banks and NBFCs) and consultants to:

1. drive further policy improvements;
2. help address challenges affecting the accelerated growth of the EV sector;
3. encourage innovation and competitiveness; and
4. help drive long term sustainability of the sector.

D. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	200			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	-	- years	- years	- %		
(ii) Subordinated loans	-	- years	- years	- %		
(iii) Equity	200	10 years	2+2 years			
(iv) Guarantees	-	- years				
(v) Reimbursable grants	-					
(vi) Grants	-					
(vii) Results-based payments	-					
(b) Co-financing information	Total amount			Currency		
	1,297			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Commercial Investors	<u>Equity</u>	<u>205</u>	<u>million USD (\$)</u>	<u>10 years</u> <u>2+2 years</u>		<u>senior</u>
Domestic/ International Banks and NBFCs	<u>Senior Loans</u>	<u>1,092</u>	<u>million USD (\$)</u>	<u>5-7 years</u> <u>Enter years</u>	<u>Market driven</u>	<u>senior</u>
Click here to enter text.	<u>Options</u>	<u>Enter amount</u>	<u>Options</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Click here to enter text.	<u>Options</u>	<u>Enter amount</u>	<u>Options</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	<u>1,497</u>			<u>million USD (\$)</u>		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	A total capital commitment of USD 405 million is expected from GCF and commercial investors. This amount will be invested over a period of 5 years across the various segments of the e-mobility value chain. <i>What is the purpose of third party debt?</i> In addition to equity capital from GCF and other investors, the SG Master Co., the SG Co, and / or the Platform will also look to raise third party debt (either INR and / or USD denominated) for future business expansion and create multiplier effect on equity investors' capital commitment. The decision to raise third party debt at either of the levels will depend on the market conditions, commercial considerations and ease of compliance. <i>How will debt be serviced?</i> Any third party debt that is drawn by the Platform will be serviced through the principal and interest payments that the Platform will receive from various lessees and borrowers.					

	<i>How will the Platform ensure financial discipline and compliance with debt servicing obligations?</i> Firstly, the maximum leverage that will be raised by the Platform will be as per industry benchmarks. Secondly, in order to secure a competitive third party credit rating (local or global) and attractive financing terms for this debt, the Platform will 1) establish robust credit and risk assessment frameworks, seeking to minimise payment defaults, 2) have a diversified portfolio by sub-sector, financing solution and credit profile of counterparties, 3) have a professional and experienced management team, 4) adopt the strong governance and risk framework generally followed by MAM managed companies, and 5) leverage on Macquarie's global expertise to overcome any initial challenges. <i>How much quantum of co-financing capital will the Platform be able to mobilise?</i> As mentioned in <i>section B.3</i>, the Programme will look to offer bespoke financial products such as operating leases, finance leases and structured loans depending upon the requirement of the counterparty. By offering such unique financing solutions, the Platform expects some level of either equity investment, mezzanine debt or secured debt to be raised by the counterparties. The exact amount of this funding may vary across each lessee / borrower.
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C.2. Financing by component

Component	Output	Indicative cost Options	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Component I: Crowding-in of capital	Output 1.1: Corporate structure establishment	<u>USD 405 million</u>	<u>USD 200 million</u>	<u>Equity</u>	<u>USD 205 million</u>	<u>Equity</u>	To be finalised once GCF is confirmed as an anchor investor.
	Output 1.2: Decarbonising India's transport sector through investments across the e-mobility value chain Click here to enter text.	<u>USD 1,092 million</u>	=	<u>Choose an item.</u>	<u>USD 1,092 million</u>	<u>Choose an item.</u>	Domestic / international banks and other financial institutions / markets to be finalised near the debt drawdown date.
Indicative total cost (USD)		<u>USD 1,497 million</u>	<u>USD 200 million</u>		<u>USD 1,297 million</u>		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☐ No ☒
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

The focus of the e-mobility Programme is to provide bespoke financial solutions to EV focussed companies in India. This Programme will create an enabling environment through the flow of capital into domestic OEMs, operators and start-ups, which have otherwise been unable to raise long term financing at attractive terms. This Programme will invest in the e-mobility sector at scale which will demonstrate new financing solutions to the market, build capacity of counterparties in developing financing solutions and support government tenders and initiatives. Furthermore, it is expected that various domestic OEMs, operators and / or start-ups who receive the capital from the Programme will:

1. recycle capital and invest either in technological advancements or expanding existing / establishing new manufacturing facilities;
2. re-invest capital in existing business to make it more sustainable;
3. invest in further development of supply chain;
4. be encouraged to participate in vertical or horizontal integration; and
5. encourage new financiers (debt and equity investors) to take additional risk.

All the above factors cumulatively are expected to help reduce the upfront vehicle and maintenance costs, thus driving down the EV TCO below that of ICE vehicles, and help in the overall development of the e-mobility sector.

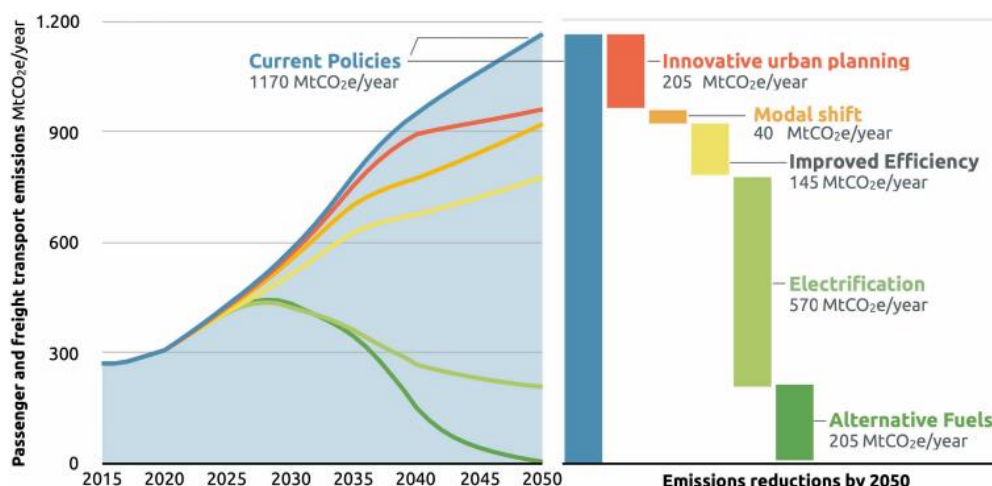
E. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

As a signatory to the Paris Agreement, India has committed to reducing the emissions intensity of its GDP by 33% to 35% below 2005 levels by 2030 (UNFCCC, 2015). While India includes a sector-specific target for the power sector in its NDC and COP26 pledges, there is no such target for the transport sector. Decarbonising the transport sector in India can help to substantially reduce the impacts related to GHG emissions and other air pollutants and has several additional potential co-benefits, such as major improvements in public health, particularly in urban areas, improved fuel savings and reduced noise pollution.

Electrification of vehicles plays an essential role in decarbonising the transport sector⁶⁹ as it offers promising GHG emission abatement potential, in comparison to other alternates, by 2050 i.e., towards 1.5°C scenario proposed by IPCC (refer to the below diagram). However, a mix of financial, behavioural and infrastructure incentives are needed to develop the e-mobility sector and associate infrastructure to achieve the promised targets.



The proposed Programme is expected to have a direct GHG reduction potential of **~2.3 MtCO₂e over the implementation period** and **~9.5 MtCO₂e over the lifetime**. Note that, to improve the additionality of the program, the Platform will seek to encourage counterparties to use natural coolants in the vehicles instead of hydrofluorocarbons. However, on a conservative basis, the impact of this has not been considered as part of the additionality analysis.

The Programme, through its implementation approach, is poised to lead the pathways to decarbonise the transport sector in India, pursuant to India's NDCs and COP26 pledges. Discussions on the Programme's leverage towards sustainable development, including contribution to 2030 Agenda for Sustainable Development, are elaborated in the subsequent section (refer to section D.3 – Sustainable development for more details).

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

The E-mobility Financing Programme aims to encourage a wide range of initiatives and aspects which are related to sustainability. For instance, the Programme aims **to support the acceleration of India's EV sector as a key mechanism in decarbonising the transport sector**. Financing solutions provided by the platform will aim to create a paradigm shift within the Indian context by supporting the private sector to drive forward the e-mobility sector, which can serve as an example for other similar markets. The Platform's ability to drive reduction of **~9.5 MtCO₂e emissions over the lifetime**, will contribute towards making India more resilient to the impacts of climate change, and provide a pathway that is consistent with national adaption strategies and plans (refer to section B.1).

The Platform will also contribute to the Indian regulatory framework and other policies launched by the Indian government and several state governments by regularly engaging with them and providing inputs on how to increase

⁶⁹ Decarbonising the Indian transport sector pathways and policies, Dec 2020, Climate Action Tracker.

the effectiveness and impact of policies / regulations enacted. For instance, the Platform will work with the government to slowly reduce subsidies and other tax incentives over time as the sector achieves maturity and substantial crowding-in of private capital, which would help achieve TCO parity across various segments of the e-mobility value chain. Additionally, the Platform will provide support to government think-tanks, such as the Niti Aayog, and other government bodies in formulating new policies leveraging MAM's global presence in leading EV markets such as China, USA, UK, and Germany, that will drive the development and success of the Indian EV ecosystem and EV financing market.

Contribution to an enabling environment for private sector investment

The EV sector is currently partially supported by an enabling policy environment, however, several barriers to investment (*as outlined in section B.2 – Theory of Change*) currently exist. The Platform aims to shift the e-mobility sector from a nascent industry to a mainstream sector which does not rely on concessional funding.

Contribute to achieving cost parity: The Platform, through its 'first-of-kind' financing solutions, aims to provide operating and finance leases to OEMs, sponsors, and operators. These financial solutions, priced at attractive rates with long tenures, will assist the counterparties to deleverage their balance sheets and recycle their limited capital, thereby encouraging further investments in technological advancements / R&D. Higher expenditure on R&D is expected to help reduce the high upfront EV cost and achieve TCO parity far before the current estimate of 2030 for different electric vehicles as provided in the diagram below.⁷⁰

Vehicles	Upfront cost differential	Expected time for TCO parity (with subsidy)	Expected time for TCO parity (without subsidy)
Two wheelers (Commercial)	0.9x – 1.1x	At parity	2023
Two wheelers (High speed)	1.5x – 2.0x	2024	2028
Three wheelers	1.2x – 1.5x	At parity	2022
Four wheelers (Commercial)	2.0x – 2.5x	2023	2025
Four wheelers (Personal)	2.0x – 2.5x	2025	2030
Bus	2.0x - 3.0x	2024	2030

Contribute to providing supportive infrastructure: By offering financial solutions to key parts of the ecosystem, the Platform is expected to support overall market development, especially in the area of charging infrastructure. For example, financial products to enable more widespread deployment of charging infrastructure assets / projects, especially in dense city areas, can reduce behavioural barriers, such as range anxiety for consumers. Supportive infrastructure such as battery swapping stations and quick charge stations would enhance EV adoption and make it more economically viable to own an EV. Strong supportive infrastructure has been shown to be a leading contributor to greater penetration growth in more mature EV markets.⁷¹

Potential for knowledge sharing and learning

The core focus of the Platform is to achieve maturation of the e-mobility market and development of EV financing market in multiple ways:

- Local and sustainable industry growth: This Programme intends to partner with local OEMs and other companies such that the e-mobility technical and market capacity is developed locally. The advantages of local development of EVs include lower costs, new job creation (*refer to section to D.3 – Sustainable Development*) and better logistical coordination. The Platform would also work with counterparties to encourage inclusive, socially responsible business practices. This would support the long-term growth of the sector and reduce the need for concessional or public funding over time.⁷² This also contributes to the Indian government's vision of an Atmanirbhar Bharat (Self-Reliant India) and 'Make in India' initiative, as the Programme encourages the growth and development of local businesses and industries.

⁷⁰ <https://assets.kpmg/content/dam/kpmg/in/pdf/2020/10/electric-vehicle-mobility-ev-adoption.pdf> and industry discussions.

⁷¹ Zero Emission Vehicles (ZEVs): Towards A Policy Framework- NITI Aayog, 2018.

⁷² Zero Emission Vehicles (ZEVs): Towards A Policy Framework- NITI Aayog, 2018.

- **Financing structures:** The focus on financing and leasing structures, rather than equity ownership of companies, through this Platform is expected to support the development of innovative business models across a broad range of local companies in India, where there are currently significant financing gaps. EV leasing is the predominate financing model in mature markets⁷³ and propagation of this structure is expected to encourage increased investment from private sector participants as the market matures. The Platform will look to offer long term leasing solutions at attractive terms (such as low interest rates, high LTV ratios, limited security and undertake asset ownership risk) which are not offered by any of the existing financial institutions in India. Below is a brief comparison of the types of products available in the market.

Parameters	E-Mobility Platform	Banks	NBFCs
EV specific product offering	Yes	Limited offering by couple of banks	Limited offering by select NBFCs
Type of products	Operating and financial leases, Structured loans	Loans	Structured loans

- Most banks and NBFCs are generally reluctant to provide financing solutions to EV companies due to the high technology risk and unproven business models (due to a lack of operating history). The Platform aims to change this paradigm as the learnings from the Programme will act as guiding precedents on how to price the product, manage various operational and financial risks, and what protection measures financial institutions need to seek to ensure appropriate risks adjusted returns on their capital investment.
- **Institutional investment:** Institutional investors generally invest in mature markets and have a lower risk appetite compared to public investors. The Platform intends to approach institutional investors to crowd-in investors to a sector / geography that they may have not previously considered due to the sector's nascent stage of development. It is intended that by providing exposure through a platform with a risk buffer provided by GCF's concessional capital, institutional investors will build their capacity and better understand the market as it matures, such that further investment can be considered with less or no reliance on concessional finance over time.

Potential for scale-up and replication

Financing solutions developed by the Platform will be designed for the Indian context, with an understanding of what has worked best in other mature e-mobility markets. Once established, the structures may have an 'example' effect whereby other financiers can replicate the products in the market to generate greater investment in the sector.

A successful electrification of India's EV market, at scale, provides a path for other geographies with similar markets to follow, including other South- and South-east Asian developing countries who are similarly faced with significant population and economic growth that will increase the contribution of emissions from the transport sector. The Platform itself could also be replicated in other markets to similarly support early-stage investment and providing financing support in-line with country ambitions.

D.3. Sustainable development (max. 500 words, approximately 1 page)

⁷³ For example, 80% of EVs in the US market are leased and not sold. KPMG, NextContinent.

Social co-benefits

Social and gender impacts

Through its partnerships with select domestic OEMs, the Programme aims to promote universal access and safety measures for women, children and people with disabilities, including proper lighting, in-vehicle cameras, vehicle monitoring systems and other safety measures, especially in buses. The Platform will look to encourage the engagement of women and men of all ages as stakeholders in the design, development and implementation of strategies and activities to be financed. A report by the World Research Institute (WRI) states that, the manufacturing and delivery of electric vehicles will require far lesser mechanical tasks that are perceived to be heavy-duty. Instead, digital and IT related skills that are more cerebral and 'strength neutral' will be needed. This opens the possibility of promoting inclusion and gender diversity in the auto industry workforce, which has historically been lopsided. Some vehicle manufacturers in India have set up women only shop floors, but mainstreaming and additional systematic efforts are required. Studies show that gender parity improves problem-solving, leading to increased profits. This is unlikely to become a reality for India without concrete actions, which may include: (i) Stronger marketing of skilling courses for women, (ii) Incentivising industry players who maintain gender balance in diverse roles across the sector, and (iii) Establishing labour and on-the-job welfare systems for equal benefits, job growth and safety for women working in the transport industry.⁷⁴ The Platform therefore will strive to maximise the impact of its funding for adaptation, mitigation and promotion of environmental, social, economic, and development benefits, taking not only a gender-sensitive, but a gender-responsive approach in line with the GCF's Gender Policy (*refer to Annex 8 – Gender assessment and programme level action plan for more details*).

Environmental co-benefits

Air pollution and health outcomes

The transport sector in India is linked with emissions of other pollutants, such as fine particulate matter (PM2.5), known to have a damaging impact on human health. This is caused by (among other things)⁷⁵ vehicles with low emission standards, compounded by increased congestion in cities, causing pollutants to be trapped. India has 21 of the top 30 most polluted cities⁷⁶ and it is estimated that over 84% of India's population are exposed to high concentrations of air pollution. This causes over 2 million deaths annually, and associated healthcare costs equalling ~5.4% of GDP.⁷⁷ India's capital, Delhi, was ranked as the worst city in the World Air Quality Report 2019,⁷⁸ and half of the 50 most polluted cities were also in India. The adoption of electric vehicles, especially in urban areas, will reduce pollution compared to adoption of ICE vehicles and, over time, replace older, polluting vehicles, thereby improving risk to the health of the population.

Economic co-benefits

Energy security

The Indian government's push for EVs is directly linked with its stated intent to enhance energy security in addition to GHG emissions reduction. Since the early 2000s, India's crude oil imports have risen steeply, of which 40% of total oil consumption comes from the transport sector.⁷⁹ Oil imports contributed USD 102 billion to the Indian trade deficit in 2020, covering over 80% of the country's oil needs.⁸⁰ An increase in the penetration of EVs directly and indirectly through this Programme will decrease India's oil import requirement, decreasing oil imports which is estimated to save USD 330 billion between 2017 and 2030, as well as corresponding GHG emissions.⁸¹

Economic impact and job creation

⁷⁴ "EV Connect" by WRI India, Issue 16, November 2019.

⁷⁵ Pollution is also linked to cooking activities, construction dust and factory emissions in cities. <https://www.bbc.co.uk/news/world-asia-india-48744163>

⁷⁶ IQAir Study, "World's most polluted cities 2019".

⁷⁷ Greenpeace report on "Air pollution from fossil fuels – Feb 2020".

⁷⁸ <https://www.iqair.com/world-most-polluted-cities/world-air-quality-report-2019-en.pdf>

⁷⁹ International Energy Agency (IEA) – World Energy Outlook 2018.

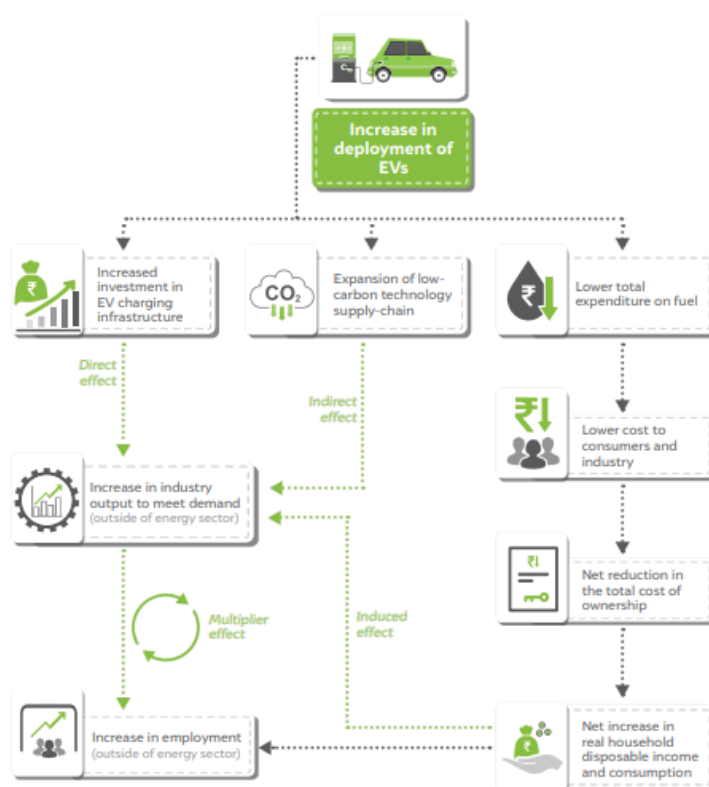
⁸⁰ Economic Times Publication on "India's crude oil import – April 2020".

⁸¹ NITI Aayog.

India's middle income class group is expected to increase from 158 million households in 2018 to 300 million households by 2030.^{82,83} During the same period, the number of urban residents is expected to increase from 473 million to 600 million.⁸⁴ This is poignant for the e-mobility Programme, which primarily aims to target public buses and corporate fleet segments which are mainly used by middle income groups and urban residents. Further, through its investments, the Programme will aim to provide a safe, reliable, comfortable, and affordable transportation system, which will increase the adoption rate among urban residents and middle-income groups, thereby accentuating the Programme's intended impact.

Additionally, as per a recent Niti Aayog report, European Climate Foundation has estimated that any reduction in oil demand by more efficient electric cars is expected to create 500,000 to 850,000 more domestic jobs by 2030. This report explains that oil production and distribution has a very low employment intensity, with just four jobs per million Euros of value added. This is far below the 24 jobs per million Euros in the general economy. Therefore, any shift in expenditure from buying imported oil to other expenditure choices would generate additional employment.⁸⁵

The below chart highlights how a transition to EVs results in cumulative benefits to the economy, primarily through increased job creation.⁸⁶



Thus, the Programme, through its partnerships with leading domestic OEMs, operators and sponsors, will have a significant impact on the local economy through new job creation in both rural and urban areas, and supporting urban mobility for economic growth.

Alignment to Sustainable Development Goals (SDGs)

The Programme represents an increasingly important linkage to provide value to society by delivering safe, secure, and sustainable mobility and associated co-benefits. It has a large potential to promote and accelerate country's

⁸² Note: Low income: < USD 4,000, Lower mid: USD 4,000-8,500, Upper-mid: USD 8,500-40,000, High income: >USD 40,000, basis income per household in real terms per annum; Projections with annual GDP growth assumed at 7.5%.

⁸³ World Economic Forum – Future of consumption in fast growing consumer markets: India.

⁸⁴ World Economic Forum – 3 biggest challenges for India's future.

⁸⁵ Zero Emission Vehicles (ZEVs): Towards A Policy Framework- NITI Ayog, 2018, pp 10.

⁸⁶ <https://www.ceew.in/sites/default/files/CEEW-India's-EV-Transition-Post-COVID-19-22Dec20.pdf>.

National Development Agenda to implement United Nations Sustainable Development Agenda 2030 as mirrored in the SDGs. The Programme relates to the following United Nations SDGs:

Key UN SDG	Contribution
 13 CLIMATE ACTION	1. Reduced emissions from greenhouse gases 2. Reduced air pollution (Lower health related issues) 3. Reduced petrol / diesel fuel use
 3 GOOD HEALTH AND WELL-BEING	1. Reduction in urban air and noise pollution 2. Improved health outcomes / lower costs^{87, 88, 89} 3. Improved ride comfort
 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	1. Accelerates shift from ICE to EV 2. Faster scale helps lower upfront costs 3. Innovative financing
 11 SUSTAINABLE CITIES AND COMMUNITIES	1. Cleaner air 2. Promoting public transport 3. Promoting financial sustainability
 8 DECENT WORK AND ECONOMIC GROWTH	1. Greater adoption of public transport 2. Improved productivity 3. New high quality manufacturing and maintenance jobs
 17 PARTNERSHIPS FOR THE GOALS	1. Promoting access to technology 2. Mobilizing international capital 3. Building capacity for green transition

D.4. Needs of recipient (max. 500 words, approximately 1 page)

India's vulnerabilities

Between 1901 and 2018, India's average temperature rose by 0.7°C, resulting in extreme weather events and putting pressure on the country's already scarce resources, thereby threatening the livelihood of millions of people. In recent times, with increases in sporadic climatic events such as cloud bursts, flash floods, unseasonal rains and prolonged droughts, crop failures have become widespread, water shortages more chronic (India already ranks as #13 on the WRI's National Water Stress Rankings Across Globe) and migration and related conflicts elevated across the country. Changes in rainfall patterns and sea level can have severe impact on coastal cities. Cyclone Amphan in 2020 in the coastal state of West Bengal was one of the strongest cyclones on record in the Bay of Bengal, killing ~100 people and causing USD 14 billion worth of damage.⁹⁰

The average temperature in India is expected to rise by 4.7°C to 5.5°C by 2100 under the Representative Concentration Pathway (RCP) 8.5 scenario (a scenario in which emissions continue to rise throughout the 21st century),⁹¹ with extreme climatic events and natural hazards rising in terms of intensity, spatial extent, and frequency across India. If there is no

⁸⁷ OECD has estimated the poor outdoor quality will negatively impact Indian GDP growth by ~1% between 2010 and 2060 in a "do nothing" scenario (~£15bn equivalent).

⁸⁸ Greenstone et al, 2015; CSTEP, 2018.

⁸⁹ Sustainable Development Department, World Bank 2013.

⁹⁰ <https://en.wikipedia.org/wiki/CycloneAmphan>

⁹¹ <https://reliefweb.int/report/india/assessment-climate-change-over-indian-region-report-ministry-earth-sciences-moes>

change to current climate mitigation policies, India may lose the equivalent of 2.8% of GDP by 2050.⁹² The most severe impact of this is expected on low and middle-income groups, women, children, and the elderly. Around 28% of the population, or 364 million people, live below the national poverty line of USD 2 per person per day.⁹³ 1.77 million people are homeless⁹⁴ and 6.5% of the population are unemployed according to the Centre for Monitoring Indian Economy (CMIE).⁹⁵ Lower GDP would mean fewer jobs, which will have a disproportionate impact on youth and women. Today, many households in the rural and urban areas do not have adequate provision of basic infrastructure and services, such as access to electricity, safe water, and mobility. Given these factors, there is an urgent need to undertake low or zero carbon and climate resilient infrastructure investments in India. According to a recent report by Lifegate, India needs ~USD 2.5 trillion of investment between 2015 and 2030, or USD 170 billion/year in-order to meet its NDCs by 2030.⁹⁶ The government's budget alone will not be sufficient to finance the required funding for climate action. While the Indian government has doubled its spending on green projects since 2016, its budget is increasingly focused on socio-economic development activities rather than climate mitigation and / or adaption activities. Furthermore, the COVID-19 pandemic, national lockdown and job losses have exacerbated the strain on the government's finances as the need to build healthcare infrastructure takes priority. In a post Covid-19 scenario, there will be an even greater need for the private sector to drive capital flow in the e-mobility sector.

Economic stimulation

The need for economic stimulation is also critical as the country faces the ongoing COVID-19 pandemic. The pandemic has triggered a shift in the priorities of the government towards the creation of an effective healthcare ecosystem, primarily focused on delivering quality primary care to citizens. Confronted with the ever-growing threat of physical climate risks, the economy requires a more focused approach to policy development and funding. This is iterated by an analysis by Oxford Economics that suggests that India's GDP could fall by 90% by 2100 if current policies are not improved and climate finance is not scaled up.⁹⁷ Additionally, a McKinsey report states India is set to face significant impacts of climate change, which will also have severe impacts on economic development.⁹⁸

Financial pathways

According to a recent report by Niti Aayog, over the next decade, India's EV transition will require a cumulative capital investment of INR 19.7 trillion (USD 266 billion) in vehicles, charging infrastructure, and batteries. This analysis suggested that, by 2030, the market size for the financing of EVs in India has the potential to be INR 3.7 trillion (USD 50 billion). As the amount of investment required for India's mobility transition is significant, mobilising the investments to finance OEMs, battery manufacturers, charging stations and end consumers will require targeted and systemic policy support, in addition to shifts in market design, business models, and financial structuring. There are financing barriers currently in place which restrict the large-scale penetration of EVs. In a recent report, Niti Aayog and Rocky Mountain Institute outline the following key challenges to scaling up finance for EVs in India:⁹⁹

1. **High interest rates:** Electric vehicles face an additional risk premium compared to ICEs when seeking financing, adding to operating costs.
2. **Low LTV ratios:** Due to the risk of borrower default and low EV resale value, low LTVs are offered in financing packages which increase the equity contribution required by operators.
3. **Specialised products:** Most financing institutions do not currently offer specialised products such as leasing for EVs. Operators face high interest rates, low LTVs and shorter repayments compared to loans for ICE vehicles. Additional collateral may also be required, which does not support innovation by newer or smaller companies.

The above issues get more pronounced in case of EVs as the upfront capital cost of acquisition of EVs are significantly higher than their peer ICE vehicles and the residual value is uncertain. Therefore, this Programme aims to provide the Indian transportation sector with a **new and focused EV financing platform** that can:

1. play a catalytic role to accelerate the **creation of India's EV leasing market** by offering customised operating and financial leases. By providing this solution, the Platform will be **willing to undertake asset ownership**,

⁹² https://www.business-standard.com/article/current-affairs/climate-change-could-cost-india-2-8-of-gdp-by-2050-says-world-bank-118062801005_1.html

⁹³ <https://www.downtoearth.org.in/blog/governance/mass-poverty-is-back-in-india-76348#:~:text=Contrast%20this%20with%20the%20situation,per%20cent%20of%20the%20population.>

⁹⁴ <https://www.indiaspend.com/1-77mn-indians-are-homeless-40-of-them-are-getting-no-lockdown-relief/>

⁹⁵ <https://www.businesstoday.in/sectors/jobs/india-unemployment-rate-drops-65-in-january-employment-rate-surges-379-cmie/story/431334.html>

⁹⁶ <https://www.lifegate.com/india-climate-finance-sdgs>

⁹⁷ <https://www.livemint.com/news/india/india-s-gdp-could-fall-90-by-2100-due-to-climate-change-analysis-11594187614729.html>

⁹⁸ "Will India get too hot to work?" – Case Study – McKinsey Global Institute, November 2020.

⁹⁹ https://niti.gov.in/sites/default/files/2021-01/RMI-EVreport-VF_28_1_21.pdf

performance and residual risks, a capability currently not available with any of the existing banks and NBFCs;

2. ensure overall development of the entire EV value chain by reducing upfront capex deployment, increasing availability of finance and offering attractive lease / credit terms by leveraging on GCF's concessional capital;
3. help mobilise the massive quantum of capital from various institutional investors towards India's EV sector and assist in reducing the need for government financial subsidy over time; and
4. bring focus to a sector which has, until now, received muted response from existing banks and NBFCs. It is important to note that, whilst select banks, NBFCs, fintechs and sustainability financiers have started providing vehicle financing loans / structured credit solutions to electric vehicles:
 - a) most banks and NBFCs are reluctant to participate in the sector due to small present market size, limited performance history and high residual value risk; and
 - b) fintechs and sustainability financiers do not have the balance sheet strength to create meaningful impact, and are primarily retail (B2C) focused with two and three wheeler customers as the target segment.

Although current policies, incentives, and financial assistance have caused some movement in the EV sector, it still is not adequately backed to overcome challenges faced by OEMs in accessing capital for expansion, weak business case for charging infrastructure and high up-front cost associated with EVs. Therefore, this Programme intends to drive climate mitigation (positive externalities) and economic stimulation (creation of a multiplier effect) by providing innovative financial solutions, investing in charging infrastructure, crowding-in institutional investors and supporting growth of the domestic industry to reduce costs.

D.5. Country ownership (max. 500 words, approximately 1 page)

This Programme directly contributes to India's NDC target of reducing carbon emissions by 35% by 2030 from 2005, including through transforming 30% of all vehicles to EVs.¹⁰⁰ EV promotion has been supported by significant policy development at both the national and state levels. The Indian government has already committed subsidies of up to USD 1.4 billion to enable an accelerated roll out of EVs under the FAME Phase II scheme. The Programme will participate in the FAME scheme and other government policies and schemes where appropriate, which includes requirements for a predefined level of domestic component manufacturing.

At least 10 states have policies that are being implemented, and another six are drafting relevant policies¹⁰¹. Although most of these policies prioritise public transport, paratransit, and job creation, each of these states have also set targets and other incentives (depending on the state) for supply-side (manufacturing) and demand-side (consumer and charging infrastructure investments).¹⁰² For example, this Programme will seek, directly or indirectly, to engage in tenders released by the states for public buses, charging infrastructure or otherwise, and aim to be involved in stakeholder discussions to support further policy development.

Stakeholder engagement

Whilst we have been in active discussions with multiple stakeholders across the value chain, including government entities such as Niti Aayog, the Ministry of Road Transport and Highway (**MoRTH**), the Ministry of Environment, Forest, and Climate Change (**MoEFCC**) and the Ministry of Power, most of the efforts have been largely concentrated towards engaging with Niti Aayog, the Indian government's think tank, which is the key driver of all e-mobility initiatives in India. It has been playing a major role in shaping industry policies, guidelines, charging standards and best practices for the Indian EV ecosystem.

In addition, several discussions have been held with other industry stakeholders, including leading domestic OEMs, bus operators, charge infrastructure providers, vehicle scrappage companies, banks, NBFCs, and independent experts. The discussions have been focused on:

- better understanding the present market dynamics;
- future growth potential of the EV sector;

¹⁰⁰ <http://www.cleanenergyministerial.org/campaign-clean-energy-ministerial/ev3030-campaign#:~:text=The%20EV30%4030%20campaign%20redefines,share%20for%20EVs%20by%202030.>

¹⁰¹ Niti Aayog and Rocky Mountain Institute. "Mobilising Finance for EVs in India". January 2021.

¹⁰² Ramanath Jha, "Preparing Indian Cities for a Shift to E-Mobility," ORF Issue Brief No. 458, April 2021, Observer Research Foundation.

- opportunities, key challenges, and constraints faced by the industry, especially in relation to high upfront vehicle costs, lack of financing solutions, slow progress on EV adoption, etc.;
- understanding how some of the key risks and constraints can be addressed; and
- ground realities, challenges, and issues in the implementation of EV related infrastructure.

The key constraint we have identified during this dialogue has been access to the right kind of capital. The proposed Programme seeks to address this constraint.

Engagement with National Designated Authority (NDA)

There has been an active engagement with NDA since the Concept Note stage, regularly updating it about the various aspects of the e-mobility programme as envisaged by MAM.

The AE secured a No Objection Letter for the Programme from Indian government on 15th September 2021 (refer to Annex 01 – No Objection Letter for more details).

Capacity and successful track record to execute investments in India

For over twenty-five years, MAM has partnered with clients, governments, and communities to finance, manage, develop, and enhance essential infrastructure. Today, MAM's portfolio is relied on by more than 100 million people every day and employs ~170,000 people globally.¹⁰³ Our team of highly experienced professionals invest in businesses that underpin economies and communities, aiming to add real and lasting value for our clients and the people our assets serve.

MAM launched its first India focused fund with ~USD 1 billion of capital commitments in 2009 in partnership with the State Bank of India and IFC.¹⁰⁴ Since then, MAM has invested via its multiple Asia funds across roads, airports, communications, energy, and renewable energy in the country.

MAM's India portfolio is managed by a team of highly experienced investment and asset management professionals in the infrastructure and impact investing space, based out of Mumbai in India. The team is well supported by MAM's global network on asset management, legal, ESG, HR, IT, finance and tax related matters.

Given MAM's presence across India through its portfolio, the investment team has a robust mechanism of stakeholder management ably supported by senior management of portfolio companies, third party consultants and advisers. This mechanism ensures that MAM India is continuously engaged with all stakeholders, such as central and state government ministries and agencies, consultants / advisers, competitors through industry forums and end users, to be abreast of the latest trends, developments and policy changes envisaged by the government. Through its strong relationships across the industry, MAM India can play a vital role in the overall development of the sector and growth of the Indian economy.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

The e-mobility sector in India requires a significant amount of capital to accelerate the transition to EVs and avoid the multi decade technology lock-in of conventional vehicles. However, as a nascent sector, commercial investors do not have the risk appetite to deploy the capital at the pace required given the real and perceived investment risks outlined in this proposal. Although there is strong government support, the market requires additional intervention from concessional capital to help the market mature and create an enabling environment for large-scale commercial investments.

Role of GCF capital

Capital provided by GCF will serve to crowd-in all other investors into the Programme that would otherwise have not explored the opportunity to invest in the e-mobility sector in India due to their risk appetite, nascency of the sector, or small investment ticket size.

The Programme aims to provide 'first of its kind' financing solutions in the market, such as operating and finance leases, through a new investment vehicle in India. GCF's capital will sit in a 'first-loss' position which provides a cushion to the

¹⁰³ <https://www.macquarie.com/assets/macq/investor/results-and-presentations/2021/macquarie-group-fy21-presentation.pdf>

¹⁰⁴ <https://www.macquarie.com/au/en/about/news/2009/sbi-and-macquarie-launch-indian-infrastructure-fund-usd1billion-raised.html>

potential downside investment risks for commercial investors. This has been sized based on the risk appetite of potential investors, including what level of downside risk can be accepted by these types of investors.

GCF's capital will be provided in the form of an equity instrument which is able to take a high-risk position in the investment structure in line with GCF's elevated risk appetite. Further, GCF's involvement will not only increase the amount of capital invested in the sector, but also allow the Platform to provide financing that is not currently available in the market (i.e. a higher risk appetite than current financiers are able to provide). Over time, current financiers / other market participants will replicate the Platform's business model and leverage its learnings to provide wide array of financing solutions to large number of counterparties, thus assisting in overall development of the EV ecosystem.

Role of the Platform

As described in this proposal, there are several interested parties and existing players (such as OEMs / operators, etc.) who are seeking to grow in the e-mobility space, however they are currently inhibited by high upfront costs and limited availability of financing solutions. This Platform, through its innovative financing solutions such as operating and financial leases, helps to mitigate these issues. Further, it generates efficiency by enabling OEMs / operators to help recycle capital which can be deployed by them to drive technological advancements, market development, growth of associated infrastructure within India.

Structure and mobilisation

Pooling of GCF and other investors outside of India establishes the waterfall of cashflow returns from the underlying Platform in a way that is familiar to international and large-scale investors (*refer to sections B.4, C.1 and C.2 for more details*).

F. LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

☒ Reduced emissions (mitigation)

☐ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	There is a growing level of investment in the e-mobility space in India, which is contributing to the penetration of EVs and ultimately reduced GHG emissions from the transport sector. However, the sector is still in its nascency and investment needs to be	<u>Low</u>	A mature market and enabling environment across multiple sub-sectors within e-mobility segment resulting in increased penetration of EVs, development of associated infrastructure, and achievement of various national adaption targets.	The programme aims to mobilise ~ USD 2.1 billion ¹⁰⁵ of capital over the implementation period by providing bespoke financing solutions (such as leasing and other structured credit solutions) to e-mobility companies and customers, thus supporting deployment of ~115,000 new EVs and charging infrastructure points over the life of the Programme (activity 2). Further, this intervention is projected to deliver ~ 9.5 MtCO₂e of lifetime GHG emission reductions and ~ 32 MT of other pollutants

¹⁰⁵ Includes redeployment of internal accruals

	scaled-up to support maturation of the market.			being avoided (~20.5 MT of Carbon Monoxide (CO) and ~11.5 MT of Nitrogen Oxides (NOx)), even before any replication effects are considered, representing a paradigm shift on emissions. Additionally, mobilisation of capital through 'first of its kind' leasing solutions is expected to drive TCO parity among electric and ICE vehicles at a much faster pace, whilst at the same time enabling OEMs / operators/ sponsors to recycle capital for further technological advancements, thus setting up the right environment for the market to mature and scale up the impact.
Replicability	There are limited financing solutions (especially leasing solutions) available in the e-mobility space in India.	<u>Low</u>	A number of financing solutions available from multiple providers (beyond the Platform funded by this GCF Program) and across e-mobility sub-sectors. These financing solutions are fit-for-purpose and apply best practices developed from the Platform's experience.	The Platform will be established by the Programme (activity 1) and provide financing to counterparties (~USD 2.1 billion¹⁰⁶ over the life of the Program) across e-mobility sectors. Financing solutions will be new to the market or build on existing best practices across different sub-sectors. Successful implementation of these new structures will promote replication both within and outside India. Further, the Platform will provide direct employment opportunities which will also help develop capacity within the sector.
Sustainability	Investment in the e-mobility space is growing, however sustainable and profitable business	<u>Low</u>	The creation of an enabling environment to support continued growth of the e-mobility sector in India, including new participants and sustainable business models. In particular, sustained	The Platform will be established as a 'going-concern' entity and will implement best in class governance, institutional and fiduciary frameworks to provide financing solutions to multiple counterparties / customers. The exit strategy for the GCF

¹⁰⁶ Includes redeployment of internal accruals

	models across various e-mobility sub-segments are yet to be established.		availability of financing for key segments of the e-mobility space where there is significant potential to reduce GHG emissions.	envisages the Platform being sold either to a new or existing market participant, who will continue to operate the Platform and therefore continue to finance the e-mobility sector. The sale of the Platform will be evidence of e-mobility market maturation and will also provide an avenue for the intervention to be continued.
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E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term ¹⁰⁷	Final ^{108,109}	
<u>MRA2 Low-emission transport</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Portfolio company data and independent verifications of inputs to GHG calculation methodology.	0	~0.3 MtCO ₂	~2.3 MtCO ₂	Methodology: For EVs (CDM Version AMS III C: Emission reduction by electric and hybrid vehicles – version 15.0) and for EV charging systems (Verified Carbon Standard (Verra) VM0038 – Methodology for electric charging systems). Project Lifetime: 20 years based on initial capital

¹⁰⁷ Mid term refers to FY27 (5 years from FAA effective date).

¹⁰⁸ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

¹⁰⁹ Final refers to FY32 (10 years from FAA effective date). Note that there is a high likelihood that GCF investment may be exited before this timeline in which case the targets will need to be suitably modified.

						deployed over 5 years and 15 year vehicle life. Annual emissions avoided: ~0.23 MtCO₂e Lifetime emission reductions: ~9.5 MtCO₂e
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E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation</u>	EV and charging technology is somewhat established in India, especially in 2W. However conversion to this new technology away from ICE is still in its nascency and further innovation is pivotal in accelerating the market towards greater penetration of EVs.	<u>medium</u>	A mature market for EVs in India where the national target of 30% (or otherwise by sector) is reached by 2030. This will involve not only conversion to existing technology, but private sector (including OEMs and other financing counterparties) investing further in research and development across the sector.	The programme will provide financing solutions to support financing counterparties choose the new technology of EV (including new models developed) over ICE, ultimately supporting greater penetration of EVs in India and supporting the paradigm shift. These financing solutions will address the significant upfront cost of a new EV, allowing for greater capital to be recycled by	<u>National level (one country)</u>

				companies into research and development.	
<u>Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level</u>	The e-mobility sector in India has significant potential, however it remains in its nascency. Despite the regulatory and policy frameworks in place, the markets in most sub-sectors remain relatively limited and best practice business models are still being developed. Increased capital, and financing options in particular, are needed to support large-scale investment that will contribute to overall market maturation.	<u>medium</u>	Strong e-mobility financing market offering a range of products to support investment of capital across the e-mobility sector, which increases the penetration of EVs and enabling infrastructure. This should, in turn, put downward pressure on the cost curve and make EVs more attractive compared to traditional fuel alternatives.	The programme will provide financing solutions to counterparties across the e-mobility space which will not only support scaling up new EV penetration, but also have the potential to be replicated by other financiers to further scale-up the market impact.	<u>National level (one country)</u>
<u>Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	Best practice and range of financing solutions available in the e-mobility space are still being developed. Many existing banks and financial institutions do not have the capacity to provide financing to the sector given the significant real and perceived risks (like unproven technology, lack of operating history, limited or negligible data on residual value, etc.).	<u>low</u>	Knowledge of financing products and best practices within each sub-sector is developed. Increased capacity among financial institutions and otherwise to provide customised financing solutions to companies / customers in the e-mobility space at attractive terms. Financiers provide knowledge and learnings to support development	The programme will provide financing solutions not currently available in the market and help develop new products that have the potential to be replicated. Counterparties will also be exposed to the implementation of good standards in environmental, social (including gender) and fiduciary elements. The programme will have	<u>National level (one country)</u>

	There are some new entrants to the market in specific segments or products, however increased capacity to deliver a range of financing solutions is important for scaling-up the penetration of EVs and the transition away from ICE vehicles.		of policies and regulations in the e-mobility space.	monitoring and evaluation systems in place to identify lessons learned. The Platform will participate in forums of knowledge sharing to support regulation and policy in the e-mobility space.	
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E.5. Project/programme specific indicators (project outcomes and outputs)

This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the “Project/programme co-benefit indicators” section in table below. Add rows as needed.

Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output I – Crowding-in of commercial capital to invest in the e-mobility sector in India	Capital raised at the platform level from commercial sources leveraging GCF capital (annual basis)	Final capital structure	0	USD 205 million	USD 205 million	This represents the total equity capital expected to be raised from commercial investors during the implementation period across the executing entities. Additional equity is not expected to be raised in the base case, however, debt financing for the Platform will be sought.

						Note that, given the 'market making' nature of the Programme, GCF will invest a minimum of USD 50 million of capital on the FAA effective date (First Close) .
	Amount of capital invested by the Platform into e-mobility value chain <i>(to be reported by segment)</i>	Platform's annual reports	0	USD 841m ¹¹⁰	USD 2,074m ¹¹¹	Total capital (including debt plus equity) to be mobilised during the implementation period. Debt capital may be raised from domestic or international banks / financial institutions depending upon the market conditions.
Output II – Electric vehicles financed	Number of e-buses financed by the Platform	The Platform's quarterly and annual reports	0	8,000	18,000	1. Continued strong government support for electrification of transport. 2. Regular government tenders for EV buses, charging infrastructure and other e-mobility sub-sectors. 3. Increased investment in EVs will displace investment in ICE vehicles.
	Number of four-wheelers financed by the Platform		0	7,000	22,000	
	Number of two / three-wheelers financed by the Platform		0	15,000	65,000	
Output III – Charging stations financed	Number of charging stations established		0	3,000	10,500	

¹¹⁰ Excludes the management fee, setup, and other operating costs.

¹¹¹ Excludes the management fee, setup, and other operating costs.

	through investment by the Platform					
Output IV – Total amount invested in other parts of the e-mobility value chain	Percentage of total investments into other parts of e-mobility value chain		0	Up to 5%	Up to 5%	
Outcome 1 – Enabling environment for EV market established	The TCO of electric buses per kilometre compared to ICE alternative	Portfolio company data	1.31x ¹¹²	1.25x	1.15x	Continued overall market growth, government support and shifting consumer attitudes.
Outcome 2 – Increased penetration of electric vehicles in India	% penetration of EVs as a proportion of total new vehicles registered	Government published data	1.7% ¹¹³	5%	15% ¹¹⁴	Not assumed to be directly attributable to the Programme but reflective of the overall growth in the sector. Conservative targets set given the Programme does not have control over outcomes of entire market. TCO representative of entire market and not just the buses financed by the Platform.
Project/programme co-benefit indicators						
Co-benefit – Decreased pollution	Tail pipe emissions of Carbon Monoxide	Portfolio company data and independent verifications	0	3,096 tonnes	20,505 tonnes	Tail pipe emissions reduced are calculated basis average of Bharat

¹¹² E-Bus at 64.46/km compared to Diesel-Lc bus at 49.08/km as per <https://wri-india.org/blog/unboxing-barriers-scaling-electric-bus-adoption-india> (Note Total Cost Ownership (TCO) will vary depending upon the bus model, opex and leverage assumptions).

¹¹³ <https://vahan.parivahan.gov.in/vahan4dashboard/>, sourced September 2021

¹¹⁴ Considering stated government target and past realisation % of the targets. https://www.niti.gov.in/sites/default/files/2021-04/FullReport_Status_quo_analysis_of_various_segments_of_electric_mobility-compressed.pdf

	(CO) avoided through Platform investments					Standard IV and BS VI emission standards across multiple vehicle segment ¹¹⁵
	Tail pipe emissions of Nitrogen Oxides (NOx) avoided through Platform investments	Portfolio company data and independent verifications	0	1,929 tonnes	11,536 tonnes	
Co-benefit – Direct job creation	Direct jobs created by the Platform	The Platform's annual reports	0	75	100	Baseline of zero given the company is not yet established. Direct jobs defined by roles created by the Platform itself and not including consultants or roles created at financing counterparties. Expected growth driven by investments and new debt capital raised increasing size of overall Platform and capacity for staff.

E.6. Project/programme activities and deliverables

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.

Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.

Activities	Description	Sub-activities	Deliverables
Activity set 1– Platform establishment and fundraising			

¹¹⁵ <https://dieselnet.com/standards/in/2wheel.php>, <https://dieselnet.com/standards/in/ld.php>, <https://dieselnet.com/standards/in/hd.php>

Activity 1.1.1 – Establishment of e-mobility program’s corporate structure	Establishment of corporate entities / structure in Singapore and India for tax efficient inflow / outflow of capital to finance India’s ‘first of its kind’ ‘e-mobility program’	Sub-activity 1.1.1.1 – Establishment of corporate structure outside India Sub-activity 1.1.1.2 – Setup of corporate structure within India Sub-activity 1.1.1.3 – Raise capital from commercial investors and financial institutions	• Setup of LP – GP structure for transfer of GCF capital into the e-mobility program. • Establishment of corporate structure and entities in India and procurement of NBFC license for the Platform company to provide financing solutions to counterparties. • Crowding-in of commercial investors, including fundraising efforts and legal execution of their investments either in LP and / or SG Master Co., SG Co. and / or the Platform. • Ensure entire corporate structure is compliant with all applicable local rules and regulations.
Activity 1.2.1 – Setup of systems and processes in each entity	Adoption of various operational and governance related frameworks and policies to ensure smooth, efficient, inclusive and transparent working of each corporate entity	Sub-Activity 1.2.1.1 – Finalise governance policies and standards including adoption of operational manual and investment procedures as agreed with GCF	• Adopt risk management, HR, cybersecurity, third party vendor, E&S and D&I, etc. frameworks and policies for each entity in alignment with GCF policies at each entity level.
Activity 1.3.1 – Setup of key management team	Hiring and appointment of management team and Board to ensure proper functioning of each entity	Sub-Activity 1.3.1.1 – Identification, hiring and appointment of Board of Directors, management team and support staff	• Appointment of Board of Directors having diverse industry and finance experience. • Ensure timely hiring of management team and relevant support staff.
Activity Set 2 – Platform Investments, Monitoring and Exit			
Activity 2.1.1 – Investment approval (pipeline development, due diligence, secure approval from Board or empower sub-committee)	The management team to source transaction opportunities across the e-mobility value chain that have strong	Sub-Activity 2.1.1.1 – Sourcing of opportunities and development of pipeline, including stakeholder engagement with local OEMs,	• Overview of origination efforts, including summary of stakeholder discussions for pipeline

	potential to help in climate impact and drive gender and social inclusion	government stakeholders, and relevant market participants Sub-Activity 2.1.1.2 - Due diligence, investment approval and transfer of funds	development. To be included in quarterly reporting. • Reporting on number of opportunities considered, screened, approved, and executed.
Activity 2.2.1 – Post investment performance monitoring	Establish robust performance monitoring and reporting protocols for the lessees or borrowers also incorporating the additional requirements for GCF compliance Collect and share knowledge based on investment experience in the EV financing market	Sub-Activity 2.2.1.1 – Monthly MIS, quarterly and annual reports to be prepared based on information received from the counterparties Sub-Activity 2.2.1.2 – Impact assessment Sub Activity 2.2.1.3 – Share knowledge developed on financing in e-mobility sector to support enabling environment	• Ongoing monitoring of financial solutions provided to counterparties, including compliance with financing covenants, meeting of agreed reporting requirements, and credit assessment of counterparty. Summary will be provided in quarterly investment reports. • The Platform will undertake a climate impact assessment of the portfolio on annual basis. This will be performed either in-house or by an external consultant. • This assessment will be separate from the two independent evaluations that will be undertaken by the Platform (mid term and final evaluations) during the implementation period.
Activity 2.3.1 – Exits	Management team to drive exit out of the underlying financing products including realising residual value, and / or overall Platform in order to achieve the target Internal Rate of Return (IRR) for Investors	Sub-Activity 2.3.1.1 – Identifying potential exit opportunities Sub-Activity 2.3.1.2 – Ensure successful transaction closure	• Management team to actively engage with potential investors in the market to explore and identify exit opportunities. Summary of any relevant exit process to be provided in quarterly investment reports. • Management team to lead all documentation discussions with the assistance of the legal adviser.

			It will ensure that appropriate sale value is received on time and subsequently distributed back to the Platform's Investors.
Activity Set 3 – Wind up of corporate structure			
Activity 3.1.1 – Wind up of corporate entities post exit	Wind up of corporate structure post sale of Platform company in India or offshore pooling vehicle in Singapore	Sub-Activity 3.1.1.1 – Distribution of capital to investors Sub-Activity 3.1.1.2 – Ensure compliance with share purchase agreements and local rules and regulations	• Distribution of capital and returns to investors as per the LPA (LP Agreement). • Ensure compliance with indemnities and warranties if any. • Wind up of corporate entities in compliance with local rules and regulations

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Investment reporting

All investors of the Platform will receive quarterly business update reports including:

1. updates for key financings, where relevant;
2. performance metrics; and
3. drawdowns and distributions during the quarter.

Financial reporting

All investors of the Platform shall receive:

1. Within 120 calendar days of the end of the relevant accounting period, annual audited accounts of the Fund for each accounting period prepared in accordance with the accounting principles agreed with the auditors, which shall include, without limitation, a balance sheet, profit and loss account, statement of changes in Investors' capital for such period, summary of the financial information received from or in respect of investments and summary of material risks and their management.
2. Quarterly reports within 60 calendar days at the end of each calendar quarter, providing an update on the performance of the Platform, a statement of drawdowns and distributions, anticipated capital calls during the next quarter if any, details of related party transactions, valuation, etc.

ESG reporting

The Platform will circulate with all investors an annual report on the ESG performance of all the counterparties to whom it has provided the financing solution. The Platform team will take inputs from the counterparties and compile the annual report for applicable reporting period as required by Investors.

Impact reporting

An annual impact reporting will be performed. The mid-term and final year verification of climate impact will be undertaken by the independent third party provider.

Monitoring and evaluation arrangements

With regards to monitoring and evaluation of the mid-term FY27 and final impact numbers (subject to modification basis exit from the Platform), an independent evaluator will be appointed in FY27.

Midterm evaluations

At the time of midterm evaluation, significant developments may not have been achieved as the Platform will take time to reach maturity. Therefore, the focus of the evaluation will be on the implementation of the investment strategy. This will also cover aspects like the quantum of credit extended by the Platform, GHG avoidance, private sector capital catalysed, and estimated number of people served.

Final evaluation

For the final evaluation, it is expected that development effects will be fully realised (adjusted to the exit timelines). Therefore, an external agency will be hired to conduct evaluation of various aspects in line with evaluation best practice such as relevance, coherence, effectiveness, efficiency, impact, and sustainability (which align to the OECD DAC evaluation criteria).

- *Relevance*: the extent to which the investments are suited to the priorities and policies of the Platform.
- *Coherence*: the interaction of the Platform with other interventions in the the e-mobility sector in India.
- *Effectiveness*: the extent to which the objectives of the Platform were achieved.
- *Efficiency*: the extent to which activities were cost-efficient and timely.

Impact: To assess the positive and negative changes driven by the Platform, directly or indirectly, intended, or unintended because of the financing solutions:

- The focus will be on local social, economic, environmental, and other development indicators as mentioned in the previous section (at output, outcome, and impact level).
- Comparison of baseline and end line data.
- Assessment of other factors which would have been influenced the impact of the Platform.

Sustainability: GCF and commercial capital will build the Platform as a high quality, long lasting business which can continue to operate and move ahead towards its objective of decarbonising the transport sector in India, long after the Investors have exited the company.

Methodology for calculation of GHG emission reduction

List of Methodologies for the Platform's key focus sectors:

Sectors methodology link

Sector	Methodology	Link
Electric vehicles	CDM Methodology AMS-III.C., version 15	CDM: Emission reductions by electric and hybrid vehicles

EV Charging Systems, including their associated infrastructure	Verified Carbon Standard (Verra) - VM0038	VM0038 - Methodology for Electric Vehicle Charging Systems, version 1.0
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Implementation and governance

The Platform's impact strategy will be implemented by the management team with access to the broader MAM platform and in accordance with MAM's governance framework and policies.

MAM has a team of dedicated professionals with expertise in impact assessment and reporting, sustainability, environmental science, and law, who will assist in the ongoing management and measurement processes throughout the Investment Period to enable reporting of impact and ESG metrics.

G. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

The risks associated with the Programme include deployment, credit / counterparty, technology, country, and foreign exchange risks inherent to all financial institutions providing leases (operating and financial), vehicle financing and other similar solutions. Whilst it is difficult to provide an exhaustive list of these risks, a summary of key risks has been detailed out below, which relate to financing by the Platform in the e-mobility sector in India. Specific risk factors will be assessed throughout the life of GCF's investment in the Programme with strict monitoring and mitigation. However, no guarantee or assurance can be given that the investment objectives of the Programme will be achieved, or that other risks not outlined below do not apply to the program.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1 Deployment Risk

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>

Description

MAM is currently in the process of building a strong pipeline which is in various stages of discussion, however, given the Platform is not viable without GCF and / or commercial capital, this risk has not been mitigated as yet. Whilst there is optimism about a robust deal flow and growth of the sector over the next few years, deployment of capital may be affected by several factors, including investment priorities of counterparties, increased competition from other financial institutions, change in government policies and volatility in macroeconomic factors which may have adverse impact on the Platform's ability to deploy the funds.

Mitigation Measure(s)

MAM believes that the Platform's financing program, once established and backed by a highly experienced management team with support from MAM senior professionals', will moderate this risk. No guarantee or representation is provided that the Platform will be able to deploy capital and / or achieve the expected financial returns. However, given the program's wide mandate, the Platform will be able to provide financing solutions to a wide range of counterparties across e-mobility value chain thereby reducing the risk of the Platform to deploy the committed capital within the agreed timeframe. The focus of the Platform will also evolve as the e-mobility market matures in India and it will be guided by the Board of Directors of the Platform as well as MAM senior professionals directly involved with this programme.

Selected Risk Factor 2 Credit and Counterparty Risk

Category	Probability	Impact
<u>Credit</u>	<u>High</u>	<u>Medium</u>

Description

There is a risk that the counterparties of the financing instruments will not be able to maintain their financing payments which could result in the instrument defaulting and ultimately a potential loss for investors.

The Platform will be primarily providing financing instruments which will be generally illiquid in nature. These financing instruments may be difficult to sell or otherwise liquidate, particularly in stressed conditions, and as such the risk of loss may be greater than the risk of listed debt securities.

Mitigation Measure(s)

The Platform will conduct extensive due diligence on potential financing opportunities, including the creditworthiness of counterparties, the impact of the financing payments compared to expected revenue from use of the asset, and performance of the financing instrument under various stress scenarios. The Platform will seek to target contracts with credible counterparties with long term visibility to cashflows (preferably over life of the investment). A risk buffer to loss from anyone financing instrument or counterparty will be provided by diversification across different financing instruments, sub-sectors, and counterparties. The Platform will apply its robust monitoring process *described in section B.3 – Programme Description* to manage the risk of default and deliver the best outcome for the Platform and ultimately

the investors. It is anticipated that any re-possessed assets will be redeployed, although the terms of redeployment may vary on a case by case basis.

Selected Risk Factor 3 Technology Risk

Category	Probability	Impact
<u>Technical and operational</u>	<u>High</u>	<u>Medium</u>

Description

The e-mobility sector is still in its nascency in most markets, particularly in India wherein technology across the entire EV value chain is still evolving. Thus, there is a substantial risk that the Platform may provide financing for EVs or other sub-sectors based on currently available technology, which is then superseded, reducing any asset value held by the Platform or by counterparties. This could reduce financial returns for investors or the credit quality of counterparties.

Mitigation Measure(s)

The Platform will apply its technical expertise and leverage MAM's experience in conducting detailed due diligence on underlying technology, monitoring the latest technical advancements and trends in the sector, and analysing any potential impact on the financing instruments before taking any exposure. Further, whilst pricing the financial solutions, particularly leases, the Platform will try to account for any technology risk through a detailed estimation of expected residual value at the end of the lease term, which will provide the Platform the necessary cushion against any accelerated obsolescence of the EV technology.

Spearately, this development in technology would also mean that there has been positive advancements in EV technology which could support higher EV penetration and help meet the Programme's objectives faster.

Selected Risk Factor 4 Country and Regulatory Risk

Category	Probability	Impact
<u>Governance</u>	<u>Low</u>	<u>Medium</u>

Description

Any detrimental change in laws and regulations relating to the EV sector, methods and rules of taxation, compliance, foreign investments, restrictions on currency conversion, imports, etc. may impact the Platform's ability to provide financing solutions at attractive terms, impact its ability to deploy capital and / or restrict it to provide financing solutions to certain e-mobility sub-sectors.

Mitigation Measure(s)

The Platform aims to provide financing instruments that are aligned to government priorities in the e-mobility sector and development of local industry. The government has put in place long-term policies to support the ongoing growth of the sector which should continue to support deployment opportunities in the sector, viability of the business models and creditworthiness of the financing counterparties. The Platform's ability to deploy across the e-mobility value chain will provide diversification against the risk of regulation or policies changing in any particular sub-sector.

Selected Risk Factor 5 Exit Risk

Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>Medium</u>

Description

There is a risk that, at the point of GCF's expected exit from the program, the Platform cannot be sold down to another investor due to poor performance of underlying financing instruments, credit worthiness of the financing counterparties, high technology and regulatory risks, unfavourable market conditions and unattractive pricing.

Mitigation Measure(s)

The Platform intends to build a diversified portfolio in terms of exposure to multiple e-mobility sub-sectors, counterparties and financing solutions, which mitigates the concentration risk if any and acts as a hedge against any foreseen risk. Apart from this, the Platform will be led by a highly experienced management with strong background in screening and evaluating potential opportunities and performance monitoring. Further, the Platform will implement best in class governance practices and will have strong oversight and support from MAM, who have significant experience in developing and executing an exit strategy for the benefit of investors. The risk is also managed by the expected

growth in the sector and strong policy alignment, which provide strong fundamentals for market development and create a case for a robust exit for the investors.

Selected Risk Factor 6 Bribery and Corruption Risk

Category	Probability	Impact
<u>Reputational</u>	<u>Low</u>	<u>Low</u>

Description

Senior management of the financing counterparties may be involved in committing fraud or the adoption of other deceptive practices which may 1) have a negative impact on Platform's reputation, 2) stifle further debt and equity capital flow into the sector, and 3) undermine the Platform's due diligence efforts and its ability to generate desired returns from the asset.

Mitigation Measure(s)

Macquarie Group has a robust Bribery and Corruption framework which will be implemented by the Platform as part of upfront establishment. Upfront and ongoing monitoring of financing counterparties will be performed, including enhanced due diligence where required. Compliance with the Bribery and Corruption policies and procedures will be reported to the Platform Board and to the Accredited Entity on an ongoing basis.

The Platform will establish a project-level grievance mechanism which will be available for the reporting of complaints and allegations of impropriety, wrong-doing or other related issues in the project. This will be made available on the GCF and AE website once established.

Selected Risk Factor 7 Foreign Exchange Risk

Category	Probability	Impact
<u>Forex</u>	<u>Low</u>	<u>Low</u>

Description

The GCF will make commitments in USD, however the Platform will receive investments and provide financing in INR terms. There is a risk that the exchange rate volatility may result in a lower amount of investment capital being available in INR terms to the Platform, or returns to GCF and commercial investors are negatively impacted at the time of exit.

Mitigation Measure(s)

No hedging instruments will be put in place for equity investments made by the investors in the Program. MAM believes that the Programme will be able to generate robust returns for investors both in INR and USD terms, and will be able to easily absorb INR's depreciation vis-à-vis USD, which historically has been in the range of 2.5-3.5% over a 5 year period.

In case, SG Master Co., SG Co. and / or the Platform undertake USD denominated currency debt then the respective entities may have to undertake a hedging instrument as per the lenders' requirements. Any such requirements / conditions will be actively managed and monitored by the Platform's management team.

Selected Risk Factor 8 AML/CTF

Category	Probability	Impact
<u>ML/FT</u>	<u>Low</u>	<u>Medium</u>

Description

There are risks associated with financing counterparties which could have links to terrorism, money laundering or subject to sanctions. Instances of these practices may undermine the Programme in achieving its aims both from an operational and reputational perspective.

Mitigation Measure(s)

The Platform will conduct the day-to-day management of Anti Money Laundering (AML) / Counter Terrorism Financing (CTF) and sanctions in line with the requirements set out in subsidiary agreements, which will flow down GCF requirements under the AMA and FAA. The AE will monitor compliance with GCF's requirements under these agreements. As entities managed by MAM, the Executing Entities will be also subject to Macquarie policies which align to international best practices. These policies were reviewed by GCF as part of the accreditation process for MAAML.

In line with Macquarie policies and the existing AMA between MAAML and GCF:

- Programme activities will not be undertaken in any jurisdiction which is subject to, or affected by, United Nations Security Council resolutions.
- No individual or entity that is listed on any UN Security Council sanctions list (including the UN Consolidated Sanctions list) will be involved in any manner with the project or its activities either as a counterparty, implementer, or beneficiary.

Selected Risk Factor 9	Environment and Social Risk		
Category	Probability	Impact	
<u>Reputational</u>	<u>Medium</u>	<u>Medium</u>	

Description

The programme will establish the Platform, which will provide financing within the e-mobility sector. There may be environmental and social risks associated with sub-projects as outlined in *section G1* and the ESMS.

Mitigation Measure(s)

The Platform will implement robust governance, policies and procedures to identify, diligence and manage E&S risks in direct operations, in the investment process and monitoring of underlying sub-projects. The Platform will assess the potential risks and impacts for each sub-project based on the IFC Performance Standards. An Environmental and Social Management Plan (**ESMP**) will be developed for each sub-project to outline any required mitigating measures against identified risks and impacts.

Selected Risk Factor 10	Gender Risk		
Category	Probability	Impact	
<u>Reputational</u>	<u>Low</u>	<u>Medium</u>	

Description

The Platform will be established and invest in sub-projects in the e-mobility sector which may have different impacts on women and men. There are identified impacts and opportunities for gender mainstreaming at a national and sector level (transport). However, there is limited sex-disaggregated data available in the e-mobility sector or transport sector more broadly to determine some potential risks or impacts. There is a risk that the Programme has an unexpected, outsized negative impact on women or that gender mainstreaming is not promoted as effectively as possible.

Mitigation Measure(s)

There are identified areas in the Gender Action Plan where gender mainstreaming activities can be implemented, including the focus areas of improving knowledge within e-mobility sector on gender sensitivity, safety to prevent gender-based violence, and opportunities to provide employment opportunities. Other actions will be identified over time through stakeholder engagement with women's groups, financing counterparties and the result of sex-disaggregated data.

H. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Governance of E&S

E&S considerations will form an intrinsic part of the Platform's operations and will be considered at all stages of the investment process. This will include consideration of E&S risks and impacts that are more prevalent in the e-mobility sector (such as materials or supply chain factors) or specific transaction types (such as heightened counterparty and procurement risks for leasing structures).

As a financing intervention, the Platform will not be able to directly implement activities regarding the operation of the EVs. However, this is mitigated through upfront diligence and structuring by the Platform as part of a transaction and by activities flowed through to the financing counterparty.

Financing counterparties will be involved in the implementation and management of E&S in sub-projects, including complying with any requirements included in the financing documentation set by the Platform or ESMP developed as part of the sub-project development. The Platform will be responsible for considering the counterparty's capacity to implement E&S management, what requirements are integrated into financing documentation to meet GCF and international standards, and overseeing that counterparties appropriately implement these requirements.

A dedicated E&S specialist will be hired by the Platform to lead the implementation of the Environmental and Social Management System (**ESMS**) at the Platform level. They will also provide or arrange training for Platform staff and, where applicable, other stakeholders such as financing counterparties.

The Platform will have oversight from MAM, which has dedicated risk and sustainability teams that are responsible for setting and implementing MAM's ESG strategy and framework, including the management and operations of the Platform.

1. MAM's risk team is responsible for ensuring all operational risks, including ESG risks, are identified and managed. Reporting to the MAM Chief Risk Officer, MAM's risk team is independent of MAM's portfolio management functions and is supported by specialist expertise within MAM, including the MAM sustainability team and Macquarie Group risk functions.
2. MAM's sustainability team is responsible for setting MAM's overall sustainability strategy and ESG framework, providing specialist expertise on environmental and certain social issues, and supporting asset management teams in harnessing ESG opportunities across the portfolio.

Supported by centralised policies and processes, MAM's risk and sustainability teams support MAM's transaction and asset management teams to effectively address ESG risks and opportunities throughout each phase of the lifecycle.

E&S risk assessment and classification

The focus of the Programme is establishing a Platform to provide financing to predominately corporate counterparties (in the initial years) for electric vehicles, building charging infrastructure and / or other investments along the e-mobility value chain. The Programme has been categorised as **Intermediation-2** to reflect that the Platform will act as an intermediary to the operation of the e-mobility investments, which will have maximum categorisation of **Category B**, based on an assessment using the Macquarie's ESR Toolkit which is based on IFC Performance Standards.

The key E&S risks for each sub-project that have contributed to this classification are outlined below, along with why their impacts are limited or mitigated. For the avoidance of doubt, this is not an exhaustive list of considerations. Each sub-project will be reviewed on a case-by-case basis to ensure E&S risks are considered holistically.

Capacity of the financing counterparties (PS1: Risk Management)	The counterparty will be involved in direct implementation of E&S management for sub-projects. There is a risk that the counterparty does not have the capacity or the governance in place to manage potential E&S risks. The Platform will review the policies, procedures and capacity of the counterparty as part of its upfront due diligence, and will monitor the implementation of E&S management as part of the ongoing monitoring of the sub-projects.
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Air quality and indirect emissions (PS3: Resource efficiency)	EVs do not have any emission from its operations as only electricity is consumed. This is a significant advantage over the petrol / diesel driven vehicles as vehicular emission contributes to a significant proportion of pollution in all major cities, including India. Indirect emissions will arise in the generation of power, which is used for charging of batteries in the EVs, given India's grid still has reliance on conventional energy sources. However, this will be mitigated through India's ambitious program to decarbonise its grid over time through the integration more renewables, which will support the decarbonisation of the transport sector alongside electrification. In this way, the impacts will decrease over time. Opportunities to encourage green-only charging points will also be explored. The Platform will use best efforts to investigate and pursue commercially feasible opportunities to reduce the indirect emissions associated with its operations. This may include installation of onsite renewables (e.g., rooftop solar), dispatchable storage, procurement of renewable electricity. The Platform will incorporate into its policies and stakeholder management program the support of and participation in government programs that support grid decarbonisation.
Hazardous waste (including e-waste) (PS3: Resource efficiency)	EVs do not require handling of hazardous materials because they do not contain engine oil and similar lubricants, which are associated with conventional vehicles. However, lithium-ion batteries that include other chemical moieties like cobalt, graphite and nickel are an important component of an EV. At the end of the battery lifespan, there is a battery waste, comprising significant amounts of chemicals such as cobalt, electrolytes, lithium, manganese oxide and nickel. These substances, if not recycled or treated adequately, can contaminate soil and groundwater. Further, lithium itself spontaneously reacts with moisture and can lead to major landfill explosions. As a nascent sector, there are limited regulatory frameworks for the end-of-life treatment or recycling of e-waste related to EVs, however these are rapidly evolving and options for re-use, recycling and safe disposal are being developed globally. Although current policies related to e-waste in India do not currently include guidance on the safe disposal of lithium-ion batteries, the Platform will align to international best practice in the disposal of the batteries for each sub-project. Regulations and best practices in the nascent e-waste management sector are rapidly evolving. It is recognised that the e-waste management approach adopted by the Platform and sub-projects will be subject to these evolving regulations and best practices. Generally, the Platform will seek that commercially feasible best practices and e-waste management principles are employed to minimise adverse environmental and social impacts associated with used batteries. An e-waste management strategy relevant for each sub-project will be developed. The Platform will develop a vehicle scrappage policy aligned with government guidelines and will require counterparties to comply with the same. Government approved scrappage facilities with focus on recycling and reusing materials recovered from the scrappage.
Supply chain and labour issues (PS2: Labour, PS4: Community)	As with any manufacturing sector, the supply chain can be prone to E&S issues if working with non-reputable counterparties, such as child labour, SEAH, inadequate working conditions, health and safety lapses and corruption. Required standards and compliance can vary from country to country and across different counterparties. There are known supply chain issues related to mining of lithium-iron and other materials for batteries, which involves extraction of metals from jurisdictions where E&S regulations may not be as strict. The Platform will conduct due diligence on counterparties, including their labour practices and supply chain, as part of the upfront assessment of a transaction which would impact on the ultimate financing decision to proceed. This will mitigate against any significant adverse social impacts that would otherwise require a Category A classification such that the Platform only works with credible counterparties with adequate policies and measure in place to manage these risks. The Platform intends to mitigate this through the following outcomes: • A Board-approved human rights policy that outlines the Platform's commitment to avoiding causing or contributing to adverse human rights impacts (including modern

	slavery) and the Platform's commitment to seek to prevent or mitigate adverse human rights impacts directly linked to the Platform's operations, products or services. The human rights policy will also support the UN Guiding Principles on Business and Human Rights, the OECD Guidelines for Multinational Enterprises and the ILO Fundamental Conventions and Recommendations. • Undertaking supply chain due diligence for select suppliers of the proposed entity. • Excluding suppliers with negative reports. • Each sub-project will have in place, commensurate with its human rights risk profile, a process to screen and undertake due diligence on suppliers to a human rights risk ascertain track record. • Actively engage in stakeholder consultations regarding regulatory developments in this sector to support greater transparency and controls. • Participating in workshops and trainings with financing counterparties regarding supplier governance and best practices in relation to this issue.
Social impacts (PS4: Community)	The addition of EVs to public transport will see new buses with better features and quality of transportation. Advanced safety features, on-board cameras, universal access, etc. are expected to benefit society in general. However, as the transition to EVs continues, there may be unintended societal impacts, such as requirements for reskilling of jobs in the conventional sector. These are not expected to be direct impacts from sub-projects as a result of the financing provided. As such, the impacts are expected to be minimal. However, the approach of the financing counterparty to managing the transition of their staff (where applicable) will be considered as part of the upfront assessment and diligence. The Platform intends to mitigate this through the following outcomes: • Consider, as part of the planning phase of each sub-project, the social impacts, including job creation for new / reskilled workers, improved amenity and aesthetics due to reduced noise and pollution, enhanced low-cost mobility and connectivity for communities and improved quality of life and security for customers. • Review any likelihood of significant retrenchment as a result of the transaction (considered unlikely). • Support or be involved in workshops and training for the e-mobility sector organised by financing counterparties.
Gender-based violence (including SEAH) (PS4: Community)	Gender-based violence and the related safety considerations in public transport are known issues in India. There is potential for gender-based violence to occur on the buses or other vehicles financed by the Platform. This is not within the control of the Platform, however there are a number of safety regulations, such as CCTV cameras, panic buttons, etc., that are required. The Platform will seek to monitor the compliance of vehicles in meeting these regulations to mitigate against significant impacts from this risk. The regulations, best practice, obligations and other mitigating features will be reviewed as part of each sub-project to limit the impacts of this risk. Mitigants for these risks are outlined in more detail in the Gender Action Plan. Some examples from the Gender Action Plan include: • the Platform will review the GBVH policies of counterparties as part of the upfront due diligence; • the Platform will monitor compliance with requirements in e-buses regarding safety features such as emergency call button, CCTV cameras, easy access for disabled; and • the Platform will support initiatives such as increasing number of woman conductors or drivers on buses or women only buses.
Land procurement	It is envisaged that, in rare cases, sub-projects that will be considered for financing may involve land procurement, such as the setting up of charging infrastructure. The footprint is likely to be small. Resettlement, significant biodiversity impacts or impacts on indigenous peoples or cultural heritage are not expected due to the nature of the size and location of charging points,

(PS5: Land resettlement, PS6: Biodiversity, PS7: Indigenous peoples, PS8: Cultural heritage)	targeting predominately urban environments where there is likely existing commercial use where charging is convenient for users. Where resettlement is required, a Resettlement Action Plan will be developed using the principles outlined in the ESMS for this Programme. Any limited biodiversity, indigenous peoples or cultural heritage impacts identified in E&S diligence will need have mitigating actions in the ESMP for that sub-project. Any major resettlement, biodiversity, indigenous peoples or cultural heritage would be classified as a Category A project or not able to be mitigated and the Platform will not invest in these activities.
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Integration into investment processing and monitoring of underlying sub-projects

E&S risks will be assessed during the due diligence process and feed through into the analysis for all transactions.

A robust E&S due diligence process will ensure that these risks are identified and appropriately incorporated into the financing decision. Integrating E&S into the process will ensure that the fundamental and credit analysis of the underlying financing captures all risks and potential impacts, to ensure the borrower or lessee's ability to repay its obligations is effectively assessed. A summary of E&S risks, mitigants, classification and actions will be prepared for each sub-project as part of the financing decision and will be approved as part of each project.

The Platform will screen opportunities using Macquarie Group's Environmental and Social Risk Assessment Tool (**ESR Tool**). The environmental and social criteria and categorisation are based on International Finance Corporation (IFC) performance standards. The system is kept up to date in line with industry best practice. The results of the ESR Tool will be used to shape the scope of due diligence (generally tailored to the location, sector, and risk profile of the opportunity) and identify material ESG risks and information gaps relevant to the financing opportunity. The ESR Tool provides a risk categorisation based on IFC performance standards as Category A, B or C. This categorisation will also be approved as part of the financing decision.

The Platform will monitor sub-projects against risks identified during upfront due diligence, in addition to against broader sectoral changes and themes, to identify new or emerging risks over the lifetime of the financing. Where (if any) a material E&S risk is identified, issues will be escalated to the Platform's Board immediately.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

MAM has completed a gender assessment study and developed a Platform level Gender Action Plan for the India e-mobility Financing Program. *Refer to Annex 8 – Gender assessment and programme level action plan for more details.*

It is understood that gender related issues in the e-mobility sector are common to those in the broader urban mobility sector in India. Issues are specific to gender-based violence and harassment, economic disparity, and limited public transport options. However, recent flow of capital in safe and well-designed transport infrastructure can increase economic empowerment of women by improving mobility and access to better avenues of employment, education, and healthcare. Introduction of new electric vehicles fleets replacing older public transportation systems will enhance the safety features due to the required design modifications under tender and concession contracts. Therefore, the growth of the e-mobility sector and sub-sectors has the potential to lead to changes in gender dimensions by supporting safer public transportation options. Moreover, as a new industry, there is potential to promote the implementation of gender inclusive design and practice, as well as applying broader workforce policies, such as bridging the pay gap, involvement of more technically qualified women and opening more avenues of employment for women.

The key issues identified in the gender assessment, and the actions the Platform will implement to address these issues, are outlined in the table below.

Key issue identified in gender assessment	Description of issue or risk	Actions to address issues identified
Employment opportunities	Limited opportunities for women to participate in the growth of the e-mobility sector through increased access to employment opportunities within the sector	Provide employment opportunities at the Platform level to grow skills within the e-mobility financing industry that is set to grow and apply fair hiring practices.
		Start an internship programme providing young women opportunities to gain experience in e-mobility financing space.
		Review relevant policies of counterparties, including statutory (e.g. anti-discrimination), company (e.g. D&I or relevant recruiting policies), and other D&I and gender related company practices.
	Limited sex-disaggregated data	Track relevant candidate gender data through application and hiring process and identify / mitigate any bias if identified.
		Track skills gaps of applicants which can feed into the internship programme design.
Gender-blindness, adequate transportation infrastructure to reach services (education, training)	Limited number of services designed for women incorporating women specific needs	Identify opportunities to work with private bus companies or corporates which support women-focused services (such as women-only buses, school services, etc), or working with counterparties to develop initiatives.
	Women's specific requirements not adequately incorporated into project-design	Upfront gender assessment to identify whether there are opportunities to optimise the placement of charging points for increased use of EVs by women wherever applicable (e.g. considering well-lit areas, public areas).
		Periodic consultation with women's groups (e.g. women drivers associations, self-help groups for women from low income backgrounds, NGOs working on women employment) to identify potential risks and opportunities.
	Limited sex-disaggregated data to improve gender sensitivity in the sector or sub-sectors	Design a robust plan for collection of relevant sex-disaggregated data for each sub-project on transport usage, and collate the data to understand areas of improvement. <i>(Note that sex-disaggregated data will be collected for relevant sub-projects relating to use of vehicles. However, as the sub-projects have not yet been identified, the actual data to be collected will be finalised as part of the investment approval of each sub-project. Any collection of sex-disaggregated data privacy regulations including collection and storage of sensitive data, as relevant and required)</i>
Gender Based Violence and Harassment (GBVH) (including SEAH)	Facilities around transport hubs and vehicles (e.g. poor lighting, unsupervised spaces, absence of emergency buttons, etc.)	Conduct due diligence of counterparties to review their existing policies and procedures for management of GBVH issues, including existing education programs, safety measures, trainings, and grievance mechanisms.
		Require projects to include adequate design and maintenance of lighting and CCTV cameras (or other relevant safety features at charging infrastructure points).
		Require reporting / representation from counterparties that the vehicles are aligned with safety requirements as mandated by authorities / relevant concession agreements.
		Reporting of GBVH incidents for vehicles financed by the Platform.

Key issue identified in gender assessment	Description of issue or risk	Actions to address issues identified
		Be involved in stakeholder consultations by public and private entities around the design of concessions or vehicles to provide views and data on relevant and improved safety features. Targets should be set on the number of consultations conducted per year on gender aspects.
	Behavioural aspects (e.g. physical violence and sexual harassment, lack of clear codes of conduct for employees, etc.)	Require counterparties to install surveillance cameras and adequate security measures.
		Review training programs periodically delivered by counterparties to drivers and transport workers on gender sensitivity and how to report and refer incidents of GBVH. Where counterparties do not have existing training programs, offer relevant training for drivers and other employees on gender sensitivity and how to manage incidents of GBVH.
		Use transport infrastructure, such as signage and tickets, to disseminate messages and create awareness around GBVH more broadly in communities and public spaces. This will be undertaken through counterparties.
Women's involvement in decision making	Due to societal and cultural practices, women are less likely to be involved in decision making, which may exacerbate gender blindness in project design	Promote women's involvement at a corporate level through: - Platform Board composition - Monitor Board composition of counterparties
		Ensure women and vulnerable groups have an active role in project and sector level stakeholder consultations.
Need for capacity building and knowledge dissemination for gender considerations in the e-mobility sector	Awareness of gender mainstreaming may be limited in the transportation, e-mobility and financing sectors	Opportunities for capacity building for both the Platform and counterparties through trainings and workshops to increase awareness of gender sensitivity. Promote multi-stakeholder discussions to integrate gender considerations across the sector.
	Limited availability of sex-disaggregated data in the e-mobility space	Ensure each sub-project collects sex-disaggregated data which can be used to identify risks, mitigants and opportunities.
	Knowledge sharing of lessons learned is limited	Promote examples of lessons learned through data collection or sub-project implementation through thought leadership or otherwise to contribute to developing best practice.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

MAM will be responsible for the monitoring and review of Platform performance. It will work closely with the Platform's management team to define their operating and financial policies and processes, and ensure compliance through industry leading **MAM asset management standards**. MAM, through Board representation, will participate in key financial, commercial and governance matters of the Platform. Further, it will seek monthly reviews with the Platform's management team to discuss business updates / progress, review financial performance, ESG updates and review of governance, commercial and other operational matters. On a quarterly basis, the investment team will also conduct a detailed formal analysis of the performance of the Platform for the purpose of reporting to the GCF and commercial investors.

MAM will seek following reporting from the Platform's management team:

- monthly reporting on financial and operational metrics;
- reporting on E&S matters;

- quarterly Board meetings;
- regular site visits;
- compliance with shareholders agreement; and
- annual audited accounts.

MAM will submit following reports to the LP Investors:

- quarterly investor report (including Platform performance and financial information at consolidated LP level);
- annual audited accounts and NAV statements;
- annual ESS report including annual climate and sustainability impact assessment; and
- GCF specific mid-term and final assessment reports.

Note that ESG will be carried out as per GCF ESS Policy and IFC Performance Standards.

Procurement

Procurement of goods and services at LP, SG Master Co., SG Co. and Platform levels will be carried out in accordance with the AE's policies and regulations as agreed under the AMA.

One time setup costs relating to LP, SG Master Co. and SG Co., and highly commoditised services requiring minimal professional expertise, will be procured on Quality Cost Based Selection basis. Various professional / consultation services required on an annual basis will be procured on Quality Based Selection basis.

G.4. Disclosure of funding proposal

The Funding Proposal (**FP**) is prepared and submitted by Macquarie Alternative Assets Management Limited (**MAAML**) to Green Climate Fund (**GCF** or **Recipient**).

Macquarie and **Macquarie Group** refer to Macquarie Group Limited (**MGL**) and its worldwide subsidiaries and affiliates including MAM and Macquarie-managed investment vehicles. Members of the Macquarie Group are the owners of the copyright material in this document unless otherwise specified. No member of the Macquarie Group accepts any liability whatsoever for a direct, indirect, consequential or other loss arising from any use of this document and / or communication in relation to this document (including its contents).

Any forward-looking statements included in this document represent our opinions, expectations, beliefs, intentions, estimates or strategies regarding the future, which may not be realised. These statements may be identified using words like “anticipate”, “believe”, “estimate”, “expect”, “intend”, “may”, “plan”, “will”, “should”, “seek” and similar expressions. The forward-looking statements reflect our views and assumptions with respect to future events as of the date of this document and are subject to risks and uncertainties. Actual and future results and trends could differ materially from those described by such statements due to various factors, including those beyond our ability to control or predict. No forward-looking statements contained in this document constitute a guarantee, promise, projection, forecast or prediction of, or representation as to, the future and actual events may differ materially. Any target returns or other financial projections (including any data related thereto) in this document are based upon subjective estimates and assumptions about circumstances and events that have not occurred yet and may never occur. If any of the assumptions used do not prove to be accurate, results may vary substantially from any target returns or financial projections contained in this document. Any target returns or other financial projections are being shown for illustrative purposes only.

This document has been prepared based by MAAML which is an entity within the Macquarie Asset Management (MAM) group of Macquarie Group. It is not a product of the Macquarie Research Department which operates outside the MAM information barrier. It is based on publicly available information and information made available to us by third parties or obtained from sources that we consider to be reliable, and is based on present circumstances, market conditions and beliefs. Where information in this document includes reference to a publicly available source or sources that can be subscribed to, the recipient should refer to and interpret the information in the context of that source. The document contains selected information and does not purport to be all-inclusive or to contain all the information that may be relevant or latest.

MAAML is not an authorised deposit-taking institution for the purposes of the Banking Act 1959 (Commonwealth of Australia) and MAAML's obligations do not represent deposits or other liabilities of Macquarie Bank Limited ABN 46 008 583 542 ("Macquarie Bank"). Macquarie Bank does not guarantee or otherwise provide assurance in respect of the obligations of MAAML. In addition, if this document relates to an investment, (a) the investor is subject to investment risk including possible delays in repayment and loss of income and principal invested and (b) none of Macquarie Bank or any other Macquarie Group company guarantees any particular rate of return on or the performance of the investment, nor do they guarantee repayment of capital in respect of the investment.

Nothing in this "Disclaimer" is intended to modify MAAML's obligations to the Green Climate Fund as set forth in the Accreditation Master Agreement between MAAML and the GCF.

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

Glossary

2W	Two wheeler
3W	Three wheeler
4W	Four wheeler
ADB	Asian Development Bank
AE	Accredited Entity
AFSTP	Alternate Fuels for Surface Transportation Program
AIF	Alternate Investment Fund
AMA	Accreditation Master Agreement
ASX	Australian Stock Exchange
AUD	Australian Dollar
AUM	Asset Under Management
B2B	Business to Business
B2C	Business to Consumer
BEV	Battery Electric Vehicle
bn	Billion
BSVI	Bharat Stage VI
CDM	Clean Development Mechanism
CGM	Commodities and Global Market
CMIE	Centre for Monitoring Indian Economy
CO	Carbon Monoxide
CO₂	Carbon dioxide
CPI	Climate Policy Initiative
CSG	Client Solutions Group
D&I	Diversity and Inclusion
DD	Due Diligence
DISCOMs	Electricity distribution companies
Ebus	Electric Bus
EE	Executing Entity
EMEA	Europe, Middle East and Africa
E&S	Environment and Social
ESG	Environment, Social and Governance
ESIA	Environmental and Social Impact Assessment
ESMP	Environmental and Social Management Plan

ESR	Environmental and Social Risk Assessment Tool
ESS	Environmental and Social Safeguards
EUM	Equity Under Management
EV	Electric Vehicle
FAA	Funded Activity Agreement
FAME	Faster Adoption and Manufacturing of Hybrid and Electric Vehicles
FCEV	Fuel Cell Electric Vehicle
First Close	The FAA effective date
FP	Funding Proposal
FRBM	Fiscal Responsibility and Budget Management Act 2003
GBVH	Gender Based Violence and Harassment
GCI	Global Competitiveness Index
GDP	Gross Domestic Product
GHG	Greenhouse Gases
GIF	Global Infrastructure Fund
GIG	Green Investment Group
GP	General Partner
GtCO₂	Giga tonnes of carbon dioxide
GRESB	Global Real Estate Sustainability Benchmark
GST	Goods and Service Tax
HR	Human Resources
IAC	Macquarie's India Advisory Company
ICE	Internal Combustion Engine
IEA	International Energy Agency
IFC	International Finance Corporation
IndAS	Indian Accounting Standards
INR	Indian Rupee
IPCC	Intergovernmental Panel on Climate Change
Investment Ratio	The ratio of commercial investors to GCF's equity capital commitment
IPO	Initial Public Offering
IREDA	Indian Renewable Energy Development Agency
IRR	Internal Rate of Return
IT	Information Technology
LCV	Light Commercial Vehicle

LP	Limited Partner
LPA	Limited Partnership Agreement
LTV	Loan to Value
LULUCF	Land Use Change and Forestry
MAAML	Macquarie Alternative Asset Management Limited
MAM	Macquarie Asset Management
MCV	Medium Commercial Vehicle
MIRA	Macquarie Infrastructure and Real Assets
MNRE	Ministry of New and Renewable Energy, Government of India
MoEFCC	Ministry of Environment, Forest, and Climate Change, Government of India
MoRTH	Ministry of Road Transport and Highway, Government of India
MGL	Macquarie Group Limited
MtCO₂e	Million tonnes of carbon dioxide equivalent
MW	Mega Watt
NAV	Net Asset Value
NBFC	Non-Banking Financial Company
NCAP	National Climate Action Plan
NCD	Non-Convertible Debentures
NDA	National Designated Authority
NDC	Nationally Determined Contribution
NEMMP	National Electric Mobility Mission Plan
NHAI	National Highway Authority of India
NOx	Nitrogen Oxides
OECD	Organisation for Economic Co-operation and Development
OEM	Original Equipment Manufacturers
PAX	Passengers
pkm	Passenger kilometers
Platform	The proposed India focused E-mobility Financing Platform
PM2.5	Particulate matter
Preferred Hold Period	The first 5 years of the Platform's operation
PRI	Principles for Responsible Investment
Programme	The proposed India focused E-mobility Financing Platform
RBI	Reserve Bank of India

RCP	Representative Concentration Pathway
R&D	Research and Development
SBI	State Bank of India
SDGs	United Nation's Sustainable Development Goals
SEAH	Sexual Exploitation, Abuse and Harassment
SEBI	Securities and Exchange Board of India
Second Close	24 months from the FAA effective date
SEP	Supplemental Environmental Projects
SGX	Singapore Stock Exchange
SRTUs	State Road Transport Undertaking
TCO	Total Cost of Ownership
TOT	Toll Operate Transfer
UK	United Kingdom
UNFCCC	United Nations Framework Convention on Climate Change
USA	United States of America
USD	United States Dollar
WRI	World Resource Institute
WWF	World Wildlife Fund

I. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) **(template provided)**
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan **(template provided)**
- ☒ Annex 5 Implementation timetable including key project/programme milestones **(template provided)**
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
(ESS disclosure form provided)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☒ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan **(template provided)**
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan **(template provided)**
- ☒ Annex 11 Monitoring and evaluation plan **(template provided)**
- ☒ Annex 12 AE fee request **(template provided)**
- ☐ Annex 13 Co-financing commitment letter, if applicable **(template provided)**
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval **(template provided)**
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information **(template provided)**
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)
- ☒ Annex 23 GHG calculations
- ☒ Annex 24 Illustrative Term Sheet for an E-bus lease

☒ Annex 25 Macquarie Group Overview

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*



No.CC-13008/57/2020-CC
15th September, 2021.

To:

The Green Climate Fund ("GCF")
Songdo Business District
175 Art Center-daero
Yeonsu-gu, Incheon 22004 (Republic of Korea)

Re: Funding proposal for the GCF by Macquarie Alternative Assets Management Limited (MAAML) regarding "India E-Mobility Financing Program".

Dear Madam/Sir,

We refer to the programme, "India E-Mobility Financing Program" in India as included in the funding proposal submitted by Macquarie Alternative Assets Management Limited (MAAML) to us on 14th July 2021.

2. The undersigned is the duly authorized representative of the Ministry of Environment, Forest and Climate Change, the National Designated Authority/Focal Point of India.

3. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

4. By communicating our no-objection, it is implied that:

- The Government of India has no-objection to the programme as included in the funding proposal;
- The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of India;
- In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

5. We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

6. We also confirm that our no-objection applies to all activities/projects to be implemented within the scope of the programme.

7. We acknowledge that this letter will be made publicly available on the GCF website.



Yours faithfully

Richa
(RICHA SHARMA)

Focal Point
National Designated Authority, India



Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	India E-Mobility Financing Program
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Private
Accredited entity	Macquarie Alternative Assets Management Limited (MAAML)
Environmental and social safeguards (ESS) category	Category I-2
Location – specific location(s) of project or target country or location(s) of programme	India
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	Friday, April 15, 2022
Language(s) of disclosure	English and Hindi
Explanation on language	Hindi is the most spoken language in India and is understood by the stakeholders concerned.
Link to disclosure	English: https://www.mirafunds.com/assets/mira/esms/ESMS-india-e-mobility-financing-program-MAAML-eng.pdf Hindi: https://www.mirafunds.com/assets/mira/esms/ESMS-india-e-mobility-financing-program-MAAML-hin.pdf
Other link(s)	N/A
Remarks	An ESMS consistent with the requirements of a Category I-2 programme is contained in the "Environment & Social Management System".

Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	Friday, April 15, 2022
Language(s) of disclosure	English and Hindi
Explanation on language	Hindi is the most spoken language in India, and is understood by the stakeholders concerned.
Link to disclosure	English: https://www.mirafunds.com/assets/mira/esms/ESMS-india-e-mobility-financing-program-MAAML-eng.pdf Hindi: https://www.mirafunds.com/assets/mira/esms/ESMS-india-e-mobility-financing-program-MAAML-hin.pdf
Other link(s)	N/A
Remarks	The IPP and RPF are annexed to the ESMS.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, April 15, 2022
Place	Hard copies of the disclosure made available at the Macquarie Group Office in Mumbai located at: 92, Level 9, 2 North Avenue, Maker Maxity, Bandra Kurla Complex, Bandra East, Mumbai 400051, India
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	TBD
Date of GCF's Board meeting	Monday, May 16, 2022

Note: This form was prepared by the accredited entity stated above.

Independent Technical Advisory Panel's assessment of FP186

Proposal name:	India E-Mobility Financing Program
Accredited entity:	Macquarie Alternative Assets Management Limited (MAAML)
Project/programme size:	Large

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential

Scale: Medium-High

- Overview** - The E-Mobility Financing Platform funding proposal (Platform) assessed here is proposed by Macquarie Alternative Assets Management Limited (MAAML) as an accredited entity (AE). This is a mitigation programme that focuses on promoting sustainable and electrified road transport in India. GCF is requested to provide USD 200 million in equity, as first loss capital to de-risk commercial investors, leading to a total commitment of USD 405 million towards an E-mobility Financing Platform. The programme aims to leverage the GCF and commercial capital to accelerate decarbonisation of the Indian economy by focusing on the electrification of India's transport sector. It will do this by providing financing solutions intended to bridge the overall cost of ownership between conventional vehicles and electric vehicles (EVs).
- Climate change is one of the biggest challenges faced by India. Between 1901–2018, India's average temperature rose by 0.7°C due to increased atmospheric concentrations of greenhouse gases (GHG). India has also seen a rapid growth in its own GHG emissions resulting from multiple factors such as high population growth and increasing levels of industrialization. It is projected that temperatures will continue to rise in India resulting in greater frequency of extreme climatic events, vector-borne diseases, food shortage, migration, and loss of livelihoods, all of which will have adverse impacts on India's social and economic development.
- In response, the Government of India has been pursuing various mitigation and adaptation activities including policy support for renewable energy and electric mobility sectors, to reduce GHG emissions. However, the scale of investment needed to help India achieve its climate goals as reflected in its nationally determined contribution (NDC) under the Paris Agreement will require the mobilization of funds beyond the public treasury. Sectors that can produce mitigation impacts such as the electric mobility sector will need to attract private capital at scale and in an accelerated manner. According to some estimates, India will need investment of approximately USD 2.5 trillion between 2015 and 2030 to meet its NDC by 2030.
- In the above context, the aim of the Platform is to facilitate the reduction of GHGs and other emissions from the road transport sector in India through accelerated adoption of electric vehicles. The funding proposal presents the creation of a Platform that would deploy targeted investments (in the form of operating and finance leases and other structured credit solutions) across the e-mobility value chain to overcome barriers that have historically limited the sector. India has the potential to be one of the largest electric vehicle markets in the world, but market growth has been hampered by high upfront capital expenditure associated with EVs, impediments around EV charging infrastructure and uncertainty around commercial EV performance. The funding proposal is intended to be complementary to Indian government efforts to accelerate its nascent e-mobility market by attracting new funding sources into the space, addressing market barriers, and financing the underlying infrastructure needed to scale up the EV sector.

5. The funding proposal seeks USD 200 million from GCF and an additional USD 205 million from commercial investors to capitalize the E-mobility Financing Platform. It will be managed by MAAML which is part of Macquarie Asset Management (MAM), one of the world's largest infrastructure asset managers, with an established presence in India. The proposal further envisages accessing local and international debt markets for approximately an additional USD 1.1 billion in debt leverage.
6. Initially the Platform will focus on select segments of the EV ecosystem such as e-buses, shared fleet and electric charging infrastructure and will over time seek to diversify its portfolio by expanding into the other segments (including retail) as more subsectors within the EV value chain gain scale and prominence. Over the implementation period, MAAML expects the Platform to mobilize up to USD 2.1 billion of capital into the sector and reduce ~9.5 MtCO₂eq over the project lifetime.
7. Impacts - The FP and the accredited entity have been able to credibly demonstrate that decarbonising the transport sector in India can help to reduce the impacts related to GHG emissions and other air pollutants. Additionally, the Platform can generate several additional potential co-benefits, such as major improvements in public health, particularly in urban areas, improved fuel savings and reduced noise pollution.
8. Electrification of vehicles plays an essential role in decarbonising the transport sector as it offers the most promising GHG emission abatement potential relative to other alternatives. In 2021, India became the world's third largest passenger vehicle market with the sector representing an increasing source of GHG emissions and other pollutants. Transportation activity reached over 6 trillion passenger kilometres, which represents three times increase from 2000 to 2020. This trend is expected to continue unabated as approximately 60 million more households are expected to enter the consuming class by 2025. At the same time, the transport sector in India is reliant on fossil fuels and contributes significantly to oil import demand. Transportation accounts for 20 per cent of the final energy use in India, and has grown by 50 per cent since 2010, comprising primarily oil products (95 per cent). India is presently the third-largest consumer of oil and fourth-largest oil refiner with well understood implications for India's energy security.
9. Transportation is the third most GHG emitting sector, contributing about 13 per cent of the total carbon dioxide emissions in India. Due to the dimension and scale of the Indian economy, its emissions from transport alone are equivalent to the total emissions of a mid-sized European country. Therefore, accelerating decarbonization of the Indian transportation sector (moving to EVs and away from fossil fuels and internal combustion engines) could significantly reduce emissions and generate co-benefits with substantial impact. However, a mix of financial, behavioural and infrastructure incentives is needed to develop the e-mobility sector and associated infrastructure. The funding proposal credibly demonstrates that the Platform could help achieve this target.
10. Currently both EV market acceptance and EV industry dynamics are at an early stage. The EV adoption rate is less than 1 per cent and the domestic Original Equipment Manufacturer (OEM) are just starting to launch EV models. For instance, although India is the world's second biggest bus market with approximately 100,000 units sold annually, over 95 per cent of them are fossil fuel reliant. At the same time, this represents a large market sector that could be converted to EVs if market financing barriers are addressed, and it is not surprising that the Platform proposes an initial focus on the passenger bus subsector of transportation.
11. The independent Technical Advisory Panel (TAP) notes that from a GHG emissions standpoint, a successful shift away from fossil fuel vehicles to EVs is dependent on the GHG intensity of the relevant electric grid. In the case of India, coal still accounts for approximately 57 per cent of the electricity mix. EV transportation will require a source of electricity for charging. The underlying source of electricity generation will affect the actual emissions avoided by the programme.

12. However, we also recognize that India has committed to an ambitious deployment programme of renewable energy resources and has already achieved its 2030 target of deriving 40 per cent of the installed capacity from non-fossil fuels. The total non-fossil based installed electricity capacity stands at 156.83 GW, which is 40.1 per cent of the total installed electricity capacity. At the UN Climate Change Conference in 2021 (COP26), India announced a goal of reaching 500 GW of non-fossil fuel capacity and meeting 50 per cent of its energy requirements from renewable sources. All indications are that India is committed to a decarbonization path for its grid, but the scale of potential impact of this funding proposal will be heavily influenced by how rapidly and how deeply India's energy generation is decarbonized. The underlying assumption in the funding proposal is that there will be increased penetration of renewables in the electricity grid and/or availability of charging sites using only renewables. The proposal intends to contribute to this decarbonization by encouraging its counterparties to use renewable energy for charging stations, by providing appropriate financial incentives linked to the carbon intensity of the energy utilized.

13. The Platform expects to finance an estimated 18,000 e-buses (72.8 per cent of the total capital deployment), 22,000 four-wheel vehicles (7.5 per cent of the total capital deployment), 65,000 three and two-wheel vehicles (3.5 per cent of the total capital deployment) and plans to invest and establish 10,500 charging stations over the life of the programme. As a result, over 10 years the Platform is expected to reduce approximately 9.5 million tons of carbon dioxide equivalent over the project lifetime from the replacement of internal combustion engine vehicles across India by the funding of comparable electric transportation vehicles and by funding electric charging infrastructure.

14. The assumptions made in the funding proposal on the pace of grid decarbonization are less aggressive than India's 2030 commitment of 500GW of energy consumption coming from renewables. If India achieves its 2030 targets, then actual impacts could be higher. Alternatively, the Platform may end up underachieving the impact targets if a lower proportion of counterparties use renewable energy for charging stations, and the decarbonization of the grid decelerates.

15. In addition to GHG mitigation impacts, millions of people are expected to directly benefit from better air quality during the lifetime of the programme and beyond.

16. Another area of independent TAP concern involves the probable scale of the funding proposal programme. The speed and scale of the programme assumes that it can leverage, in addition to GCF funds, USD 205 million equity co-financing to be mobilized from commercial investors and that the platform will also raise substantial third-party debt (primarily from domestic sources) in the ratio of 1:1 (debt:equity). This amount of co-financing will be vital for the Platform to achieve its intended climate impact. Based on the AEs demonstrated experience, early market soundings and its market feasibility study, we are sufficiently satisfied that these targets for raising capital can be achieved (with the catalysing effect of GCF funding), however, final realization may be difficult or delayed, due to unforeseeable market factors and shortage of liquidity in the debt capital market.

17. We also note that the speed and scale proposed in the funding proposal rely on continued support from the Government of India for the EV sector and the continuing programme of regular government tenders for EV buses, charging infrastructures and other e-mobility subsectors. Thus, continued government support is critical for the programme and any major policy change on the part of the Government could be detrimental to expected impacts. However, we have been sufficiently satisfied that support of the e-mobility sector is embedded in India's long-term policy.

18. Overall impact potential is assessed as Medium-High. The realistic possibility of materially contributing to the accelerated decarbonization of the Indian transport sector allows for a high rating. However, the relatively high percentage of GCF funding required to mobilize other equity co-investment and uncertainties about the pace of India's efforts to decarbonize the power grid pushes this funding proposal into the territory of Medium-High.

1.2 Paradigm shift potential

Scale: High

19. The E-mobility Financing Platform aims to support the acceleration of India's EV sector as a key mechanism in decarbonizing the transport sector. By enhancing the availability of credit and by introducing new financial solutions in the EV market that accept operational performance and other risks, the programme seeks to make EV ownership more attractive, help OEMs recycle capital faster and cause electric vehicles to become the preferred choice and mode of transport among fleet operators.

20. Financing solutions provided by the Platform will aim to create a paradigm shift within the Indian context by supporting the private sector to drive forward the e-mobility sector which can serve as an example for other similar markets. The Platform is expected to directly cause 9.5 MtCO₂eq in emissions reductions. Moreover, by demonstrating the commercial viability of the EV sector and financing critical EV infrastructure, the Platform can help catalyse a sector that would result in even greater emission reductions outside the Platform envelope and support a growth pathway that is consistent with national climate strategies. In this way, the paradigm shift potential is embedded in the potential for scaling up the sector and replication of the EV financing tools across India, and possibly other developing countries that could benefit from e-mobility gains.

21. By 2027, India will be the most populous country on the planet with nearly 1.5 billion people. At the same time India's middle class is set to grow rapidly from around 80 million today to 580 million people by 2025. This growth is expected to drive significant demand for transportation capacity, creating a substantial opportunity to meet this growth with lower carbon options to the combustion engine. This opportunity is reflected at the passenger, commercial and mass transport subsectors of mobility. This transition will require the supply of finance targeting EVs and finance for the infrastructure of e-mobility. According to the funding proposal, the estimated size of the EV finance market is expected to reach USD 50 billion in 2030. The Platform could serve a paradigm-shifting role by demonstrating financial viability, attracting new sources of capital to the sector and funding infrastructure improvements.

22. The Platform also has high potential for replicability. The proposal aims to develop a leasing solution to address the existing barriers. Leasing as a model for the EV sector in India has several advantages. First, the residual value and other perceived risks of EV ownership are absorbed by the Platform as opposed to end users. This could accelerate take-up of leasing but more importantly from a paradigm-shifting perspective, demonstrate for customers the difference between actual financial risk and perceived risk. Second, leasing allows EV acquisition costs to be spread over a longer tenor. In this way, leasing mitigates EV acquisition barriers associated with the higher upfront cost of most EVs.

23. The Platform also intends to contribute to greater knowledge-sharing in India by the accredited entity regularly engaging with Indian government institutions (national and state level) and providing inputs on how to increase the effectiveness and impact of policies/regulations enacted. According to MAAML, they along with various MAM and Platform representatives will work with the Government to slowly bring down the subsidies and other tax incentives over time as the sector achieves maturity and substantial crowding in of private capital which would help achieve parity across various segments of the e-mobility value chain.

24. The Platform will also work with several large OEMs, to provide leasing solutions to the EV operators. These financial solutions priced at attractive rates along with long tenors will assist counterparties in de-leveraging their balance sheets and recycling their limited capital, thereby encouraging further investments in local technological advancements. The Platform will also provide capital to other actors in the EV value chain, as the market evolves, especially in the areas of charging infrastructure.

25. By supporting the long-term growth of the EV sector, the Platform should contribute to reducing the need for concessional or public funding over time. For instance, the financing of accelerated deployment of charging infrastructure can reduce behavioural barriers (such as

range anxiety for consumers). Supporting key infrastructure has been shown to be a leading contributor to greater EV market penetration in more mature markets.

26. By raising capital in local markets and engaging with local financial institutions as well as international and national equity providers, the Platform could contribute to familiarizing and embedding new finance providers in the e-mobility sector in India.

27. The proposed financing platform also brings to India MAM's best practices on incorporating climate due diligence in the financing of the e-mobility sector. From the perspective of managing climate-related physical (e.g., flooding, elevated temperatures) and/or transition risks which can affect their subprojects under the financing programme, MAAML is aware of such risks and together with the broader Macquarie Asset Management has the experience to manage it. When the independent TAP asked MAAML for more information, they responded that "MAM incorporates climate-related physical and transition risk in its diligence of each of its portfolio companies to ensure that these risks are integrated into business planning."¹ More broadly, the Macquarie Group reports under the Task Force on Climate-related Financial Disclosures (TCFD).²

28. Overall paradigm shift potential is assessed as High. There are of course risks regarding an actual paradigm shift around transport in India, particularly relating to behavioural biases toward EV purchases and the need for continued government support. However, considering the size and scale of opportunity for India and the realistic possibility of pushing this large, high GHG emitting sector onto a low carbon pathway we can attribute an assessment of High.

1.3 Sustainable development potential

Scale: Medium-High

29. Social and Gender benefits: The funding proposal states that gender policy will be embedded at the Platform level. The policy will include commitment to gender analysis in the assets/projects funded by the Platform. For instance, the programme aims to promote universal access and safety measures for women, children and people with disabilities including proper lighting, in-vehicle cameras, vehicle monitoring systems and other safety measures, especially in buses. According to the funding proposal, the Platform will seek to encourage the engagement of people of all ages as stakeholders in the design, development and implementation of strategies and activities to be financed. The policy will also outline how gender metrics will be monitored, evaluated, and assessed. The Platform intends to collaborate with local and international stakeholders and partners to address systemic and structural barriers to gender equality in local communities.

30. Health benefits: The Indian transport sector has a high percentage of vehicles with low standards for emissions and suffers from significant levels of traffic congestion. For these reasons (among others) the transport sector in India is linked with emissions of pollutants such as fine particulate matter known to have a damaging impact on human health. The elevated levels of emitted air pollutants have resulted in India having 21 of the world's top 30 most polluted cities. It is estimated that over 84 per cent of India's population is exposed to high concentrations of air pollution resulting in over 2 million deaths annually and significant associated healthcare (approx. 5.4 per cent of GDP). The transition to electric vehicles, especially in urban areas, can reduce air pollution. This is particularly important considering the expected growth of the overall transportation sector. If future growth is concentrated around internal combustion engine technologies, then the already dangerous levels of pollution can significantly worsen health outcomes.

31. Economic co-benefits: The Platform is expected to create an estimated 100 new direct jobs at the Platform level. Greater job creation is expected to come from increased investments in the e-mobility space which should indirectly lead to new jobs across the sector. According to

¹ [A summary of Macquarie's supplier governance policy.](#)

² [A summary of Macquarie's supplier governance policy.](#)

Indian government statements, the EV sector is expected to create 10 million jobs in the coming decades. Additionally, as per a recent Niti Aayog report, the European Climate Foundation has estimated that any reduction in oil demand by more efficient electric cars is expected to create 500,000 to 850,000 more domestic jobs by 2030. The report explains that oil production and distribution have exceptionally low employment intensity of just four jobs per million euros value added compared to 24 jobs per million euros in the general economy. Therefore, any shift in expenditure from buying imported oil to other expenditure choices would generate additional employment.

32. India's middle income class group is expected to increase to 300 million households by 2030 (from 158 million) and the number of urban residents overall is expected to top 600 million. The e-mobility Platform will primarily target the public buses and corporate fleet segments which are mainly used by middle income groups and urban residents. Through its investments, the programme aims to improve the safety, reliability and affordability of the transportation sector leading to an increased adoption rate among urban residents and middle-income groups.

33. The Indian government's push for EVs is linked with its stated intent to enhance energy security as well as reduction of GHG emissions. Since the early 2000s, India's crude oil imports have risen steeply with 40 per cent of total oil consumption coming from the transport sector. Oil imports contributed USD 102 billion to the trade deficit in 2020 covering over 80 per cent of the country's oil needs. An increase in the penetration of EVs directly and indirectly through this programme will decrease India's oil import requirement as well as corresponding GHG emissions.

34. Social and environmental co-benefits: The independent TAP clarified if the funding proposal has a negative list of suppliers or if it has undertaken an assessment of potential risk on EV battery component sources (i.e., lithium, cobalt, magnesium, etc.) that are said to be engaged in child labour, human rights abuse, and environmentally unsustainable extraction activities. The AE has responded that they "have have not done a detailed assessment of the source of battery materials imported or sourced in India as there are a number of suppliers and given the nascency of the market, the profile is not static," and that they "do not have an existing list of suppliers that would be automatically excluded. The risk identified is to highlight the known risk in the industry that should be considered as part of our ESS [environmental and social safeguards] risk assessment for each sub-project and supplier. MAM and Macquarie have existing, robust policies and procedures on upholding supplier governance that will be applied at the Platform."³

35. Alignment to Sustainable Development Goals: The funding proposal programme has the potential to promote and accelerate the country's National Development Agenda to implement United Nations Sustainable Development Agenda 2030 as mirrored in the Sustainable Development Goals.

36. The independent TAP assesses this programme as having a medium-high sustainable development potential with significant social, economic and health co-benefits.

1.4 Needs of the recipients

Scale: Medium-High

37. India's average temperature rose by 0.7°C since the last century resulting in extreme weather events and putting pressure on the country's already scarce resources thereby threatening the livelihood of millions of people. India's average temperatures are projected to continue to rise under the relevant Representative Concentration Pathway scenarios resulting in increased frequency and intensity of climate events. These events are expected to determinately impact the economic and social well-being of India if there is no change to

³ [A summary of Macquarie's supplier governance policy.](#)

current climate mitigation policies. The most severe impacts are expected to be borne by low and middle-income groups, women, children, and the elderly. Around 28 per cent of the population or 364 million people live below the national poverty line of USD 2 per person per day.

38. Considering the above, there is a recipient need to undertake low or zero carbon and climate resilient infrastructure investments. According to the funding proposal, India needs ~USD 2.5 trillion of investment between 2015 and 2030 to meet its NDC by 2030. Public funds will not suffice to finance the required climate action. Initiatives such as the Platform are needed to fill the finance gap by directly mobilizing private sector funds into climate-friendly sectors like e-mobility and funding a more rapid maturation of the sector to indirectly unlock additional private sector finance. In a post COVID-19 scenario, there will be an even greater need for the private sector to drive capital flow in the e-mobility sector.

39. India has made a commitment to growing the e-mobility sector and has put in place several incentive and financial support programmes. However, according to the funding proposal, India's EV transition will require a cumulative capital investment of USD 266 billion in the coming decade to fund vehicles, charging infrastructure, and batteries. This analysis suggested that by 2030 the market size for financing EVs in India has the potential to be USD 50 billion. It will be necessary to attract a large amount of finance that will be critical to address existing barriers to finance flows and to demonstrate the investability of the sector. The Platform will help overcome some of these financing barriers including (1) high upfront costs; (2) unwillingness of banks and financial institutions to take EV asset risks; and (3) low loan to value ratios (among other barriers).

40. The independent TAP assesses this programme as rated Medium-High in terms of needs of recipients.

1.5 Country ownership

Scale: High

41. MAAML and the broader MAM teams have had extensive engagement with India's national designated authority regarding the proposed e-mobility programme. It secured the no-objection letter for the funding proposal from the Government of India on 15 September 2021.

42. This programme directly contributes to India's NDC targets of reducing carbon emissions from 2005 to 2030 by 35 per cent including through transforming 30 per cent of all vehicles to EVs. EV promotion has been supported by significant policy development at both the national and state levels. The Government of India has already committed subsidies of up to USD 1.4 billion to enable the accelerated roll-out of EVs under the Faster Adoption and Manufacturing of Hybrid and Electric Vehicles (FAME) Phase II scheme. Where appropriate the Platform will participate in the FAME scheme and other government policies and initiatives which include requirements for a pre-defined level of domestic component manufacturing.

43. The Platform is aligned with India's climate priorities and strategy. It directly contributes to India's NDC GHG emissions reduction targets. It is complementary to and aligned with publicly funded programmes to increase the e-mobility sector, including India's FAME I and II.

44. The funding proposal provided evidence of support across various government entities and other relevant stakeholders including leading domestic OEMs, bus operators, charge infrastructure providers, and Non-Banking Financial Institutions.

45. We assess the country ownership of the funding proposal as High.

1.6 Efficiency and effectiveness

Scale: Medium-High

46. The nascent e-mobility sector represents a significant opportunity to reduce GHG emissions at scale in India and place the country on a transportation development path that avoids locking in fossil-fuel based technologies for many decades. This transition to EVs will require a significant amount of capital and a willingness for early investors to address infrastructure barriers and demonstrate the financial viability of the sector. Notwithstanding country buy-in reflected in strong government support, the market requires additional intervention from concessional capital to help the market mature to where it can readily attract large-scale commercial investments. Commercial investors currently do not have the risk appetite to deploy the capital at the pace required given the real and perceived investment risks outlined in the funding proposal.

47. The Platform aims to provide financing solutions in the market such as operating and finance leases through a new investment vehicle in India. GCF capital will sit in a 'first-loss' position which provides a cushion to the potential downside investment risks for commercial investors. The ratio of GCF and non-GCF funding and the amount of GCF funding has been sized based on the perceived risk appetite of potential investors including what level of downside risk can be accepted by these types of investors. The funding proposal provides that GCF will contribute the first USD 50 million in equity at first financial close with the remaining GCF funds being committed alongside and subject to third-party commercial investor commitments. GCF's capital will be provided in the form of an equity instrument which is able to take a higher risk position in the investment structure in line with GCF's risk appetite. Further, the involvement of GCF will not only increase the amount of capital invested in the sector but also allow the Platform to provide financing solutions not currently available at scale in the market (i.e., a higher risk appetite than current financiers are able to provide).

48. The Platform is expected to be funded by a combination of equity and debt. From our review of the funding proposal and discussions with the AE and Secretariat team, we have a reasonable level of comfort that there is commercial investor interest in the Platform as designed and that MAM has the capacity to raise the commercial funds alongside the GCF contribution. Capital provided by GCF will serve to crowd into the Platform commercial investors that would otherwise have overlooked the opportunity to invest in the e-mobility sector in India due to their risk appetite and nascency of the sector, among other reasons.

49. In terms of efficiency of impact, the funding proposal has set a fundraising target of USD 1,478 million in commercial debt and equity. This represents USD 1,278 million in additional co-financing to GCF financing. The Platform is expected to have a positive catalytic effect with a co-financing ratio of GCF funds of at least 1:6.4. In other words, each GCF dollar invested should mobilize at least 6.4 dollars in co-financing through commercial and private investors. The leverage effect beyond the anticipated investments of the programme through capacity-building is likely to lead to an even higher mobilization. The leasing solution is still nascent in India. If the Platform proves that this business model can work in the country, then private sector investors will be able to see the opportunity and enter the market.

50. The estimated cost per tonne of carbon dioxide equivalent is USD 21.0 based on the requested GCF amount of USD 200 million.

51. The funding proposal is assessed with a rating of Medium-High in terms of efficiency and effectiveness.

II. Overall remarks from the independent Technical Advisory Panel

52. Based on the assessment above, the independent TAP endorses this funding proposal.

53. We note that GCF funds will be pooled into a limited partnership through the accredited entity (MAAML), which will be the investor of record. The success of the investment proposal and its climate impact will depend on multiple factors including:

- setting up a financial organization in India with a robust management team;
- securing all remaining regulatory approvals from the India's Central Bank to provide finance lease and other credit solutions;
- successfully raising commercial equity into the Platform and being able to access/raise debt at the levels described in the funding proposal to leverage its Platform equity; and
- The Platform products finding market acceptance and sufficient investable opportunities to support the target returns of commercial investors and GCF.

54. The success of the Platform will depend on the above factors and others identified by the Secretariat. The independent TAP encourages the Secretariat to build in enough legal and governance flexibility in final deal documentation to be able to adapt to commercial realities if the expected parameters turn out to be materially different. Similarly, based on these numerous factors, the funding proposal programme will require close ongoing monitoring during the implementation phase.



Gender Assessment for E-Mobility Financing Program

Macquarie Infrastructure & Real Assets
India Pvt. Ltd.

Final Report

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Signature Page

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Gender Assessment for E-Mobility Financing Program

Final Report

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1. INTRODUCTION

Macquarie Alternative Assets Management Limited (hereinafter referred to as '**MAAML**' or '**AE**'), has developed a Funding Proposal ('**FP**') for submission to the Green Climate Fund ('**GCF**'), to seek their investment in a dedicated electric mobility (e-mobility) program focusing on India (hereinafter referred to as the '**Program**'). AE is part of Macquarie Asset Management ('**MAM**'), one of the 4 operating business groups of Macquarie Group.

MAM is a global specialist asset manager with assets under management at 31 December 2021 of AUD 750.1 billion, aiming to deliver positive impact for everyone. MAM provides investment solutions to clients across a range of capabilities, including infrastructure & renewables, real estate, agriculture & natural assets, asset finance, private credit, equities, fixed income and multi-asset solutions

ERM India Private Limited (ERM) has been appointed by MAM, to conduct a Gender Assessment Study and to develop a Gender Action Plan. The Gender Assessment Study and Gender Action Plan for the Program forms part of GCF's requirements for all Funding Proposals. This document presents the **Gender Assessment**.

1.1 A brief overview of the 'Dedicated E-Mobility Program'

MAM is looking to partner with GCF and other co-investors to establish an India-focused electric mobility program to facilitate decarbonisation of the road transport sector through rapid scale-up of the electric vehicles sector in the country.

The Program will look to provide financing to private sector counterparties across the e-mobility value chain (BEV and FCEV) such as e-buses, 2/3/4 wheeler segments, supporting infrastructure & services like charging infrastructure, vehicle scrapping and after-sales services.

The Program will comprise of three main activities as outlined in the Funding Proposal presented to the GCF:

- 1) Establishment of a financing platform in India (hereinafter referred to as the '**Platform**'), including its legal entity, policies, procedures and broader governance.
- 2) Providing financing by the Platform to corporate counterparties in the e-mobility space, which involves upfront due diligence and structuring as well as ongoing monitoring.
- 3) Sale of the Platform (to facilitate GCF and other investors exit) with the expectation that the Platform will continue to operate in the market.

For the purposes of this Program, 'sub-projects' will refer to each transaction at the Platform level where a commitment of financing is made to a counterparty in the e-mobility space. Please see the Gender Action Plan for more details.

1.2 Objective and Scope of work

The objective of the assignment is to undertake a Gender Assessment and formulate a Gender Action Plan for the Program with consideration of its targeted sectors. The scope of the exercise includes:

- assessment of general social baseline conditions in India in which the Platform will operate;
- focused interventions that are in practice for the transport and e-mobility sectors from a gender perspective;
- identification of areas that can be best targeted by the Platform for gender mainstreaming activities; and
- formulation of a Gender Action Plan for the Program that highlights key performance indicators and timelines.

The reference framework for the exercise was taken to be the GCF Guidance Note (Form 09 updated 1 February 2019).

1.3 Applicable Reference Framework

The applicable reference framework for the assignment is presented below:

- applicable national legislation and regulations pertaining to social and gender aspects;
- GCF Manual on Mainstreaming Gender in Green Climate Platform Projects¹; and
- GCF Gender Toolkit and Guidance Note (Form 09).

1.3.1 Need for Gender Assessment and Gender Action Plan

Gender mainstreaming is central to the GCF's objectives and guiding principles. The aim is to address it continually by engaging women and men of all ages as stakeholders in the design, development and implementation of strategies and activities to be financed. The GCF Governing Instrument states that: "The Fund [GCF] will strive to maximize the impact of its funding for adaptation and mitigation promoting environmental, social, economic, and development benefits, taking not only a gender-sensitive, but a gender responsive approach."

Gender mainstreaming, as defined by the United Nations Economic and Social Council, refers to a strategy for making women's, as well as men's, concerns and experiences integral dimensions of policies and programs in all political, economic and social spheres. Gender mainstreaming is essential to ensure that women and men benefit equally from a certain system, and inequality is not perpetuated.

There is still a heavy linking of the term "gender" with "women's issues"; however, the evolved understanding on gender and gender mainstreaming is that Gender represents equality of both the sexes.

However, GCF has established on the basis of prevalent practices and past experiences that women and girls are disproportionately affected by the impacts of climate change and associated severe weather events. Women's mortality from climate-related disasters is higher than that of men, and climate-induced impacts due to natural disasters like floods and earthquakes exacerbate women's poverty and unpaid domestic and care work burdens. Gender inequality, exacerbated by climate change, is linked, as are other development areas, to vulnerability and risks. The greater vulnerability of women to climate change stems from societal norms and discrimination that result in the imbalanced division of labour, lower incomes, and lesser livelihood opportunities; less access and control over land and other productive assets; fewer legal rights; lesser mobility and lesser political and professional representation.

1.3.2 GCF's Gender Policy

The GCF Board adopted the Gender Policy and Action Plan in March 2015 and an update to this policy was issued in February 2019, to respond to a fast-changing landscape related to gender in the context of climate change. Its rationale is rooted in the GCF's mandate to encourage a paradigm shift towards low-emission and climate-resilient development pathways. The updated policy emphasizes gender responsiveness rather than gender sensitivity. Being gender-responsive means that instead of only identifying gender issues or ensuring a "do no harm" approach, a process will substantially help to **overcome historical gender biases**. Gender responsive approaches are essential to this, including decisions and actions by women and men.

GCF's Gender policy aims to:

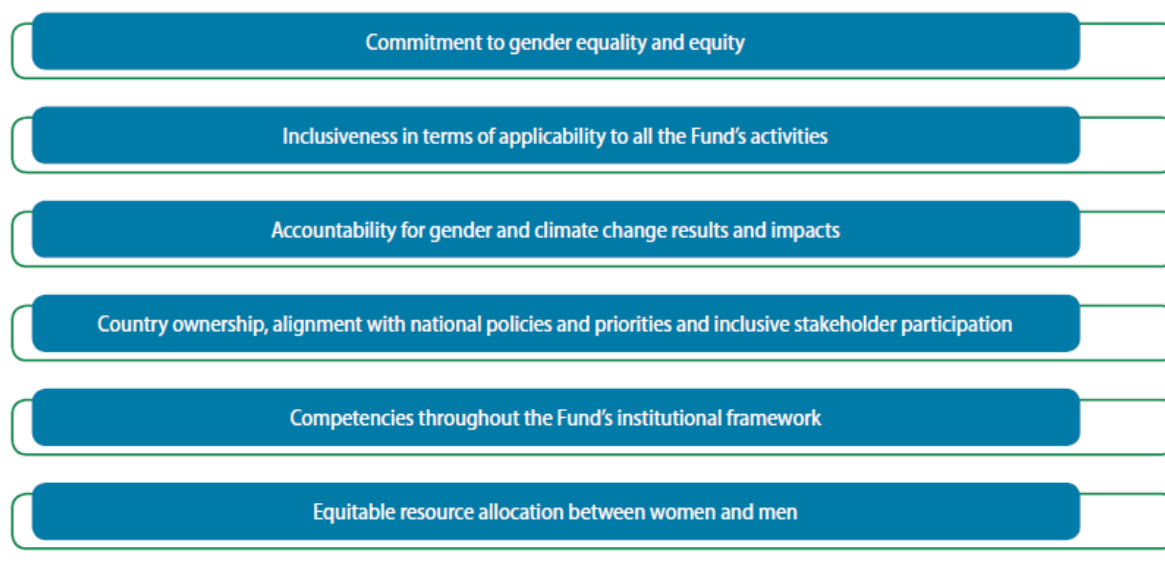
¹ Published in 2017

- Achieve greater, more effective, sustainable, and equitable climate change results, outcomes, and impacts, through the adoption of a gender sensitive approach
- Build resilience to climate change equally for men and women, as well as to ensure that men and women equally contribute and benefit from activities supported by the Fund
- Address or mitigate risks for women and men associated with adaptation and mitigation activities financed by the Fund; and
- Reduce the gender gap of social, economic, and environmental vulnerabilities exacerbated by climate change.

For the purposes of this Gender Equality Policy, GCF requires that women and men, including marginalized and vulnerable groups, will be provided with an equal and equitable opportunity to be fully and effectively engaged in meaningful consultations and decision-making throughout the project cycle.

The GCF's Gender Policy is based on the following six Fundamental principles.

Figure 1.1 Principles of GCF's Gender Policy



Source: Green Climate Platform Proposal Toolkit 2017

The GCF's paradigm shift is rooted in the acknowledgement of women's critical role in climate responses, as agents of change sharing in the benefits, and, in several cases, disproportionately affected by its negative impacts. Marginalised communities also bear disproportionate negative outcomes and are also included in this shift.

According to GCF, "Gender assessment refers to methods used to understand relationships between men and women, their access to resources, their activities, and the constraints they face relative to each other. A comprehensive gender analysis/assessment entails the examination of the different roles, rights, needs, and opportunities of women and men, boys and girls in a given project/context. Gender analysis needs to take place early in the planning process so that an understanding of gender roles and power relations is built into the project."

1.4 Methodology

In order to achieve the above-mentioned scope, the following methodology has been adopted.

1.4.1 Kick off call with MAM

ERM organised an Inception call with the Client to understand the current investment context, get a general understanding of the planned working of the Platform and expected investments. ERM also discussed the role of MAM in the Program as the investment manager, applicability of an ESMS, a Gender Policy or any other ongoing/proposed interventions that will be undertaken by the Program that may particularly focus on addressing gender related aspects.

1.4.2 Detailed Gender Assessment

A desk-based research of aspects related to women in India and specially in urban mobility has been undertaken to understand the nature and extent of data and studies capturing aspects like female urban ridership, challenges faced by urban females due to limited mobility and while using public transport, general scenario of employment of women and specifically in the urban transportation sector.

1.4.2.1 National Context guided by GCF Guidance Note

An assessment of the nationwide gender related baseline scenario and associated impacts, risks, opportunities and trends in this sector was undertaken using secondary data.

Key steps followed for the assessment are:

- Profiling the social and development status of women and men (education, literacy, work force participation, unemployment, role in formal/informal economy, division of labour, access to resources and opportunity, decision making roles and capacity, as relevant) based on secondary data at the national level
- An assessment of the legal status, beliefs, perceptions, prevalent gender stereotypes, as relevant for the key sectors
- An assessment of relevant issues like gender-gap, inequality, gender-specific vulnerabilities (with respect to climate change adaptation, social, socio-economic factors, technical capacity, and resource access etc.) has been undertaken based on relevant secondary literature and published research.

Please note: In the absence of latest data on gender issues such as employment and literacy in the transport sector? the report has relied on Census of India, 2011 data and has supported the findings, wherever possible with other latest available studies.

1.4.2.2 Establishing a Baseline Scenario for Gender Mainstreaming in E-mobility

The assessment has sought to identify a baseline scenario for gender issues in the e-mobility sector and ultimately identify where the Program could best implement gender-based interventions.

Given the lack of data on gender and e-mobility sector, especially in developing countries, the report has relied on secondary review, data and analysis available for gender and transportation sector in general. It should also be noted that there is a general lack of gender disaggregated data on urban mobility and hence also in the e-mobility sector.

Consequently, there is little available research on strategies, interventions and case studies on specific preferences (of women and men, young and old, or those with disabilities) regarding urban public transport usage, sector employment trends, health and safety, ergonomics, or accident risks. Sex-disaggregated data has been provided where available.

Whilst this is one limitation of the Gender Assessment, collecting sex-disaggregated data is also an identified area for improving gender sensitivity in this sector.

1.4.2.3 Stakeholder Engagement

Specific stakeholder engagement as part of this gender assessment has not been completed. This is primarily because the proposed Platform has not yet been established and the broad scope of the proposed transactions involve multiple e-mobility sectors and are not based in a specific region in India. It is proposed that targeted, gender-focused stakeholder engagement is completed once the Platform is established to build upon the desk-based assessment and interventions outlined in the Gender Action Plan. Please refer to the Gender Action Plan for more details.

1.4.3 Formulation of the Program-level Gender Action Plan

The Gender Action Plan has been formulated to include strategies and action items identified for the Platform. The action plan includes key indicators and timelines, for effective monitoring of outcomes.

1.4.3.1 Limitations in development of Action Plan

- The Platform is not yet established, including the defined policies, procedures and governance. Actions have been identified to integrate gender mainstreaming frameworks as part of this establishment.
- There is limited ability to set baseline scenarios given the lack of data available in the sector, the broad mandate across sectors and geography, and the Platform is not yet established so there is no benchmarking applicable of its existing practices. Activities have been outlined to support establishment of baseline.

2. THE BROADER GENDER LANDSCAPE IN INDIA

The Constitution of India conveys a powerful mandate for equality and rights of women in its Preamble, Fundamental Rights, and Duties and also provides for specific provisions for affirmative actions. India is also a signatory to a number of UN Conventions, primarily Convention on Elimination of all Forms of Discrimination against Women (CEDAW), Beijing Platform for Action and Convention on Rights of the Child where the commitment of the nation to protect and empower its women and girls is quite pronounced. The recent endorsement by India, of the ambitious 2030 Sustainable Development Goals (SDGs) will further change the course of development by addressing the key challenges such as poverty, inequality, and violence against women, which is critical for the global success of the goals as well.

However, according to the World Economic Forum's Global Gender Gap Report 2021², India has seen its gender gap widen from 66.8% in 2019 to 62.5% in 2020. It now ranks at 140 out of 156 countries. Similarly, according to the Human Development Index (HDI) report prepared by United Nations Development Programme in 2020, India was placed at 123rd position in the Gender Inequality Index. This is largely due to women's inadequate representation in politics, technical and leadership roles, decrease in women's labour force participation rate, poor healthcare, lagging female to male literacy ratio and income inequality. However, like most of the developing countries, India struggles with gender inequality issues beyond just equal economic growth and access to educational resource opportunities. Gender inequality exists in the form of socially constructed, predefined gender roles firmly anchored in India's sociocultural fabric that has deep cultural and historical roots. Sociocultural influences have spill-over effects across all domains, including workforce ratio, and social and political contexts.

Further, the COVID-19³ pandemic has raised new barriers to build inclusive and prosperous economies and societies. In addition to the pressures of providing care in the home, the crisis has halted progress towards gender parity in several economies and industries. As per the Global Gender Gap Report 2021, preliminary evidence suggests that the health emergency and the related economic downturn have affected women more severely than men, partially re-opening gaps that were being closed.

This section provides a broad baseline at the national level to set up a context for the gender scenario in the country at large and bring out the differences with respect to overall development indicators (demography, education, and health for example) and various policies introduced by Government of India to address the differences. Specific aspects of the focus sector, comprising of urban public transport, ridership, safety and risk, worker participation, other travel needs, gaps, and related aspects are included in the subsequent section.

2.1 Demographic Profile

In India, as per Census 2011, the total population is 1,210.8 million with a decadal population growth rate of 17.7%. The decadal population growth of males has been registered as 17.1% and for females has been registered as 18.3%. It has also been established through the Census 2011 data that about 31.1% of Indian population lives in urban areas; while the rest live in rural areas.

The sex ratio (number of females per 1,000 males), in the country as per Census 2011 is 943 and has improved from a figure of 933 in 2001. In rural areas, the sex ratio has increased from 946 females per 1,000 males to 949 females per 1000 males and the corresponding increase in urban areas has been of 29 points from 900 to 929 females per 1000 males. The gender balance in the country is

² This report benchmarks the evolution of gender-based gaps among four key dimensions of Economic Participation and Opportunity, Educational Attainment, Health and Survival, and Political Empowerment and tracks progress towards closing these gaps over time.

³ Coronavirus disease (COVID-19) is an infectious disease caused by a newly discovered coronavirus. The World Health Organization declared the outbreak a Public Health Emergency of International Concern on 30 January 2020, and a pandemic on 11 March 2020.

skewed towards males as depicted from the sex ratio figures and the general socio-cultural environment in the country. One of the reasons for this difference can be attributed to the trend of migration of males in search of employment from villages to nearby towns and cities, while their families continue to stay in their native place.

The share of youth in the total population has been continuously increasing from 30.6% in 1971 to 34.8% in 2011⁴. The percentage of male youth population to total male population follow the pattern of overall youth and was also on a steady rise till 2011, but the share of female youths to total female population showed a slight decrease in 2011 which may have been due to a declining sex ratio during this period.

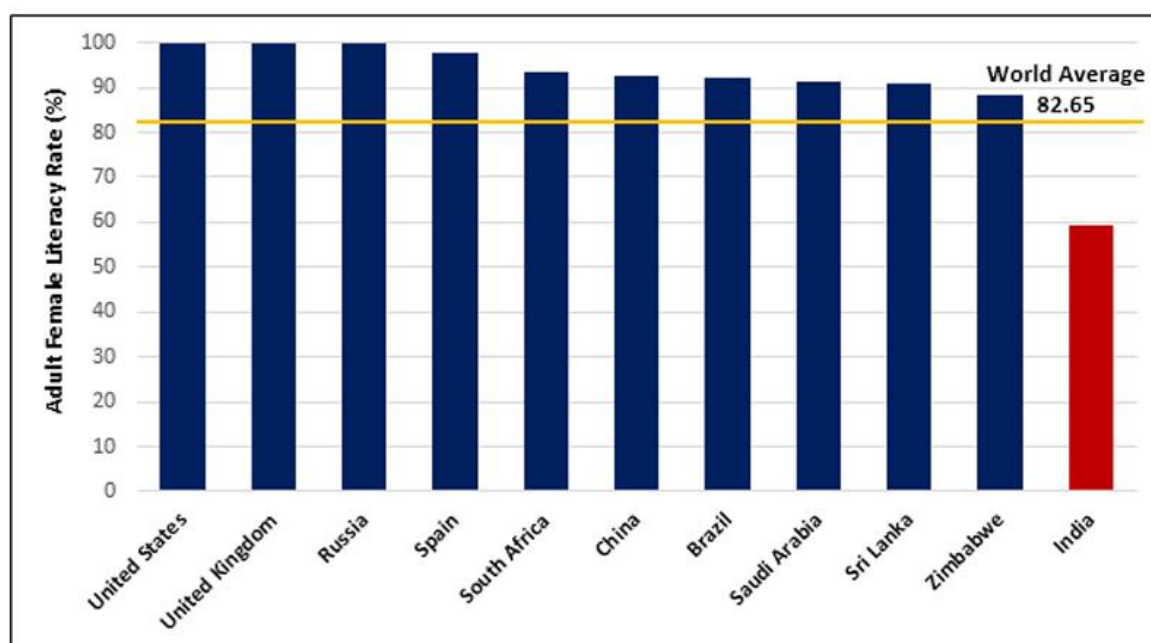
2.1.1 Educational Attainment

According to the 2011 Census of India, the overall adult literacy rate was 73%. The literacy rate for females and males were 64.63% and 80.9% respectively. According to the Global Gender Gap Report 2021, almost one third of women are illiterate (34.2%) compared to 17.6% of men in India. The trend continues despite intervention by the government such as Right of Children to Free and Compulsory Education Act, 2009. This gap is attributable to a combination of social, economic, and cultural factors. Even as the benefits of female education will lead to a more productive workforce, lower fertility and lower infant mortality rates, the costs such as tuition fees, transport, and school supplies as well as opportunity costs of forgone child labour, which are privately borne by households, continue to influence this choice. The underinvestment in women's schooling results in lower literacy and lower educational attainment, and further, to lower employability for semi-skilled and skilled jobs. Social restrictions on women's mobility also prevents an educated woman from entering the labour force and supporting the household financially.

The figure below shows the glaring difference in India's female literacy, not only with the developed countries but also with middle- and lower-income countries like Sri Lanka and Zimbabwe. According to this, female literacy in India is 22 percentage points below the world average⁵. These numbers are a reminder that India is still a long way from meeting Goal 4 of the UN SDGs (Sustainable Development Goals) of ensuring "inclusive and equitable quality education" and "lifelong learning opportunities for all" by 2030.

⁴ http://mospi.nic.in/sites/default/files/publication_reports/Youth_in_India-2017.pdf

⁵ <https://www.firstpost.com/india/indias-female-literacy-has-gone-up-but-still-22-percentage-points-behind-world-average-education-among-young-women-rising-7197631.html>

Figure 2.1 Adult Female Literacy rate (India and the rest of the world)

Source: Gender Disparity in Literacy, world Bank. <https://www.firstpost.com/india/indias-female-literacy-has-gone-up-but-still-22-percentage-points-behind-world-average-education-among-young-women-rising-7197631.html>

2.1.2 Enrolment and Dropout Statistics

According to a study conducted by the Statistics division of Department of Higher Education, Ministry of Human Resource Development in the year 2015-16, the Gross Enrolment Ratio⁶ status of population in India as per the various levels of education are presented in **Table 2.1** below.

Table 2.1 GER for males and females in India, 2015-16

Gender	Primary	Upper Primary	Elementary	Secondary	Higher Secondary
Boys	97.83	88.72	94.53	79.16	55.95
Girls	100.69	97.57	99.59	80.97	56.41
Total	99.21	92.81	96.91	80.01	56.16

Source: District Information System for Education Study – 2015-16

The Gross Enrolment Ratio (GER) for girls at the primary level stood at 100.6 compared with nearly 97.8 for boys. At the upper primary level, the GER for girls has been reported as 97.5 against 88.72 for boys. The higher GER for girls again reflects the positive change in the country where increasing number of families are signing up for the education of girls at least up to elementary level. The GER and educational attainment figures reduce significantly at the secondary and higher secondary levels and the major reason for non-enrolment at higher levels as been reported as "not interested in education" for rural males & females, whereas for urban India, "financial constraints" as the major reasons for non-enrolment⁷.

⁶ The Gross Enrolment Ratio (GER) for a class-group is the ratio of the number of persons in the class-group to the number of persons in the corresponding official age-group.

⁷ http://www.mospi.gov.in/sites/default/files/reports_and_publication/statistical_publication/social_statistics/Chapter_3.pdf

Supporting this is a survey report from Teen Age Girls (or the TAG report)^{8,9}, released by Nanhi Kali in 2018, a project by the Naandi Foundation, which works with adolescent girls. This report noted that one in five teenage Indian girls is currently not studying and the dropout rate increases with age. Nearly every second teenage Indian girl believes boys have better opportunities to pursue education and work, and only one in five believes that boys can do as much household work as themselves.

For higher education, the student enrolment statistics given by All India Survey on Higher Education (AISHE) 2014-15, shows that the highest number of students are seen to be enrolled at Undergraduate level. Out of the total students enrolled, 79% students are enrolled in Undergraduate level. Second to the Undergraduate, 11% students are enrolled in Post Graduate level¹⁰.

Table 2.2 Educational attainment data per 1000 distribution of population above 5 years

Level of Education	Rural		Urban		Total	
	Male	Female	Male	Female	Male	Female
Literate without Schooling	8	6	5	6	7	6
Up to Primary	377	336	285	280	349	319
Upper Primary	173	130	157	143	168	134
Secondary	118	80	150	133	128	96
Higher Secondary	71	45	114	102	85	62
Diploma	11	5	30	14	17	8
Graduation	37	18	126	96	64	42
Post-Graduation and above	8	4	43	35	18	14

Source: http://www.mospi.gov.in/sites/default/files/publication_reports/Women%20and%20men%20in%20india-2017%20Mail.pdf

2.1.3 Workforce Participation

According to the Periodic Labour Force Survey (PLFS) (2017-18), Ministry of Statistics and Implementation, India, India's workforce has fewer women than it did six years ago: no more than 18% of women in rural areas are employed, compared to 25% in 2011-12. However, in urban areas, the percentage of women in salaried jobs has increased from 35.6% in 2004 to 52.1% in 2017 but continue to be under-represented compared to their presence in self-employed or casual work¹¹.

Figure 2. presents participation of women in workforce from 1993 to 2017.

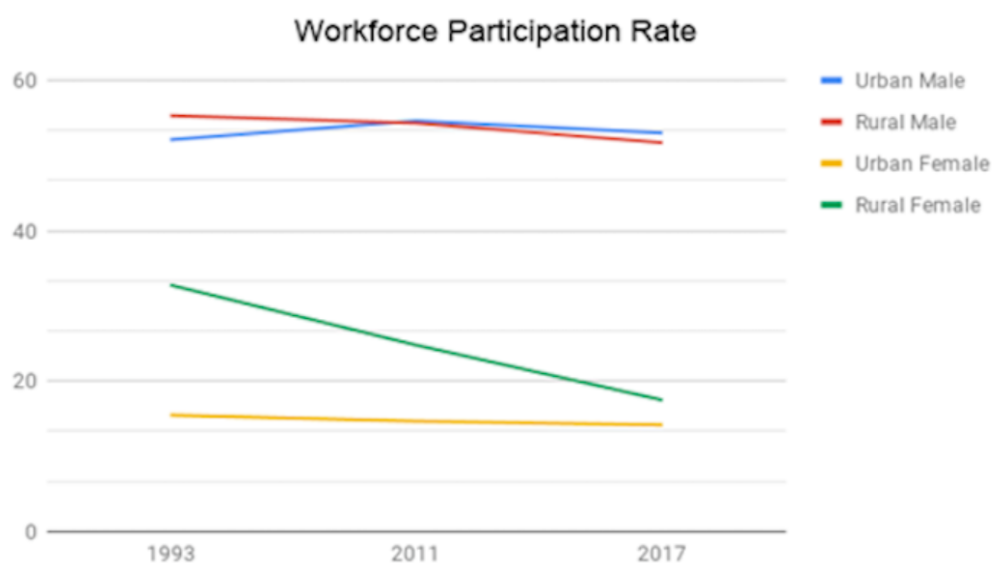
⁸ <https://www.nanhikali.org/pdf/TAG-Report.pdf>

⁹ The survey asked girls aged 13-19 years questions on nine topics including educational and health status, basic life skills, agency and empowerment within and outside the home and aspirations.

¹⁰ http://mospi.nic.in/sites/default/files/publication_reports/Youth_in_India-2017.pdf

¹¹ Women Workers in India: Labour Force Trends, Occupational Diversification and Wage Gaps Bidisha Mondal, Jayati Ghosh, Shiney Chakraborty and Sona Mitra, March 2018.

Figure 2.2 Fewer Women Are Participating in the Paid Workforce



Source: Periodic Labour Force Survey (PLFS) 2017-18

As per literature reviews, there are various reasons for this decline including social and cultural barriers to women's employment and household work expectations from women (particularly mothers). Notably, another key reason is the lack of reliable, safe, and cheap public transportation, as per the study by the Institute of Transportation and Development and the NGO, Jagori, shows. This is further discussed in detail in Section 3 below.

Further, various government policy interventions have also not been very helpful (See section 2.3). Women also withdraw from the workforce due to an 'income effect', that is an increase in the husband's income which pushes up the household income. It is also understood that women are withdrawing from the workforce to pursue higher education, while others contend that there is a lack of good jobs for women^{12,13}.

In addition, the burden of unpaid work falls disproportionately on women in India because tasks such as cooking, cleaning, fetching water and firewood are highly gendered. Patriarchal norms dictate that women also perform care work, validate men's failure to assume domestic responsibilities and thus entrench women's unequal social status¹⁴

According to Organisation for Economic Co-operation and Development (OECD) data, Women in India currently spend up to 352 minutes per day on domestic work, 577% more than men (52 minutes) and at least 40% more than women in South Africa and China.

According to the same report Periodic Labour Force Survey (PLFS) (2017-18), 24.5% of women were part of the labour force, which means they are either employed or looking for work, as compared to 75.5% of men. However, as per the survey by Business Standard (2020)¹⁵, after lockdown was imposed, 13% fewer women than a year ago were employed or looking for jobs, compared to 2% fewer men. The lockdown affected urban women more than rural women.

¹² Why is female labour force participation declining so sharply in India? / Steven Kapsos, Evangelia Bourmpoula, Andrea Silberman ; International Labour Office.2014.

¹³ Female labour force participation in India and beyond. Ruchika Chaudhary and Sher Verick October 2014

¹⁴ Oxfam India's second India Inequality Report, 'Mind The Gap – The State of Employment in India', 2019.

¹⁵ https://www.business-standard.com/article/current-affairs/covid-impact-women-workforce-disappearing-most-affected-in-urban-india-120121500259_1.html

2.2 Health

Health is one of the important indicators defining the status of development of a nation. Its improvement is characterised by lower incidences of deaths and diseases and an improvement in the overall health indicators, including maternal mortality rates and infant mortality rates.

In India, infant mortality rate¹⁶ (disaggregated by sex and place of residence) has been reported as 37 per 1000 live births which is comparatively higher than the urban total of 25 per 1000 live births and lower than the rural total 41 per 1000 live births. In case of females, the infant mortality is higher in the rural area with a figure of 43 per 1000 live births than in urban areas of 28 per 1000 live births. The female infant mortality has been higher than male mortality rate which highlights a certain bias prevalent in the society, more so in the rural areas. The percentage change in the average level of infant mortality between 2004-06 and 2014-16 at the national level has declined by 36.7%.

Reproductive, maternal, new-born, child and adolescent health services and national programs to curb incidences of diseases such as polio, HIV, TB, and leprosy have played pivotal roles in improving health outcomes for the Indian population. There are noteworthy improvements in health indicators such as life expectancy, infant mortality rate (IMR) and maternal mortality rate (MMR) due to increasing penetration of healthcare services across the country, extensive health campaigns, sanitation drives, increase in the number of government and private hospitals in India, improved immunisation, and growing literacy. Initiatives such as Janani Shishu Suraksha Karyakarm, Janani Suraksha Yojana¹⁷. During 2011 to 2015, the percentage of urban mothers who received medical attention at the time of delivery has increased from 87.9% to 93.5% and from 60.7% to 74.2% for rural mothers.

2.3 Gender Institutional Architecture and Government Schemes targeted at Women's development

At the national level the Ministry of Women and Child Development acts as a nodal agency for working towards realization of constitutional and international commitments to gender equality and social justice in India. State Resource Centres for Women under the National Mission for Empowerment of Women (NMEW) provides support to the respective State/UT Women and Child Development Department for coordination and implementation of the various policies and regulations. They are further supported by Panchayats (in rural areas) and Urban local bodies. Civil society organizations, associations, and federations also provide inputs and help in implementation and monitoring of policies and programmes impacting women.

There are various gender focused legal regulations in India that aim to address the vulnerabilities and cultural biases associated with both the genders, through provision of certain measures that support a safe and equal professional platform for women. These regulations are presented below.

- Equal Remuneration Act, 1973 provides for payment of equal remuneration to men and women workers for the same work of similar nature without any discrimination. In order to ensure social security to the workers including women in the unorganised sector, the Government has enacted the Unorganised Workers' Social Security Act 2008.
- The Maternity Benefit Act, 1961 and The Maternity Benefit (Amendment) Act, 2017 regulate employment of women in certain establishments for a certain period (26 weeks) before and after childbirth and provides for maternity and other benefits.
- The Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act (POSH), 2013 has been enacted, which covers all women, irrespective of their age or employment status and protect them against sexual harassment at all workplaces both in public and private sector, whether organised or unorganised.

¹⁶ Infant Mortality Rate is defined as the infant deaths (less than one year) per thousand live births

¹⁷ <http://www.indiaenvironmentportal.org.in/files/file/NHP%202018.pdf>

The Government of India has been launching programmes like Beti Padhao, Beti Bachao campaign (launched in 2015) for improving the Gender Parity Index (GPI)¹⁸ by promoting survival, protection, and education of girls. It aims to address the issue of declining child sex ratio through a mass campaign targeted at changing social mind set and creating awareness about the criticality of the issue. Furthermore, Mahila E-Haat has been launched by the Government in 2016 for promoting women entrepreneurs, with the objective to provide an e-marketing platform by leveraging technology for showcasing products made, manufactured and sold by women entrepreneurs, SHGs and/or NGOs.

Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA), 2005¹⁹ is one of the important schemes launched in India which ensures participation by women in the economic activity by stipulating minimum 33 per cent participation by women.

2.4 Climate Change Impacts on Women in India

One of the biggest challenges that we are facing in 21st century is climate change. As per the latest report of the United Nations' Intergovernmental Panel on Climate Change (IPCC), human activities have already caused warming of 1.0 degree Celsius as compared to pre-industrial times. By 2030, or latest by mid-century, global warming is likely to reach 1.5 degrees Celsius. Thus far, with a 1°C rise in global temperatures, India has already experienced extreme weather events such as floods in Kerala, wildfires in Uttarakhand and heat waves in the north and the east, demonstrating its vulnerability. Its impacts vary among regions, generations, age, classes, income groups, and gender²⁰. However, climate change exacerbates existing structural and social gender inequalities in ways that has led women and girls to experience climate change the worst and first.

The IPCC (International Panel on Climate Change) found that gender inequalities are further exaggerated by climate-related hazards, and they result in higher workloads for women, occupational hazards indoors and outdoors, psychological and emotional stress, and higher mortality compared to men.

It is not only in India but worldwide, women have less access than men to resources such as land, credit, agricultural inputs, decision-making structures, technology, training and extension services that would enhance their capacity to adapt to climate change²¹. As per one of the reports by United Nations²², women represent a high percentage of poor communities that are highly dependent on local natural resources for their livelihood, particularly in rural areas where they shoulder the major responsibility for household water supply and energy for cooking and heating, as well as for food security.

2.4.1 Impacts of Pollution

According to the World Health Organization (WHO)²³, air, noise pollution is a global environmental health emergency and India sits on the frontline of this global crisis. Fourteen of the world's 20 most polluted cities are still found in India and an estimated 99.9 percent of India's population live in areas where air pollution exceeds WHO air quality guidelines. In 2019, air pollution was estimated to be

¹⁸ Gender Parity Index (GPI) in education is a valuable indicator which reflects the discrimination against girls in access to educational opportunities.

¹⁹ Mahatma Gandhi Employment Guarantee Act 2005, is an Indian labour law and social security measure that aims to guarantee the 'right to work'

²⁰ <https://www.un.org/en/chronicle/article/womenin-shadow-climate-change>

²¹ Aguilar, L., 2008. "Is there a connection between gender and climate change?", International Union for Conservation of Nature (IUCN), Office of the Senior Gender Adviser.

²² https://www.un.org/womenwatch/feature/climate_change/downloads/Women_and_Climate_Change_Factsheet.pdf

²³ https://www.who.int/health-topics/air-pollution#tab=tab_1

responsible for 1.6 million premature deaths in India and \$95 billion USD in economic losses (3 percent of India's GDP²⁴).

2.4.1.1 Transportation, Pollution and Gender

Vehicle transport is a key source of air and noise pollution and greenhouse gas emissions in Indian cities, and one in which power dynamics, occupational differences, as well as socioeconomic and cultural differences related to gender shape exposure for women and men.

Mobility Patterns

As per various studies,²⁵ gender inequalities often shape women's and men's mobility and use of transport globally and in India. If a family owns a car, the male head of the household is more likely to use it, leaving women and children to rely on alternatives, mainly walking and public transport. As per census of India, 2011²⁶ a gendered comparison of five large cities in India showed, on average, 37 percent of women walked to work compared with 27 percent of men. Data from the Delhi morning rush hour demonstrates the implications of walking near roadways for exposure to increased levels of air pollution.

Occupational Differences

As per a study by WHO²⁷ in India (especially in rural areas), women are more vulnerable to roadway emissions because they have reduced lung function due to indoor air pollution from domestic duties, such as the use of biomass fuels for cooking. As per a study conducted in 2013²⁸ cited found the risk of chronic respiratory morbidity among street sweepers was 4.24 times higher than that in the comparison group, and the risk increased significantly with increasing length of service. Women in the informal sector often also work in street vending or minding small retail shops that are frequently located in high-traffic roadsides where they have greater exposure to emissions. These women may be forced to choose between the increased incidence of respiratory disorders if they vend on busier streets or the increased risk of gender-based violent attacks on quieter streets.

²⁴ <https://www.forbes.com/sites/dishashetty/2020/12/22/air-pollution-cost-india-368-billion-in-2019/?sh=57a3a5405c70>

²⁵ <https://www.itdp.org/publication/women-transport-indian-cities/>

²⁶ https://www.censusindia.gov.in/2011census/B-series/B_28.html

²⁷ <https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health>

²⁸ Chronic Exposure to Dust and Lung Function Impairment: A Study on Female Sweepers in India
S Smilee Johncy¹, G Dhanyakumar¹, Kanyakumari¹, T Vivian Samuel²

3. SECTOR SPECIFIC ASSESSMENT

As discussed above, the focus of the Program will be to provide financing to counterparties across the e-mobility value chain such as e-buses, 2/3/4 wheeler segment, support infrastructure & services like charging infrastructure, vehicle scrapping and after sales services.

Through review of secondary literature and experience of projects within e-mobility space, it is understood that gender related issues in the e-mobility sector are common to those in the broader mobility sector in India, including gender-based violence and harassment, economic disparity, and appropriate and safe transport options.

The recent flow of capital in safe and well-designed transport infrastructure can help increase economic empowerment of women by improving mobility, access to better avenues of employment, education, and healthcare. For example, introduction of new electric bus fleets over recent years is helping state governments enhance the safety measures for women and elderly on such buses by mandating inclusion of specific safety features such as provision of first aid kits and fire extinguishers or ensuring proper in-bus lighting at all times in the concession agreements with operators.

As a result, there is a strong belief that growth of the e-mobility sector can lead to changes to gender dimensions by supporting safer public transportation options. Moreover, as a new industry there is potential to promote the implementation of gender inclusive design and practice, as well as applying broader workforce policies such as bridging the pay gap, involvement of more technically qualified women, opening more avenues of employment for women.

However, there is lack of study and research in gender disaggregated use of sustainable transportation. For example, there is less data to understand if women use more electric vehicles than traditional fuel vehicles. This study therefore relies on data on general use of transportation and employment in the transportation sector.

Subsequent sections describe in detail the gender issues prevalent in the urban mobility and transportation sectors.

3.1 Gender Dimensions of Urban Transport

The recently announced Green Urban Mobility Scheme (GUMS) expects to invest around INR 700,000 million over 2018–2023 to promote use of sustainable transportation. The central government has initiated schemes to invest in urban transport and infrastructure and created indicators and service level benchmarks to establish a city's baseline and goal for improvement²⁹.

Transportation planning and design are commonly regarded as “gender neutral.” It is assumed that transport projects equally benefit both men and women and that there are no significant differences between travel needs and patterns of either gender³⁰. However, for these same reasons urban transport investments are largely “gender blind” with a limited understanding of the interrelationships between gender and transport inequities. This is intensely rooted in community-driven gender roles with economic, social and livelihood influences in the Indian society where women are invariably responsible for household chores and taking care responsibilities of children and elders, making the mobility needs of women and men heterogeneous.

Men and women have also been found to use public transportation very differently, noted in a December 2017 report, *Women and Transport in Indian Cities*, by the Institute for Transportation & Development Policy (ITDP) and Safetipin. For instance, women tend to combine multiple destinations

²⁹ Women and Transport in Indian Cities. A Policy Brief. December 2017. ITDP. Safetipin

³⁰ <https://thecityfix.com/blog/why-it-is-key-to-include-gender-equality-in-transport-design-jyot-chadha-vishal-ramprasad/>

within one trip, for example to pick up groceries on the way back from office or drop a child off to childcare on the way to work. While men use transport services largely to travel to work and return. The same study points out that, in metropolitan cities in India, women travel shorter distances, with most preferring to walk or take the bus to work. Men seem to have greater choice across modes of transport including bicycles, two-wheelers, and four-wheelers. Further across Indian cities, women form only 22% of people who travel for work.

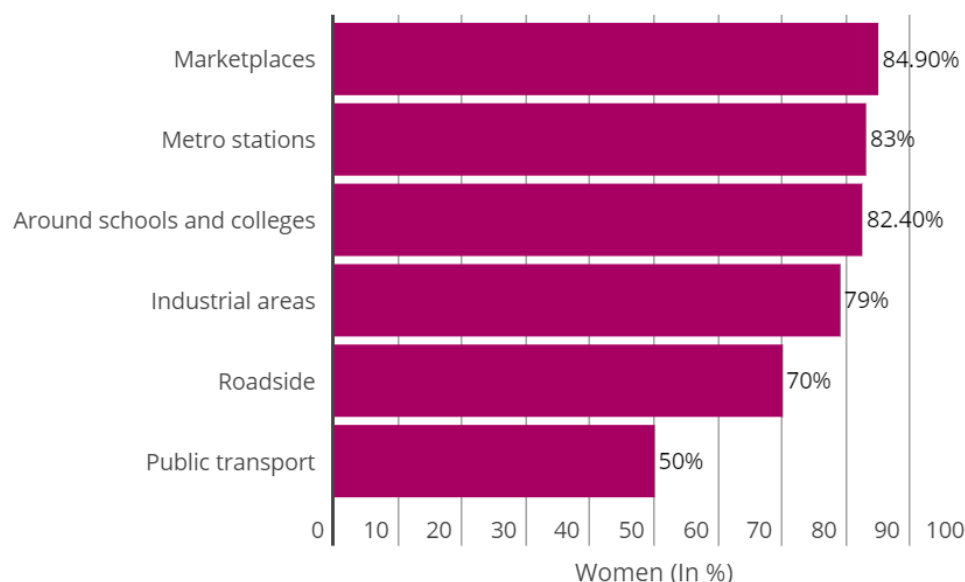
According to the National Sample Survey Office (NSSO) 2016 data, more than 60% of rural and urban households use the bus as their main mode of public transportation, followed by auto-rickshaw, taxi, railways, and cycle rickshaw. While women's bus transport modal shares vary from 25% in Mumbai to 37% in Bhopal, women are more dependent on public transport than men are, especially when they are from lower-income groups. In Mumbai, women undertook 45% more trips by bus than by train, which increased to 67% for households with incomes less than INR 5,000 per month, as per a study by World Bank in 2011. In Delhi, 34% of women commuted to work by bus compared to 25% of men, whereas the corresponding figures for Chennai were 34% for women and 22% for men as per Census 2011 data.

3.2 Women safety in Public Transport

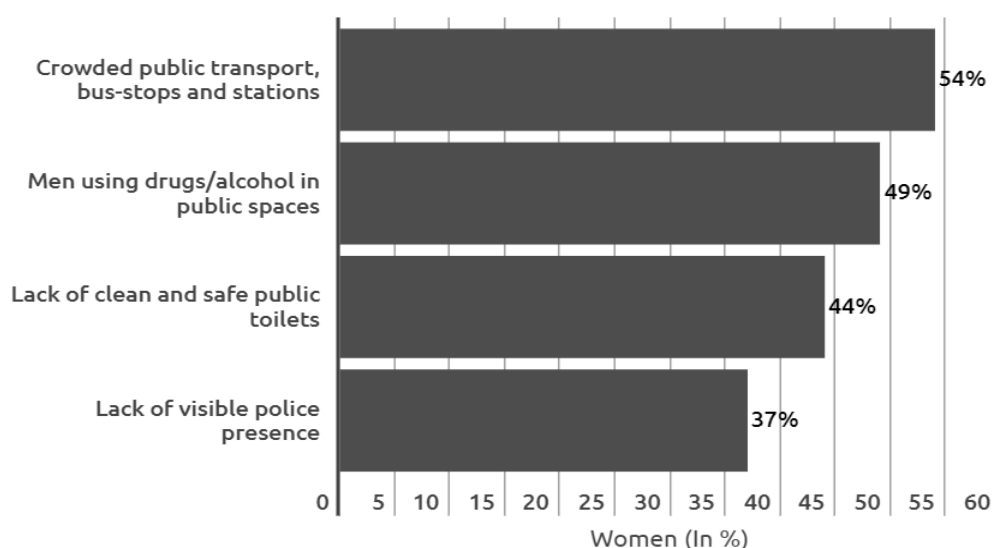
Sexual harassment in public places is rampant and based on a 2010 study of 5,010 women and men in Delhi by the NGO Jagori an overwhelming concern of 85.4% women fear a violent physical attack and even rape. Almost two out of three women reported facing sexual harassment between two to five times over the previous year. The harassment occurred day and night, in places secluded and crowded with most being reported from buses, public transport and roadsides.

Another report by ActionAid, released in 2016, found that 79% of women surveyed had experienced some form of harassment in public.

Figure 3.1 Where Women Reported Sexual Harassment Takes Place



Source: http://www.jagori.org/wp-content/uploads/2011/03/Baseline-Survey_layout_for-Print_12_03_2011.pdf

Figure 3.2 What Makes Women Feel Unsafe

Source: http://www.jagori.org/wp-content/uploads/2011/03/Baseline-Survey_layout_for-Print_12_03_2011.pdf

Ola Mobility Institute conducted a survey of 43,486 people in 20 cities around India in 2018 along 52 criteria. 91% of women felt that public transport was very unsafe or somewhat unsafe. Though gruesome incidents of violence against women have received attention in mainstream media, this data suggests that most women may face harassment such as verbal abuse, staring, groping, catcalls, whistling and molestation. When women use public transport, 96% prioritize affordability, coverage, frequency, safety, and comfort³¹.

The lack of safety in public transportation limits women's access to better educational and professional opportunities, ultimately reducing their quality of life as well as having a negative impact on the economic and social growth³².

In India, the challenge of ensuring women's safety using urban transport came to fore after the death of a woman in Delhi in December 2012, which is termed as 'Nirbhaya Case'. The Government of India created the Nirbhaya Fund between 2013–16 to implement schemes for improving women's security. Further, the Cabinet Committee on Economic Affairs has approved the setting up of a unified system at the national and state levels (City Command and Control Centre) for Global Positioning System (GPS) tracking through emergency buttons and video recording in public transport vehicles in 32 cities³³. It has been understood that majority of the funding by Government of India under the 'Nirbhaya Scheme' was spent on CCTV cameras for improved surveillance. However, the limited success-rate of such measures has brought out other specific measures to improve women's safety, such as better lighting at bus stops and terminals and gender-specific planning of infrastructure and services. It has been highlighted that there is a need to include women's issues within the general decision-making framework and not as an additional feature or cost item for the transit agency³⁴.

The Ministry of Road Transport has mandated all taxis, buses, and public transport vehicles (barring three-wheelers) to install GPS tracking and panic buttons mandatorily from 1st April 2018.

³¹ https://olawebcdn.com/ola-institute/ola_women_and_mobility.pdf

³² <https://unleash.org/solutions/asha-2/>

³³ https://www.researchgate.net/publication/322330719_Women_and_Transport_in_Indian_Cities

³⁴ <https://india.uitp.org/articles/women-in-public-transport-urban-mobility-india-2018>

3.3 Impact on Female Workforce Participation due to Public Transportation

According to a 2017 study by the Institute for Transportation and Development Policy (ITDS), women in urban India turn down better employment opportunities further away from home in favour of lower-paid local opportunities as the public transport system is often unreliable or unaffordable. In a developing country like India, women often rely on some form of public transport as their only choice of motorised transport. It is therefore important that they perceive public transport as being safe and secure.

In over two decades preceding 2013, female labour force participation in India fell from 34.8% to 27%.³⁵ As per a 2016 study by FIA Foundation,³⁶ women are more affected than men when access to employment, education or basic services are located far away from their residences.

Women are more dependent on public transport than men, especially when they are from lower-income groups, according to the ITDS report. Across cities, on average, 37% of women, compared to 27% of men, walk to work and 30% of women as compared to 25% of men use public transport.³⁷

3.4 Gender and Employment in the Transportation Sector

According to the International Labour Organization, transport is one of several sectors that has traditionally been regarded as having limited scope for women to participate in the workforce and largely considered a masculine profession. There is limited gender disaggregated data on the sectoral employment data which reflects that limited thought is being given towards improving the participation of women in the sector. The latest study from 2005 showed that 6.85% of women were employed in the transportation sector in India compared to 19% of men³⁸. According to a World Bank report³⁹, women constituted only 12.5% of the Brihanmumbai Electric Supply & Transport (BEST) Undertaking Committee and 1% of its engineers in 2010. When BEST attempted to induct female bus conductors, it was recorded that all of them requested to be shifted to desk jobs. Similarly, the Delhi Transport Corporation (DTC) has 245 women conductors and only one female driver⁴⁰.

Women face multiple challenges including accessing vocational training specific to transport sector jobs. For example, the requirements pertaining to driving licenses are passed on informally from older male family members to younger ones. Moreover, the upfront investment associated with training programs may be a barrier for resource-poor women. In DTC, drivers are required to hold a valid heavy motor vehicle license for at least three years and a public service vehicle (PSV) badge issued by the licensing authority, which is a major entry barrier. Additionally, when women are qualified for the jobs, insidious gendered assumptions regarding their suitability for the job can prevent them from breaking into the transport sector (Azad Foundation and University of Western Ontario 2014)⁴¹.

Harassment and threats by commuters and colleagues, and lack of public toilet facilities are additional barriers discouraging them to participate in the transport sector. Encouraging women in non-traditional occupations requires institutions to create a conducive environment. Simultaneously, women's presence at different levels in public transport authorities has the potential of mainstreaming gender within the organization by bringing women's issues to the fore in its services and infrastructure.

Under-representation of women in the transport sector leads to a low visibility of women's perspectives, and the differences in transport needs and patterns do not become apparent. Higher

³⁵ <https://www.indiaspend.com/why-indian-workplaces-are-losing-women-our-nationwide-investigation-begins-53927>

³⁶ <https://www.fiafoundation.org/media/224027/safe-and-sound-report.pdf>

³⁷ <https://www.indiaspend.com/in-indian-cities-women-travel-slow-men-in-a-rush-32171>

³⁹ A GENDER ASSESSMENT OF MUMBAI'S PUBLIC TRANSPORT. MUMBAI, JUNE 2011

⁴⁰ (Kaul and Shrivastava, Safety of Women in Public Spaces in Delhi: Governance and Budgetary Challenges 2017).

⁴¹ https://www.researchgate.net/profile/Sonal_Shah11/publication/322330719_Women_and_Transport_in_Indian_Cities/links/5b7f9f134585151fd12e7a0d/Women-and-Transport-in-Indian-Cities.pdf?origin=publication_detail

participation of women in the planning and design process is therefore increasingly necessary. A commitment to gender mainstreaming must include gender awareness and capacity-building programs for policy and decision makers to completely understand and appreciate gender-related issues⁴².

3.5 Good Practice Examples in Transportation Sector

The following provides some good practice examples in transportation sectors and various government interventions in gender and transportation. It should be noted that these interventions have been implemented directly by companies or governments and not by intermediaries (like the Platform) and so the activities cannot be copied by the Platform. However, these provide examples of where key issues have been identified for interventions.

Box 3.1 Good Practice Examples in Transportation Sector – Safetypin

Description: SafetiPin is a map based mobile phone application, designed to make communities and cities safer by providing safety related information collected by users and by trained auditors. At the core of the app is the Women's Safety Audit (WSA), a participatory tool for collecting and assessing information about perceptions of urban safety in public spaces. The audit is based on nine parameters:

- Level of lighting
- Openness (sight lines)
- Visibility (overhanging trees, high walls etc)
- Level of crowding or emptiness
- Security (agents)
- Walk/ footpath
- Availability of public transport,
- Usage of the public space (for instance is it male dominated or mixed) and
- Feeling/ emotional response to the space (some time also called “eyes on the street”).

Safetypin conducted audits to assess the safety of the city of Delhi since August 2013 and collected more than 7,000 data points that can be used by the city to improve the safety, such as replace lightings that no longer work, repairing broken footpaths and cutting overhanging trees.

Box 3.2 Good Practice Examples in Transportation Sector – Gender Sensitization Trainings, India

Gender Sensitization Trainings, India

Description: Sensitization for women's issues among public transportation employees as transportation plays an essential role for changing behaviour.

An example, as described by ITDP (2017), is the training conducted with the Delhi Transport Cooperation that includes role-plays as a basis for discussions about gender issues in transport, as well as After Trainings Support. Since 2012, a mental health organization in Delhi conducts those trainings and has by now trained over 100,000 transportation employees.

Box 3.3: Initiative by Government in India to Improve Public Transportation

- In Kerala, She-Taxi, the all-woman taxi service is like other taxi aggregators but only hires women drivers. The customer care centre tracks the vehicle with a GPS device. Police,

⁴² <https://thecityfix.com/blog/why-it-is-key-to-include-gender-equality-in-transport-design-jyot-chadha-vishal-ramprasad/>

relatives, and friends of the passenger can also monitor the cab using the system. There are separate switches in the car for drivers and passengers to send distress signals to the centre. The service is initiated by the state social justice department, which started it under Gender Park, a new institution launched by the department to empower women through livelihood creation. The drivers own their vehicles. (She Taxi 2013). Shimla City in Himachal Pradesh has new low-floor buses with each bus having three CCTV cameras. Two of them face two entrances from inside and one outside on the rear of the bus. The buses have been made more spacious with wider passageways, wider windowpanes, and lower entry/exit doors.

- In 2016, Maharashtra introduced the Tejaswini buses, which were ladies' special buses running during peak hours with women drivers and conductors
-

Box 3.4: Driving employment opportunities – Safeway Concessions

Safeway Concessions is currently a MAM-managed company acquired in 2018 by MAM investors from National Highways Authority of India (NHAI).

Safeway Concessions owns and operates a portfolio of 9 toll roads (687 kms) across the states of Andhra Pradesh and Gujarat in India.

Last year, it piloted a program to recruit women from local areas to work as toll collectors at toll plazas, a profession which has primarily been male dominated. Under the pilot program, more than 40 females have been employed who operate during day shift at three of the ten toll collection plazas. The initiative has played pivotal role in empowering local women both in terms of providing access to additional income and improving their livelihood.

In hiring this new demographic workforce, the company considered the following:

- Put in place an all-female shift organized around daytime hours for enhanced safety
 - Male toll collectors were redeployed to comparable jobs to create open roles for the daytime shift.
 - The all-female shift ensured the new hires felt appropriately supported by female colleagues in a largely male dominated environment.
 - New facilities (e.g., female bathroom and changing rooms) were put in place to accommodate the new workforce demographic and is now a standard facility in the designs of new toll plaza constructions for the company.
-

4. KEY AREAS OF INTERVENTION

Based on this Gender Assessment, the following areas for gender mainstreaming have been identified, taking into consideration:

- the key issues affecting women both generally and specifically within the e-mobility/ transport sector as outlined in this assessment;
- the intermediation role of the Platform; and
- the proposed investment sectors of the Platform.

These may be updated over time as the sector develops and the Platform is active in identifying further areas to incorporate gender mainstreaming.

Table 4.1 Key Areas of Intervention

Key issue identified in gender assessment	Description of issue or risk	Actions to address issues identified
Employment opportunities	Limited opportunities for women to participate in the growth of the e-mobility sector through increased access to employment opportunities within the sector	Provide employment opportunities at the Platform level to grow skills within the e-mobility financing industry that is set to grow and apply fair hiring practices.
		Start an internship program providing young women opportunities to gain experience in e-mobility financing space.
		Review relevant policies of counterparty including statutory (e.g. anti-discrimination) and company (e.g. D&I or relevant recruiting policies), and other D&I and gender related company practices.
	Limited sex-disaggregated data	Track relevant candidate gender data through application vs hiring process and identify/mitigate any bias if identified.
		Track skills gaps of applicants which can feed into the internship program design.
Gender-blindness, adequate transportation infrastructure to reach services (education, training)	Limited number of services designed for women incorporating women's specific needs	Identify opportunities to work with private bus companies or corporates which support women-focused services (such as women-only buses, school services etc), or working with the counterparties to develop initiatives.
	Women's specific requirements not adequately incorporated into project-design	Upfront gender assessment to identify whether there are opportunities to optimise the placement of charging points for increased use of EVs by women wherever applicable (e.g. considering well-lit areas, public areas).
		Periodic consultation with women's groups (e.g. women drivers associations, self-help groups for women from low

Key issue identified in gender assessment	Description of issue or risk	Actions to address issues identified
		income background, NGOs working on women employment) to identify potential risks and opportunities.
	Limited sex-disaggregated data to improve gender sensitivity in the sector or sub-sectors	Design robust plan for collection of relevant sex-disaggregated data for each sub-project on transport usage, and collate the data to understand areas of improvement. <i>(Note: sex-disaggregated data will be collected for relevant sub-projects relating to use of vehicles. However, as the sub-projects have not yet been identified, the actual data to be collected will be finalised as part of the investment approval of each sub-project. Any collection of sex-disaggregated data privacy regulations including collection and storage of sensitive data, as relevant and required)</i>
Gender Based Violence and Harassment (GBVH)	Facilities around transport hubs and vehicles (e.g. poor lighting, unsupervised spaces, absence of emergency buttons, etc.)	Conduct prior due diligence of counterparties to review their existing policies and procedures for management of GBVH issues, including existing education programs, safety measures, trainings, and grievance mechanisms.
		Require projects to include adequate design and maintenance of lighting and CCTV cameras (or other relevant safety features at charging infrastructure points).
		Require reporting/representation from counterparties that the vehicles are aligned with safety requirements as mandated by authorities/relevant concession agreements.
		Reporting of GBVH incidents for vehicles financed by the Platform.
		Be involved in stakeholder consultations by public and private entities around the design of concessions or vehicles to provide views and data on relevant and improved safety features. Target should be set on number of consultations conducted per year on gender aspects.
	Behavioural aspects (e.g. physical violence and sexual harassment, lack of clear codes of conduct for employees, etc.)	Require counterparties to install surveillance cameras and adequate security measures.
		Review training programs periodically delivered by counterparties to drivers and transport workers on gender sensitivity and how to report and refer incidents of GBVH. Where counterparties do not have existing training programs, offer relevant training for drivers and other employees on gender sensitivity and how to manage incidents of GBVH.

Key issue identified in gender assessment	Description of issue or risk	Actions to address issues identified
		Use transport infrastructure such as signage and tickets to disseminate messages and create awareness around GBVH more broadly in communities and public spaces. This will be undertaken through counterparties.
Women's involvement in decision making	Due to societal and cultural practices, women are less likely to be involved in decision making which may exacerbate gender blindness in project design	Promote women's involvement at a corporate level: ■ Platform Board composition ■ Monitor Board composition of counterparties Ensure women and vulnerable groups have an active role in project and sector level stakeholder consultations.
Need for capacity building and knowledge dissemination for gender considerations in the e-mobility sector	Awareness of gender mainstreaming may be limited in the transportation, e-mobility and financing sectors	Opportunities for capacity building at both the Platform and for counterparties through trainings and workshops to increase awareness of gender sensitivity. Promote multi-stakeholder discussions to integrate gender considerations across the sector.
	Limited availability of sex-disaggregated data in the e-mobility space	Ensure each sub-project collects sex-disaggregated data which can be used to identify risks, mitigants and opportunities.
	Knowledge sharing of lessons learned is limited	Promote examples of lessons learned through data collection or sub-project implementation through thought leadership or otherwise to contribute to developing best practice.

APPENDIX A GENDER ASSESSMENT OF SUB-PROJECTS

The Platform will undertake specific gender assessment of sub-projects as part of the counterparty and sub-project due diligence (included in ESMS) prior to financing. The gender assessment will include review against equal access and benefits, identifying risks and mitigants, identifying opportunities for gender mainstreaming, what sex-disaggregated data will be collected and assessment of the counterparty.

During screening and due diligence gender-related data will be collected and analysed to determine whether the sub-project qualifies under the Platform's gender policies and identify business performance gender opportunities and risks. Gender-related data and information will be gathered from the potential counterparty for analysis to determine whether gender gaps present opportunities that may impact performance and affect the sub-project operations and financials.

The following checklist is to be completed by counterparties during the due diligence stage. Its aim is to enable the Platform to understand the current level of internal gender diversity and inclusion, to identify initiatives that support women entrepreneurs, and acknowledge efforts to serve women clients and users.

For the avoidance of doubt, any data collection, storage and utilisation of sex-disaggregated data will be compliant with local data privacy laws.

Table A1 Gender specific due diligence questionnaire

S.N	Parameter	Questions	Response	Remarks
1.	Leadership	Does the board and/or senior management have strong oversight and a clear governance process to support gender diversity and inclusion? Are any initiatives in place to promote women in leadership?	Yes ☐ No ☐	
		What is the gender composition of the Board?		
2.	Workforce	Are there specific policies and programmes to support inclusiveness of both men and women at work? If so, what are they? For example, specific HR policies that go beyond basic regulatory requirements (e.g. childcare support, flexible work, gender targets for construction and maintenance, standardised pay rates for each role).	Yes ☐ No ☐	
		Building on recent and/or existing data, are there differences between the roles/responsibilities/functions/levels/grades undertaken by men and women in the workforce, including permanent, contracted, seasonal staff as well as contractor construction and maintenance staff?	Yes ☐ No ☐	
		Does the company collect and monitor gender-disaggregated staff data (e.g. average salary, turnover, absenteeism, retention, and promotion)? How is the data analysed and to what extent is data used	Yes ☐ No ☐	

S.N	Parameter	Questions	Response	Remarks
		for decision-making on gender-related efforts (e.g. to better serve women clients)?		
		What training or skills development schemes are in place to address existing or emerging skills gaps?	Yes ☐ No ☐	
3.	Gender based violence and harassment governance	Has the company identified GBVH risks and are these reflected in the company's risk management system? Has GBVH been reflected in company policy, including a clear definition of GBVH (either as a separate policy or integrated into wider policies)? Has the company developed clear codes of conduct that prohibit all forms of GBVH? Are GBVH policies and codes of conduct available to workers and external stakeholders in accessible formats? Is mandatory training on GBVH and company policies and procedures provided to all workers of all grades, including contractors and security personnel, as well as induction training for new recruits?	Yes ☐ No ☐	
4.	Grievance mechanisms and reporting	Are confidential grievance mechanisms in place where reports and concerns related to GBVH can be raised? Are accessible reporting channels available to workers, community members and service users, including anonymous options? Are procedures in place to respond to reports of GBVH when they are made, including clear investigation procedures that focus on the safety and wellbeing of survivors? Have trained investigators been identified? Have reports of GBVH been made and how have they been handled? Is additional specialised training provided to members of staff with specific responsibilities for GBVH prevention and response?	Yes ☐ No ☐	

S.N	Parameter	Questions	Response	Remarks
		Has company leadership publicly committed to taking reports seriously?		
5.	Equal opportunity	Are recruitment procedures in place, with interview panels staffed by at least two people? Are candidates' identities checked at interview and are references requested? Are written procedures in place for performance appraisals, promotions and any performance-related pay (if applicable)?	Yes ☐ No ☐	
6.	Project design	Have gender issues been identified as part of the project development? Have these issues been incorporated into project design and implementation plan?	Yes ☐ No ☐	
7.	Safety features and requirements	Have all required and safety features been incorporated in to the EVs, charging infrastructure or project design?	Yes ☐ No ☐	
8.	Stakeholder consultations	Have women and vulnerable people been involved in the stakeholder consultation process for this project? Have women had the opportunity to raise potential issues in the project's design and implementation?	Yes ☐ No ☐	
9.	Dedicated services	Does the company offer any community engagement programme supporting women's adoption and use of the company's infrastructures or services? Has the counterparty identified any opportunities to provide services dedicated to supporting women or gender mainstreaming?	Yes ☐ No ☐	
10.	Procurement	Are clauses included in contracts committing contractors and suppliers to adhere to company codes of conduct? Is information provided to contractors and suppliers about company grievance mechanisms for reporting GBVH? Is GBVH included in workplace safety assessments, including worker accommodation and transportation?	Yes ☐ No ☐	

S.N	Parameter	Questions	Response	Remarks
11.	Products/ services	Does the company collect and monitor gender-disaggregated data on demand, service usage and/or customer segmentation? For example, related to access, availability, affordability, feasibility, and satisfaction.	Yes ☐ No ☐	
		If demand, service usage and/or customer segmentation data is collected, how is this data analysed and to what extent it is used for decision-making related to planning, design, marketing, sales, pricing, service(s) and distribution?	Yes ☐ No ☐	

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Gender Action Plan for E-Mobility Financing Program

Final Report

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Macquarie Infrastructure & Real Assets
India Pvt. Ltd.

Signature Page

13 April 2022

Gender Action Plan for E-Mobility Financing Program

Final Report

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1. INTRODUCTION

Macquarie Alternative Assets Management Limited (hereinafter referred to as '**MAAML**' or '**AE**'), has developed a Funding Proposal ('**FP**') for submission to the Green Climate Fund ('**GCF**'), to seek their investment in a dedicated electric mobility (e-mobility) program focusing on India (hereinafter referred to as the '**Program**'). AE is part of Macquarie Asset Management ('**MAM**'), one of the 4 operating business groups of Macquarie Group.

MAM is a global specialist asset manager with assets under management at 31 December 2021 of AUD 750.1 billion, aiming to deliver positive impact for everyone. MAM provides investment solutions to clients across a range of capabilities, including infrastructure & renewables, real estate, agriculture & natural assets, asset finance, private credit, equities, fixed income and multi-asset solutions

ERM India Private Limited (ERM) has been appointed by MAM, to conduct a Gender Assessment Study and to develop a Gender Action Plan. The Gender Assessment Study and Gender Action Plan for the Program forms part of GCF's requirements for all Funding Proposals. This document presents the **Gender Action Plan**.

1.1 A brief overview of the 'Dedicated E-Mobility Program'

MAM is looking to partner with GCF and other co-investors to establish an India-focused electric mobility program to facilitate decarbonisation of the road transport sector through rapid scale-up of the electric vehicles sector in the country.

The Program will look to provide financing to private sector counterparties across the e-mobility value chain (BEV and FCEV) such as e-buses, 2/3/4 wheeler segments, supporting infrastructure & services like charging infrastructure, vehicle scrapping and after-sales services.

The Program will comprise of three main activities as outlined in the Funding Proposal presented to the GCF:

- 1) Establishment of a financing platform in India (hereinafter referred to as the '**Platform**'), including its legal entity, policies, procedures and broader governance.
- 2) Providing financing by the Platform to corporate counterparties in the e-mobility space, which involves upfront due diligence and structuring as well as ongoing monitoring.
- 3) Sale of the Platform (to facilitate GCF and other investors exit) with the expectation that the Platform will continue to operate in the market.

For the purposes of this Program, 'sub-projects' will refer to each transaction at the Platform level where a commitment of financing is made to a counterparty in the e-mobility space. Please see the Gender Assessment Report for more details.

1.2 Objective and Scope of work

The objective of the assignment is to undertake a Gender Assessment and formulate a Gender Action Plan for the Program with consideration of its targeted sectors. The scope of the exercise includes:

- assessment of general social baseline conditions in India in which the Platform will operate;
- focused interventions that are in practice for the transport and e-mobility sectors from a gender perspective;
- identification of areas that can be best targeted by the Platform for gender mainstreaming activities; and
- formulation of a Gender Action Plan for the Program that highlights key performance indicators and timelines.

The reference framework for the exercise was taken to be the GCF Guidance Note (Form 09 updated 1 February 2019).

1.3 Applicable Reference Framework

The applicable reference framework for the assignment is presented below:

- applicable national legislation and regulations pertaining to social and gender aspects;
- GCF Manual on Mainstreaming Gender in Green Climate Platform Projects¹; and
- GCF Gender Toolkit and Guidance Note (Form 09).

1.3.1 Need for Gender Assessment and Gender Action Plan

Gender mainstreaming is central to the GCF's objectives and guiding principles. The aim is to address it continually by engaging women and men of all ages as stakeholders in the design, development and implementation of strategies and activities to be financed. The GCF Governing Instrument states that: "The Fund [GCF] will strive to maximize the impact of its funding for adaptation and mitigation promoting environmental, social, economic, and development benefits, taking not only a gender-sensitive, but a gender responsive approach."

Gender mainstreaming, as defined by the United Nations Economic and Social Council, refers to a strategy for making women's, as well as men's, concerns and experiences integral dimensions of policies and programs in all political, economic and social spheres. Gender mainstreaming is essential to ensure that women and men benefit equally from a certain system, and inequality is not perpetuated.

There is still a heavy linking of the term "gender" with "women's issues"; however, the evolved understanding on gender and gender mainstreaming is that Gender represents equality of both the sexes.

However, GCF has established on the basis of prevalent practices and past experiences that women and girls are disproportionately affected by the impacts of climate change and associated severe weather events. Women's mortality from climate-related disasters is higher than that of men, and climate-induced impacts due to natural disasters like floods and earthquakes exacerbate women's poverty and unpaid domestic and care work burdens. Gender inequality, exacerbated by climate change, is linked, as are other development areas, to vulnerability and risks. The greater vulnerability of women to climate change stems from societal norms and discrimination that result in the imbalanced division of labour, lower incomes, and lesser livelihood opportunities; less access and control over land and other productive assets; fewer legal rights; lesser mobility and lesser political and professional representation.

1.3.2 GCF's Gender Policy

The GCF Board adopted the Gender Policy and Action Plan in March 2015 and an update to this policy was issued in February 2019, to respond to a fast-changing landscape related to gender in the context of climate change. Its rationale is rooted in the GCF's mandate to encourage a paradigm shift towards low-emission and climate-resilient development pathways. The updated policy emphasizes gender responsiveness rather than gender sensitivity. Being gender-responsive means that instead of only identifying gender issues or ensuring a "do no harm" approach, a process will substantially help to **overcome historical gender biases**. Gender responsive approaches are essential to this, including decisions and actions by women and men.

GCF's Gender policy aims to:

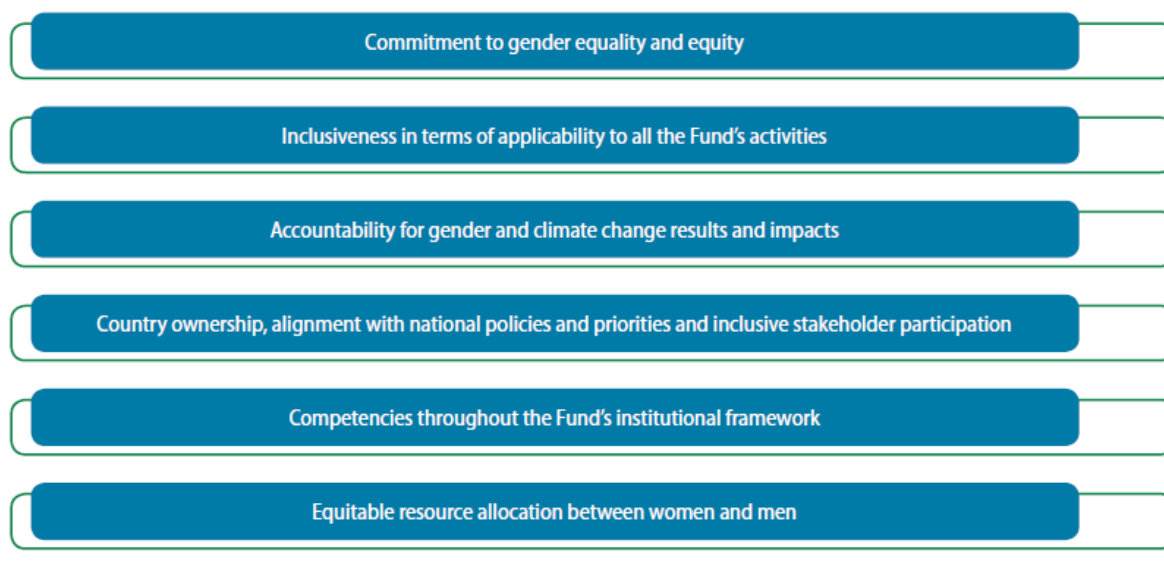
¹ Published in 2017

- Achieve greater, more effective, sustainable, and equitable climate change results, outcomes, and impacts, through the adoption of a gender sensitive approach
- Build resilience to climate change equally for men and women, as well as to ensure that men and women equally contribute and benefit from activities supported by the Fund
- Address or mitigate risks for women and men associated with adaptation and mitigation activities financed by the Fund; and
- Reduce the gender gap of social, economic, and environmental vulnerabilities exacerbated by climate change.

For the purposes of this Gender Equality Policy, GCF requires that women and men, including marginalized and vulnerable groups, will be provided with an equal and equitable opportunity to be fully and effectively engaged in meaningful consultations and decision-making throughout the project cycle.

The GCF's Gender Policy is based on the following six Fundamental principles.

Figure 1.1 Principles of GCF's Gender Policy



Source: Green Climate Platform Proposal Toolkit 2017

The GCF's paradigm shift is rooted in the acknowledgement of women's critical role in climate responses, as agents of change sharing in the benefits, and, in several cases, disproportionately affected by its negative impacts. Marginalised communities also bear disproportionate negative outcomes and are also included in this shift.

According to GCF, "Gender assessment refers to methods used to understand relationships between men and women, their access to resources, their activities, and the constraints they face relative to each other. A comprehensive gender analysis/assessment entails the examination of the different roles, rights, needs, and opportunities of women and men, boys and girls in a given project/context. Gender analysis needs to take place early in the planning process so that an understanding of gender roles and power relations is built into the project."

1.4 Methodology

In order to achieve the above-mentioned scope, the following methodology has been adopted.

1.4.1 Kick off call with MAM

ERM organised an Inception call with the Client to understand the current investment context, get a general understanding of the planned working of the Platform and expected investments. ERM also discussed the role of MAM in the Program as the investment manager, applicability of an ESMS, a Gender Policy or any other ongoing/proposed interventions that will be undertaken by the Program that may particularly focus on addressing gender related aspects.

1.4.2 Detailed Gender Assessment

A desk-based research of aspects related to women in India and specially in urban mobility has been undertaken to understand the nature and extent of data and studies capturing aspects like female urban ridership, challenges faced by urban females due to limited mobility and while using public transport, general scenario of employment of women and specifically in the urban transportation sector.

1.4.2.1 National Context guided by GCF Guidance Note

An assessment of the nationwide gender related baseline scenario and associated impacts, risks, opportunities and trends in this sector was undertaken using secondary data.

Key steps followed for the assessment are:

- Profiling the social and development status of women and men (education, literacy, work force participation, unemployment, role in formal/informal economy, division of labour, access to resources and opportunity, decision making roles and capacity, as relevant) based on secondary data at the national level
- An assessment of the legal status, beliefs, perceptions, prevalent gender stereotypes, as relevant for the key sectors
- An assessment of relevant issues like gender-gap, inequality, gender-specific vulnerabilities (with respect to climate change adaptation, social, socio-economic factors, technical capacity, and resource access etc.) has been undertaken based on relevant secondary literature and published research.

Please note: In the absence of latest data on gender issues such as employment and literacy in the transport sector? the report has relied on Census of India, 2011 data and has supported the findings, wherever possible with other latest available studies.

1.4.2.2 Establishing a Baseline Scenario for Gender Mainstreaming in E-mobility

The assessment has sought to identify a baseline scenario for gender issues in the e-mobility sector and ultimately identify where the Program could best implement gender-based interventions.

Given the lack of data on gender and e-mobility sector, especially in developing countries, the report has relied on secondary review, data and analysis available for gender and transportation sector in general. It should also be noted that there is a general lack of gender disaggregated data on urban mobility and hence also in the e-mobility sector.

Consequently, there is little available research on strategies, interventions and case studies on specific preferences (of women and men, young and old, or those with disabilities) regarding urban public transport usage, sector employment trends, health and safety, ergonomics, or accident risks. Sex-disaggregated data has been provided where available.

Whilst this is one limitation of the Gender Assessment, collecting sex-disaggregated data is also an identified area for improving gender sensitivity in this sector.

1.4.2.3 Stakeholder Engagement

Specific stakeholder engagement as part of this gender assessment has not been completed. This is primarily because the proposed Platform has not yet been established and the broad scope of the proposed transactions involve multiple e-mobility sectors and are not based in a specific region in India. It is proposed that targeted, gender-focused stakeholder engagement is completed once the Platform is established to build upon the desk-based assessment and interventions outlined in the Gender Action Plan. Please refer to the Gender Action Plan for more details.

1.4.3 Formulation of the Program-level Gender Action Plan

The Gender Action Plan, as presented in this document, has been formulated to include strategies and action items identified for the Platform. The action plan includes key indicators and timelines, for effective monitoring of outcomes.

1.4.3.1 Limitations in development of Action Plan

- The Platform is not yet established, including the defined policies, procedures and governance. Actions have been identified to integrate gender mainstreaming frameworks as part of this establishment.
- There is limited ability to set baseline scenarios given the lack of data available in the sector, the broad mandate across sectors and geography, and the Platform is not yet established so there is no benchmarking applicable of its existing practices. Activities have been outlined to support establishment of baseline.

2. GENDER ACTION PLAN

This Gender Action Plan has been developed based on understanding developed through review of existing baseline scenario in India on gender related aspects, study of relevant research done in the field of urban public transport, especially from a gender perspective and discussions with MAM.

The Action Plan is at Platform level and covers components of the program at high-level. As the Program is still in proposal stage and the electric urban mobility sector is also set to evolve rapidly from its current, nascent stage, there will be need for measures to adapt and lessons to be incorporated efficiently in subsequent stages.

As a strategy, where secondary research has brought out the issue of lack of data on useful aspects, the Action Plan recommends generation of good quality data to make subsequent phases and projects in this area more impactful.

2.1 Implementation of Gender Action Plan

The overall outcome of the Gender Action Plan is to develop greater knowledge of gender sensitivity and mainstreaming activities within the e-mobility sector, including greater skills developed by Platform staff, and improved attitudes towards integration of gender-related interventions by other stakeholders within the sector.

The Plan will be implemented based on the timelines identified in **Table 2.3**.

2.1.1 Impact & Outcome Statement

Impact Statement:

E-mobility Financing Program aims to leverage the GCF and commercial capital to accelerate decarbonisation of the Indian economy by focusing on electrification of the road transport sector. This will be enabled by supporting faster adoption of electric vehicles and growth of associated infrastructure in the country. The focus of the program will be to provide capital to private sector business opportunities across the e-mobility value chain such as e-buses, 2/3/4 wheeler segment, support infrastructure & services like charging infrastructure.

The Program aspires to:

- Creation of new jobs and stimulate economic activity with a corollary of empowering women and reducing gender bias.
- Enhancement of economic activities, livelihoods, and empowerment of women through increased generation of participatory and inclusive employment opportunities.
- Improved access to safer transportation options for women due to design, adoption of inclusive and gender-responsive public transport systems and related solutions.

Outcome Statement:

Through the current proposed systems and mechanisms, the Platform will seek to contribute to achieving the following gender-responsive outcomes with gender-disaggregated sub-project level impacts:

- Formal employment and improved livelihoods for women through generation of jobs by the Platform and counterparties at sub-project level line in with the growth of the e-mobility.
- Empowerment of local communities, including women and girls, through training, awareness, and capacity building, to build more gender-inclusive and responsive sub-projects (transportation systems).

- Building self-reliance among women and girls through increased use of public transportation by addressing safety concerns around using mass transport options and improving 'gender blind' transportation systems.
- Reductions in gender violence, increase gender equity, and economic activities for women and girls through the Platform Level Environmental and Social Management System (ESMS), mandatory sub-project safeguards, gender assessment and action plan, and gender-responsive project designs.

2.1.2 Definition of Sub-Projects and Counterparties

The Platform will provide financing to counterparties in the e-mobility space (also referred to as a 'transaction') across the value chain including buses, four-wheelers, two- and three-wheelers, charging infrastructure and others. Key characteristics of the sub-projects include:

- physical assets in the e-mobility space are involved (e.g., financing a set number of electric buses).
- vehicles or charging points will be operated by the counterparty.
- the relationship between the Platform and the counterparty will be governed by a financing agreement that will place certain obligations on both parties.
- vehicles financed may not be dedicated to a specific route but rather used by the counterparty generally. This may limit site specific diligence.

For the avoidance of doubt:

- the Platform is designed to be a financing intervention. Sub-projects will not involve the design or development of broader transportation interventions except (where relevant) for providing input on the e-mobility sector development from a financing perspective (outside the scope of a sub-project).
- The Platform will not be involved in the day-to-day operation of vehicles or charging infrastructure.
- the Platform will not be involved in the design or manufacturing of vehicles.

Given the financing nature of the Platform it will not be responsible for the primary development and implementation of each sub-project. Instead, the Platform will provide financing (governed by financing documentation such as lease or loan contracts) to Counterparties, defined as entities who will be responsible for the day-to-day operation of sub-projects.

Examples of counterparties include:

- A bus operating company which is bidding for a new electric bus tender issued by a state government for the provision of new e-buses to service an urban area,
- A private bus company operating a route between two major cities,
- A company constructing a number of new charging points to support the increased penetration of e-vehicles in a new area,
- A large corporate company providing a fleet of cars to their employees.

2.1.3 Limitations

- Baseline data: There is no baseline provided as most activities require the Platform to be established. For other activities, there is limited data available in the sector and the mandate of the program is broad in geographic and sector scope across e-mobility making data available very general and not context specific. There are activities in the action plan which relate to increasing data availability and, where applicable, setting sex-disaggregated baselines and targets. For the avoidance of doubt, any data collection, storage, and utilisation of sex-disaggregated data will be compliant with local data privacy laws.

- **Stakeholder engagement:** The Action Plan is designed based on the desk-based Gender Assessment. It is intended that stakeholder engagement will be undertaken once the Platform is established basis which further actions may be included in the Plan.
- **Intermediary role:** As a financing intervention, the Platform will not be able to directly implement activities regarding the operation of the electric vehicles. However, this is mitigated through upfront diligence and structuring by the Platform as part of a transaction and by activities flowed through to the financing counterparty.
- **Sub-Projects not yet defined:** The Platform has not yet identified specific sub-projects that will be financed by the Platform given it is not yet established. As a result, actions for sub-projects cannot be fully defined at this stage (such as gender assessment for each sub-project for which sex-disaggregated data will be collected).
- **Existing skills, attitudes and practices:** The impact of some activities may be impacted by the existing context which may limit the outcomes. For example, there may not be sufficient number of women skilled or interested in roles in the e-mobility sector due to cultural attitudes towards women working in male-dominated sectors. Whilst the Action Plan can implement activities to promote more opportunities and equality, the outcomes may be impacted.

2.1.4 Roles & Responsibilities

The primary responsibility of implementation of the Gender Action Plan is with the Gender Specialist who will report to the Platform CEO and Board. However, other stakeholders will be involved in different parts of the Action Plan. This is outlined at a high level below however responsibilities for specific actions are outlined in **Table 2.3**.

Table 2.1 GAP Implementation – Roles and Responsibilities

Role	Responsibility
Gender specialist	■ Lead implementation of Gender Action Plan within Platform staff
Accredited Entity	■ Review and monitor the implementation of the Gender Action Plan through reporting received from the Executing Entities (including Platform). ■ Review progress against targets and any significant updates to the Gender Assessment or Action Plan
Investment Manager (MAM) and Macquarie India Advisory co (IAC)	■ Provide guidance with expertise from MAM where required to develop and implement best practices based on experience in other jurisdictions, sectors or transactions
Platform Board	■ Responsibility for causing the implementation of the Gender Action Plan ■ Responsibility for establishing and maintaining governance frameworks ■ Monitor implementation of Gender Action Plan from management reporting
Financing counterparties	■ Financing counterparties will be required to meet obligations in their financing documentation, including those that relate to gender-mainstreaming interventions and reporting obligations outlined in this Gender Action Plan

2.1.5 Costs & Budgets

The budget required to support implementation of the gender action plan is based on indicative estimates summarized below. These are included in the financial analysis of the Programme.

Other items in the Action Plan are considered to be transaction specific costs (e.g. environmental and social assessments, gender assessments) will be borne by the financing counterparty in line with other transaction costs (such as legal fees etc.). These will be included once the sub-project level plans have been concretized and activities have been identified.

Table 2.2 Indicative Budget for Implementation of GAP

SI No.	Cost Head	Budget (INR)	Budget (USD)
1.	Employment of Gender specialist, including time spent on various activities outlined in Gender Action Plan	50,00,000 p.a.	66,667 p.a.
2.	Capacity building training programs on annual basis (for 10 years) for Platform staff and counterparties	20,00,000 p.a.	27,000 p.a.
3.	Stakeholder engagement, consultations, survey, data collection and analysis at Platform and sub-project level on annual basis	30,00,000 p.a.	40,500 p.a.
4.	Internship program (including on-costs and training)	24,00,000 p.a.	32,000 p.a.

2.1.6 Grievance Mechanism

An activity level grievance mechanism will be established once the Platform is established. The means of access will be published on the GCF and AE's websites.

Grievance mechanisms are also available through the Accredited Entity and GCF:

Accredited Entity	Macquarie Integrity Office https://www.macquarie.com/au/en/disclosures/whistleblower-policy.html integrityoffice@macquarie.com
Green Climate Fund	File a complaint Independent Redress Mechanism Green Climate Fund

#	Activity	Actions	Indicators	Targets	Timeline	Responsibility
1.1	Due diligence of gender-related practices of financing counterparty	Review relevant policies of counterparty including statutory (e.g. anti-discrimination) and company (e.g. D&I or relevant recruiting policies), and other D&I and gender related company practices.	Number of counterparties with policies meeting required practice	100% of counterparties	Assessment completed before each sub-project (see 6) Investment Procedures of this action plan)	Gender specialist
1.2	Capacity building of financing counterparties	Provide capacity building workshops to counterparties to identify ways to incorporate gender considerations into company policies, practices, product design and operations.	Percentage of counterparties involved in workshops provided	All major counterparties to attend the workshop at least once during their relationship with the Platform	Capacity building workshops offered annually to counterparties	Gender specialist
1.3	Platform hiring	Fair and gender inclusive hiring practices applied in line with MAM and GCF's policies, and learnings captured to promote further actions.	Sex-disaggregated candidate data on applications vs. hired to identify/mitigate against any bias but also identify any skills gaps	Metrics captured on 100% of candidates	First data reporting to Board within 2 years of Platform establishment (to allow for data to be collected)	Gender specialist and Platform Management
			Percentage of women hired at the Platform	25% of staff employed at the Platform are women	Within 5 years of Platform establishment	Gender specialist and Platform Management
1.4	Skills development	Implement an internship program providing young women opportunities to gain experience in the e-mobility finance space. Use data collected on hiring above to support focus areas for the program.	Number of interns completing the program	At least five over five years (1/year), 100% women	5 years	Gender specialist

#	Activity	Actions	Indicators	Targets	Timeline	Responsibility
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2) Gender Blindness in Transportation infrastructure

Impact: Contribute towards improving gender sensitivity in project design such that transportation the e-mobility sector reduces gender-blindness and promotes gender mainstreaming. Projects to contribute to providing adequate and safe infrastructure to reach essential services (education, health and training) to improve women's participation in society.

2.1	Women focused services	Identifying opportunities to work with private bus companies which support women-focused services (such as women-only buses, school services etc), or working with the counterparties to develop initiatives.	Number of initiatives supported across e-mobility segment	2 dedicated initiatives (each initiative likely to involve multiple vehicles)	Within first 4 years of Program implementation	Gender specialist
2.2	Optimising charging infrastructure locations	Upfront gender assessment to identify whether there are opportunities to optimise the placement of charging points for women's use of EV (e.g. considering well-lit areas, areas visited most by women, institutions for women, etc.)	Proportion of charging infrastructure projects with gender considerations identified and incorporated into project design	100% of sub-projects in the charging infrastructure sub-sector	Gender assessment completed prior to investment approval of sub-project	Gender specialist

#	Activity	Actions	Indicators	Targets	Timeline	Responsibility
2.3	Consultation with women's groups	Periodic consultation with women's groups (e.g. women drivers associations, self-help groups for women from low income background, NGOs working on women employment) to identify potential risks and opportunities in the e-mobility space and reduce gender blindness in project design. These consultations can also be held relating to specific sub-projects in coordination with counterparties, during project planning stage.	Meeting or other communication with women's groups	1 / year with each of at least 4 women's groups with a national remit / focus (and connections to regional and other organisations), plus additional consultations with regional and/or sector focused groups as determined by sub-projects	Engagement with national groups within 6 months of Platform establishment	Gender specialist
2.4	Involvement in regulation and policy development	Platform to use knowledge and learnings from due diligence, data collection and sub-projects to be involved in stakeholder consultations driven by governments for regulations or new tenders in the e-mobility space to include gender-specific actions such as highlight the need for the safety features to be included or scaled-up in public bus concessions and the need for the design of transport routes to incorporate gender sensitivity (including collection and release of data on women's transport patterns).	Gender specific issues identified when responding to broader stakeholder consultations	100% responses developed by Platform to include gender-specific issues	Ongoing throughout the Platform life	Gender specialist

#	Activity	Actions	Indicators	Targets	Timeline	Responsibility
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3) Gender-based violence and harassment (GBVH)

Impact: Improve and maintain e-mobility projects and vehicles to mitigate against gender-based violence and harassment, particularly on public transport and in public spaces through implementing safety features and building capacity and awareness amongst stakeholders. Increased safety should lead to greater use of public transportation amongst women and girls which build self-reliance.

3.1	Standards of counterparties	Conduct prior due diligence of counterparties to review their existing policies and procedures for management of GBVH issues, including existing education programs, safety measures, trainings and grievance mechanisms.	Number of counterparties with policies meeting best practice	100% of counterparties	Assessment completed before each sub-project investment approval (see 6) Investment Procedures of this action plan)	Gender specialist, Platform staff
3.2	E-vehicles meet safety standards aiming to combat GBVH	Require reporting/representation from counterparties that the vehicles are aligned with safety requirements as mandated by authorities/relevant concession agreements, such as surveillance cameras, panic buttons etc.	Reporting or representations received from counterparties that the safety requirements are in place. Requirement to be included in financing documentation.	100% vehicles financed by the Programme to be in compliance with safety features annually	Annually	Counterparties to provide reporting Platform management to monitor compliance

#	Activity	Actions	Indicators	Targets	Timeline	Responsibility
3.3	Safety features incorporated into project design	Request information from counterparties as part of upfront due diligence on how safety features which address GBVH have been incorporated in the project design. For example, require projects to include adequate design and maintenance of lighting and CCTV cameras (or other relevant safety features) at charging infrastructure points)	Sub-project gender assessment including details on features addressing GBVH.	100% relevant sub-projects to incorporate adequate safety features	Assessment completed before each sub-project investment approval (see 6) Investment Procedures of this action plan)	Gender specialist, counterparties involved in project design
3.4	Training and capacity of employees of counterparties	Review training counterparties deliver periodically to drivers and transport workers on gender sensitivity and how to report and refer incidents of GBVH. Where counterparties do not have existing training, offer relevant training for drivers and other employees on gender sensitivity and how to manage incidents of GBVH.	Training provided to employees of counterparties.	100% counterparties to provide or be provided with training for employees on gender sensitivity and GBVH at least once.	Once in the lifetime of a relationship with the counterparty	Gender specialist
3.5	Reporting on incidents	Annual reporting of GBVH incidents for vehicles financed by Platform.	Reporting covenant included in financing documentation	100% compliance from counterparties to provide reporting	Annually for each sub-project during financing term	Counterparties to provide reporting Platform management to monitor compliance
3.6	Reporting on usage of safety features	Where possible and relevant, require the collection of data on usage of panic buttons or other safety features.	Reporting covenant included in financing documentation	100% compliance from counterparties to provide reporting	Annually for each sub-project during financing term	Counterparties to provide reporting Platform management to monitor compliance

#	Activity	Actions	Indicators	Targets	Timeline	Responsibility
3.7	Policy development (combined with 2.4)	Be involved in stakeholder consultations by public and private entities around the design of concessions or vehicles to provide views and data on relevant and improved safety features.	Gender specific issues identified when responding to broader stakeholder consultations	100% responses developed by Platform to include gender-specific issues	Ongoing throughout the Platform life	Gender specialist
3.8	Public awareness	Use transport infrastructure such as signage and tickets to disseminate messages and create awareness around GBVH more broadly in communities and public spaces.	Materials disseminated in vehicles or infrastructure financed by the program	25% of relevant vehicles financed by the Platform	Reach target within 5 years of Platform establishment	Gender specialist, financing counterparties to support implementation

4) Women's involvement in decision making

Impact: More women in decision making roles in the e-mobility space and involved in project design to increase gender sensitivity into the sector and specific projects

4.1	Governance of the Platform	Women to be involved in key decisions of the Platform	Sex-disaggregated data on Board of the Platform	At least 1 at all times from Board establishment	From inception	Accredited Entity
4.2	Governance at financing counterparties	Requesting the split of women / men in Board in the organisation.	Reporting of sex-disaggregated data on Board of each financing counterparty included in financing documentation	100% of financing counterparties have at least 1 woman on their Board	Within 5 years of financing provided	Counterparties to provide reporting

#	Activity	Actions	Indicators	Targets	Timeline	Responsibility
4.3	Stakeholder consultation	Require stakeholder engagement activities at project level through project information meetings, ESIA consultations, etc., with records of attendees maintained by gender and ensure that consultations by counterparties, reports voices risks and benefits of men, women and vulnerable groups.	Stakeholder engagement summarised for each sub-project	Involvement of 50% women and vulnerable groups in each stakeholder consultation	Prior to approval of each sub-project	Gender specialist

5) Capacity Building & Knowledge Dissemination

Impact: Increased awareness of gender-based issues through capacity building, increased sex-disaggregated data to develop potential learnings and opportunities. Promote gender mainstreaming within the e-mobility space amongst all stakeholders.

5.1	Platform staff capacity building	Mandatory training on implementing of gender policy, gender mainstreaming and sensitivity.	Training module developed and disseminated	100% of employees attend a relevant training	Within 12 months of setting up the Platform	Gender specialist
5.2	Capacity building of counterparties	Platform to conduct in-house workshops offered to all counterparties to develop sensitivity and responsiveness policies and procedures to identify actions they could take within their business, especially for smaller counterparties.	Training module developed and disseminated	100% of counterparties to attend a training during life of their financing	During Platform life, first training delivered within 12months of first transaction	Gender specialist

#	Activity	Actions	Indicators	Targets	Timeline	Responsibility
5.3	Increase data available	Design robust plan for collection of relevant sex-disaggregated data for each sub-project on use of the transport, and collate the data to understand areas of improvement. This could include data on travel preferences, usage of transport in the e-mobility space, impact on health outcomes, cost to access services or otherwise. <i>(Note: sex-disaggregated data will be collected in 100% of sub-projects relating to use of the vehicles. However, as the sub-projects have not yet been identified, the actual data to be collected will be finalised as part of the investment approval of each sub-project)</i>	Number of sub-projects where sex-disaggregated data is collected	100% sub-projects to collect sex-disaggregated data	Data confirmed to be collected included in Gender Action Plan for each sub-project (prior to investment approval) and included as a reporting covenant (at investment execution)	Gender specialist, counterparty to collect the data
5.4	Data collection and maintenance	Database established and maintained at the Platform level based on regular inputs of sex-disaggregated data from counterparties. Data collected can be used to develop other actions from the Platform or counterparties, refine due diligence of counterparties, develop thought leadership or otherwise to improve gender sensitivity in sub-projects and e-mobility.	Database established	1 comprehensive database established	Within 12 months of Platform establishment	Gender specialist

#	Activity	Actions	Indicators	Targets	Timeline	Responsibility
5.5	Sectoral guidelines	Develop sectoral guidelines on gender mainstreaming for Platform investments for use by staff on key issues and potential interventions that could be implemented by the Platform in upfront due diligence or financing documentation	Initial guidelines to be established informed by gender assessment and stakeholder consultations	Guidelines developed for 4 sub-sectors	Within 12 months of Platform establishment	Gender specialist
5.6	Investor reporting	Annual reporting to investors to include uniform gender reporting indicators for the Platform. Gender reporting based on sex-disaggregated data to also be included on a sub-project basis.	Annual investor reports	100% compliance with incorporation of gender reporting indicators to investors	Metrics developed within 12 months of Platform establishment	Gender specialist
5.7	Knowledge sharing – workshops	Workshops bringing together e-mobility industry participants to discuss opportunities to improve gender in e-mobility	Workshop organised by Platform with multi-stakeholder involvement	50% of attendees to be women	Within 4 years of Platform establishment (to allow for development of underlying data to support)	Gender specialist
5.8	Thought leadership	Aim to organise or participate in thought leadership (published article or otherwise) related to gender, climate and e-mobility.	Thought leadership published	2 thought leadership pieces	First within 2 years of Platform establishment (to allow for development of underlying data to support)	Gender specialist

6) Investment Procedures

Impact: Integrated of gender considerations into the investment process of the Platform to identify gender-related risks, mitigants and opportunities in sub-projects

#	Activity	Actions	Indicators	Targets	Timeline	Responsibility
6.1	Gender policy	Adopt a Gender Policy at the Platform. The policy will reflect the MAAML-GCF Gender Policy which will include alignment to the GCF Gender Policy such as commitment to gender analysis in investments under the program, monitoring and evaluation of gender equality where possible, collection of sex disaggregated data where possible, and collaboration with local and international organizations beneficiaries where possible. The Gender Policy shall be communicated to all the Program's employees.	Board adoption of Gender Policy	Communication to 100% of the employees subject to approved Gender Policy by the Board	Within 9 Months of setting up the Platform	The Platform Board will have ultimate responsibility for the establishment and maintenance of the Gender Policy Gender Specialist to be involved in development of Gender Policy
6.2	Sub-project Gender assessment	The investment due-diligence process by the Platform will include gender sensitivity assessment for each financing instrument. (Appendix A – Gender Assessment).	Gender Assessment approved as part of Investment approval for relevant sub-projects	100% sub-projects to have a Gender Assessment	At investment approval of each relevant sub-project	Gender specialist, Platform staff
6.3	Sub-Project Gender Action Plan	Gender Action Plan for each sub-project to be developed based on the Gender Assessment. This will include at a minimum the sex-disaggregated data that will be collected, any reporting covenants that are expected, and/or trainings to be provided.	Gender Action Plan approved as part of Investment approval for each sub-project	100% sub-projects to have a gender action plan	At investment approval of each relevant sub-project	Gender specialist, Platform staff

#	Activity	Actions	Indicators	Targets	Timeline	Responsibility
6.4	Stakeholder Engagement Plan	Drive preparation of project level Stakeholder Engagement Plan (SEP) and Grievance Redressal Mechanism (GRM), with women as separate stakeholder category. SEPs to ensure that consultations are culturally appropriate and conducted in gender responsive manner, with vulnerable groups also represented.	Stakeholder Engagement Plan approved as part of Investment approval for each sub-project	100% sub-projects to have a stakeholder engagement	At investment approval of each relevant sub-project	Gender specialist
6.5	Board reporting	Platform Board to receive annual report on implementation of Gender Policy and this Gender Action Plan, including compliance of counterparties as well as costs vs. budget of implementing the plan to ensure sufficient resource allocation.	Annual report to Platform Board	100% compliance with reporting obligations to Board	Annually (first report within 12 months of Platform establishment)	Gender specialist, Platform Board

APPENDIX A GENDER ASSESSMENT OF SUB-PROJECTS

The Platform will undertake specific gender assessment of sub-projects as part of the counterparty and sub-project due diligence (included in ESMS) prior to financing. The gender assessment will include review against equal access and benefits, identifying risks and mitigants, identifying opportunities for gender mainstreaming, what sex-disaggregated data will be collected and assessment of the counterparty.

During screening and due diligence gender-related data will be collected and analysed to determine whether the sub-project qualifies under the Platform's gender policies and identify business performance gender opportunities and risks. Gender-related data and information will be gathered from the potential counterparty for analysis to determine whether gender gaps present opportunities that may impact performance and affect the sub-project operations and financials.

The following checklist is to be completed by counterparties during the due diligence stage. Its aim is to enable the Platform to understand the current level of internal gender diversity and inclusion, to identify initiatives that support women entrepreneurs, and acknowledge efforts to serve women clients and users.

For the avoidance of doubt, any data collection, storage and utilisation of sex-disaggregated data will be compliant with local data privacy laws.

Table A1 Gender specific due diligence questionnaire

S.N	Parameter	Questions	Response	Remarks
1.	Leadership	Does the board and/or senior management have strong oversight and a clear governance process to support gender diversity and inclusion? Are any initiatives in place to promote women in leadership?	Yes ☐ No ☐	
		What is the gender composition of the Board?		
2.	Workforce	Are there specific policies and programmes to support inclusiveness of both men and women at work? If so, what are they? For example, specific HR policies that go beyond basic regulatory requirements (e.g. childcare support, flexible work, gender targets for construction and maintenance, standardised pay rates for each role).	Yes ☐ No ☐	
		Building on recent and/or existing data, are there differences between the roles/responsibilities/functions/levels/grades undertaken by men and women in the workforce, including permanent, contracted, seasonal staff as well as contractor construction and maintenance staff?	Yes ☐ No ☐	
		Does the company collect and monitor gender-disaggregated staff data (e.g. average salary, turnover, absenteeism, retention, and promotion)? How is the data analysed and to what extent is data used	Yes ☐ No ☐	

S.N	Parameter	Questions	Response	Remarks
		for decision-making on gender-related efforts (e.g. to better serve women clients)?		
		What training or skills development schemes are in place to address existing or emerging skills gaps?	Yes ☐ No ☐	
3.	Gender based violence and harassment governance	Has the company identified GBVH risks and are these reflected in the company's risk management system? Has GBVH been reflected in company policy, including a clear definition of GBVH (either as a separate policy or integrated into wider policies)? Has the company developed clear codes of conduct that prohibit all forms of GBVH? Are GBVH policies and codes of conduct available to workers and external stakeholders in accessible formats? Is mandatory training on GBVH and company policies and procedures provided to all workers of all grades, including contractors and security personnel, as well as induction training for new recruits?	Yes ☐ No ☐	
4.	Grievance mechanisms and reporting	Are confidential grievance mechanisms in place where reports and concerns related to GBVH can be raised? Are accessible reporting channels available to workers, community members and service users, including anonymous options? Are procedures in place to respond to reports of GBVH when they are made, including clear investigation procedures that focus on the safety and wellbeing of survivors? Have trained investigators been identified? Have reports of GBVH been made and how have they been handled? Is additional specialised training provided to members of staff with specific responsibilities for GBVH prevention and response? Has company leadership publicly committed to taking reports seriously?	Yes ☐ No ☐	

S.N	Parameter	Questions	Response	Remarks
5.	Equal opportunity	Are recruitment procedures in place, with interview panels staffed by at least two people? Are candidates' identities checked at interview and are references requested? Are written procedures in place for performance appraisals, promotions and any performance-related pay (if applicable)?	Yes ☐ No ☐	
6.	Project design	Have gender issues been identified as part of the project development? Have these issues been incorporated into project design and implementation plan?	Yes ☐ No ☐	
7.	Safety features and requirements	Have all required and safety features been incorporated in to the EVs, charging infrastructure or project design?	Yes ☐ No ☐	
8.	Stakeholder consultations	Have women and vulnerable people been involved in the stakeholder consultation process for this project? Have women had the opportunity to raise potential issues in the project's design and implementation?	Yes ☐ No ☐	
9.	Dedicated services	Does the company offer any community engagement programme supporting women's adoption and use of the company's infrastructures or services? Has the counterparty identified any opportunities to provide services dedicated to supporting women or gender mainstreaming?	Yes ☐ No ☐	
10.	Procurement	Are clauses included in contracts committing contractors and suppliers to adhere to company codes of conduct? Is information provided to contractors and suppliers about company grievance mechanisms for reporting GBVH? Is GBVH included in workplace safety assessments, including worker accommodation and transportation?	Yes ☐ No ☐	
11.	Products/ services	Does the company collect and monitor gender-disaggregated data on demand, service usage and/or customer segmentation? For example, related to	Yes ☐ No ☐	

S.N	Parameter	Questions	Response	Remarks
		access, availability, affordability, feasibility, and satisfaction.		
		If demand, service usage and/or customer segmentation data is collected, how is this data analysed and to what extent it is used for decision-making related to planning, design, marketing, sales, pricing, service(s) and distribution?	Yes ☐ No ☐	

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